

**MOLECULAR MAPPING OF DRY ROOT ROT
RESISTANCE GENE(s) IN CHICKPEA (*Cicer
arietinum* L.)**

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BY

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**DEPARTMENT OF GENETICS AND PLANT BREEDING,
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CERTIFICATE

This is to certify that the thesis entitled “**MOLECULAR MAPPING OF DRY ROOT ROT RESISTANCE GENE(s) IN CHICKPEA (*Cicer arietinum* L.)**” submitted by **Miss. ASHWINI KARADI** for the degree of **DOCTOR OF PHILOSOPHY** in **GENETICS AND PLANT BREEDING** to the University of Agricultural Sciences, Raichur is a record of research work done by her during the period of her study in this University under my guidance and supervision and the thesis has not previously formed the basis for the award of any degree, diploma, associateship, fellowship or other similar titles.

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Affectionately dedicated
to My Beloved Parents

Smt. Ganga S.K

Shri. Sadashiv B.K

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LIST OF ABBREVIATIONS

DRR	:	Dry root rot	RIL	:	Recombinant inbred line
%	:	Per cent	HSW	:	100 seed weight
ANOVA	:	Analysis of variance	Kg	:	Kilo grams
BLUPs	:	Best Linear Unbiased Predictors	ha	:	Hectares
CD	:	Critical difference	MAX	:	Maximum
cm	:	Centimetres	MIN	:	Minimum
CV	:	Coefficient of variation	MSS	:	Mean sum squares
df	:	Degrees of freedom	<i>et al</i>	:	And others
PCV	:	Phenotypic coefficient of variation	SNP	:	Single nucleotide polymorphism
GA	:	Genetic advance	Viz.	:	Namely
GAM	:	Genetic advance as percent of mean	SS	:	Sum of squares
GCV	:	Genotypic coefficient of variation	MSS	:	Mean sum of square
gm	:	Grams	σ^2_g	:	Genotypic variance
h^2	:	Heritability	σ^2_p	:	Phenotypic variance
DF	:	Days to 50 per cent flowering	DM	:	Days to maturity
PH	:	Plant height	HSW	:	Hundred seed weight
GY	:	Grain yield	BM	:	Biomass
HI	:	Harvest index	MAS	:	Marker assisted selection
R	:	Resistant	S	:	Susceptible
DI	:	Disease Incidence	QTL	:	Quantitative trait loci

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Introduction

I. INTRODUCTION

Chickpea (*Cicer arietinum* L.) is the world's second most important food legume, next to common bean, grown in over 50 countries with 90% of its area in the developing countries. It is grown mainly in arid and semi-arid regions of Asia and Sub-Saharan Africa. Chickpea is a self-pollinated diploid ($2n=2x=16$) cool season leguminous crop with an estimated genome size of 738 Mb (Varshney *et al.*, 2013). Though chickpea is grown in over 50 countries globally, South and South-Eastern Asia account for 79.2 % of the global chickpea production. India is the largest chickpea producing country, growing the crop in an area of 9.6 m ha with 8.8 m tonnes production and productivity of 920 kg ha⁻¹. The other major chickpea producing countries are Australia, Pakistan, Turkey, Myanmar, Iran, Ethiopia, Canada and USA.

Chickpea originated in the region of South-Eastern Turkey and adjoining areas of Syria (Singh, 1987). The progenitor of cultivated chickpea is *Cicer reticulatum*. Cultivated chickpea is highly self-pollinating legume crop that belongs to genus *Cicer*, tribe *Cicereae*, subfamily *Faboideae* and family *Fabaceae*. The main types of chickpea grown in the world are '*Kabuli*' (large ram-shaped, cream coloured) and '*Desi*' (small angular and dark coloured) (Van der Maesen, 1972). It plays an important role in human nutrition for large population in the developing world and is considered as a healthy food in many developed countries (Jodha and Subbarao, 1987). In addition to its high protein content (22-28%) and carbohydrates (~60%), chickpea is a good source of essential amino acids such as tryptophan and lysine, dietary fiber, minerals, unsaturated fatty acids and β -carotene (Awasthi *et al.*, 1991 and Jukanti *et al.*, 2012). Compared to other pulse crops, anti-nutritive components are very less in chickpea (Williams and Singh, 1987). Chickpea is known to have low glycemic index or glycemic load resulting in gradual rise in blood sugar levels.

The average yield of chickpea is low due to various production constraints. This is because the crop is largely grown on residual moisture in the post-rainy season and it is also affected by biotic and abiotic stresses (Gaur *et al.*, 2008). Terminal drought is the most important constraint to yield in chickpea accounting for about 50% yield reduction globally (Ryan, 1977).

Among biotic stresses, insects and diseases play a major role. Several soil-borne diseases cause yield losses in grain legumes worldwide and are considered prime targets for control by genetic resistance through plant breeding (Lenne and Allen, 1998). The production of chickpea is largely constrained by *Fusarium* wilt. However, recent reports indicated that dry root rot (DRR) has emerged as major disease in rainfed ecologies worldwide (Pande *et al.*, 2010; Sharma *et al.*, 2010 and Ghosh *et al.*, 2013), where 10-25 per cent loss in yield is due to wilt and DRR diseases (Mahendra Pal, 1998).

DRR was not of much significance in chickpea earlier, however, is gaining importance under the changing scenario of climate particularly on the account of longer drought spells. Higher temperature and soil moisture depletion during crop growth period particularly at post-flowering stage is predisposing chickpea to DRR in the semi-arid tropics of Ethiopia and in central and southern India (Sharma and Pande, 2013). It causes considerable yield losses that vary from 5-50% and may cause 100% losses in susceptible cultivars under favorable conditions. Although DRR is found in all chickpea growing areas of India, it is becoming severe in central and southern India, where the crop is grown mostly under rainfed conditions causing 30-40 per cent loss in yield.

DRR of chickpea is a soil borne disease caused by *Rhizoctonia bataticola* (*Macrophomina phaseolina*). Maximum ambient temperatures (above 30°C) and moisture stress (dry conditions) at the reproductive stages favour the disease development. Symptoms include a sudden drying of the plants in the field at flowering and podding, where the leaves and stems of affected plants remain straw coloured. Roots dry and become brittle exhibiting shredding of bark and break easily. Sometimes when the rest of the plant is dried, the top most leaves are chlorotic. The leaves and stems of affected plants are usually straw coloured. Dark, minute sclerotial bodies can be seen on the roots.

There is an urgent need to develop chickpea varieties with resistance to DRR for sustainable increase in yield in dry areas. Development of disease resistant varieties requires screening and evaluation of the available germplasm for resistance to diseases, and understanding the genetics of disease resistance. The identification of molecular markers closely linked to dry root rot resistance genes is important for resistance breeding, as it allows selection based on markers rather than resistance phenotype. So, molecular markers can greatly facilitate breeding for development of DRR resistant

cultivars in chickpea. This study was aimed to identify molecular marker for DRR resistance, which would help the breeders in breeding for DRR resistance through the marker-assisted selection.

In the framework of an effort to breed dry root rot resistance, the present study has been proposed with the following objectives:

1. To study the genetics of dry root rot resistance.
2. To identify molecular markers linked to dry root rot resistance genes.
3. To understand the relationship of dry root rot resistance with yield and other agronomical traits.

Review of Literature

II. REVIEW OF LITERATURE

The present study concerned primarily to study the genetics of dry root rot resistance, development of genetic linkage map, identification of genomic regions dry root rot resistance genes in chickpea and association studies. The available literature relevant to this study in chickpea and other legumes is reviewed and presented briefly under the following sections.

2.1 Chickpea

2.2 Major constraints in chickpea production

2.3 Genetic variability and genetic parameters for grain yield and yield related traits

2.4 Correlation and path analysis

2.5 Screening of chickpea genotypes for DRR resistance

2.6 Genetics of DRR resistance

2.7 Understand the relative comparison of resistant and susceptible RILs for yield along with other agronomical traits and DRR disease incidence

2.8 DNA markers linked to DRR resistance genes

2.1 Chickpea

Chickpea (*Cicer arietinum*) is a major grain legume in the semi-arid tropics. This cool season food legume is self-pollinated, annual and diploid ($2x = 2n = 16$) in nature. It belongs to the genus *Cicer*, tribe *Cicereae*, sub-family Papilionaceae and family Leguminosae. The genus *Cicer* includes 43 species, nine of which are annual, 33 are perennial and one with unspecified life cycle (Van der Maesen, 1987). The species *C. arietinum* is the only cultivated species of this genus. Van der Maesen (1987) categorized the genus *Cicer* into four sections. Eight annual species, namely *C. arietinum*, *C. reticulatum*, *C. echinospermum*, *C. pinnatifidum*, *C. bijugum*, *C. judaicum*, *C. yamashitae* and *C. cuneatum* were placed in section Monocicer, *C. chorassanicum* and *C. incisum* (perennial species) in section Chamaecicer, 23 perennial species in section Polycicer and

seven woody perennial species in section *Acanthocicer*. It was found that two wild annual *Cicer* species (*C. echinospermum* and *C. reticulatum*) native to eastern Turkey are closely related to the cultivated chickpea. Later, *C. reticulatum* was proposed as the immediate wild progenitor of domesticated chickpea based on cytogenetic analyses (Ladizinsky and Adler, 1976a and 1976b).

The cultivated chickpea is a crop of ancient origin and is one of the first grain legumes domesticated in the old world about 7000 years ago (Van der Maesen, 1987). It is believed to have originated in south eastern Turkey and the adjoining northern region of Syria (Van der Maesen, 1987). Chickpea can be easily distinguished into two subtypes: *Kabuli* (also known as macrosperma) and *Desi* (also known as microsperma) differing in their geographical distribution, seed and plant characteristics. The *desi* types predominately found in Central Asia and in the Indian subcontinent are characterized by pink coloured flowers, small, angular, brown seeded, having anthocyanin pigmentation on most of the plant parts. While, the *kabuli* types mostly found in the Mediterranean region do not show anthocyanin pigmentation in any part of the plant and they have large, cream colored, smooth seeds with white flowers. Between the two subtypes, *desi* type is believed to be originated first, followed by *kabuli* type, developed by selection and mutation (Singh, 1997). From the linguistic evidence, it is found that *Kabuli* type reached India via Kabul, the capital of Afghanistan about two centuries ago and acquired the name as kabuli chana in Hindi (Van der Maesan, 1972). In India, *desi* types occupy nearly 90% of total area under chickpea cultivation. Across the world, the ratio of *desi* and *kabuli* chickpea production is 3:1. Recently, the chickpea draft genome sequence was published and it was found that the genome size of cultivated chickpea (*C. arietinum*) is 738 Mbp which is only 1.5 times higher than *Medicago truncatula* (Varshney *et al.*, 2013a).

2.2 Major constraints in chickpea production

Despite being a hardy, deep-rooted dry land crop, the productivity of chickpea is not up to its yield potential. Globally, biotic and abiotic stresses are the major constraints limiting chickpea production. More than 50 pathogens have been reported to affect chickpea, but only a few devastate the crop. Fusarium wilt (FW) [caused by *Fusarium oxysporum f.sp. ciceri*], dry root rot [caused by *Rhizoctonia bataticola*] and collar rot

[caused by *Sclerotium rolfsi*] are the important root diseases of chickpea in areas where the chickpea growing season is dry and warm e.g., Southern and Eastern Asia (Central and Southern India) and Eastern Africa. While, Ascochyta blight (AB), [caused by *Ascochyta rabiei* (Pass.) Labr.] and botrytis grey mold (BGM) [caused by *Botrytis cineria* Pres.] are the important foliar diseases in the areas where the chickpea growing season is cool and humid. AB is important in the west and central Asia, North Africa, North America, Australia, the northern part of India and Pakistan, while BGM is important in Nepal, Bangladesh, northern India and Australia (Gaur *et al.*, 2010).

Drought and heat are considered the most important abiotic stresses affecting chickpea production globally. It is estimated that 50% of the yield losses caused by abiotic stresses are due to drought and heat (Gaur *et al.*, 2014a). Chickpea is grown mostly as a post-rainy season crop on residual soil moisture from the preceding rainy season. Apart from this, recently a large shift of chickpea cultivation from cooler long growing seasons to short growing seasons were observed in India. Also, the efforts of farmers to increase the cropping intensity forcing chickpea cultivation to relatively warmer environments. The crop is, therefore, often exposed to terminal drought and heat stress causing a heavy yield loss.

Pod borer (*Helicoverpa armigera* Hubner.) is the most important pest of chickpea worldwide. It is a highly polyphagous pest and can feed on various plant parts such as leaves, tender shoots, flower buds, and immature seeds. The extent of global losses to chickpea by this pest is estimated at over US\$ 500 million (Ryan, 1997 and Gaur *et al.*, 2012). The viral diseases, rust (*Uromyces ciceris-arietini*), root nematodes (*Meloidogyne* sp.), Phytophthora root rot (*Phytophthora medicaginis*), powdery mildew (*Oidiopsis taurica* Lev.), cutworm (*Agrotis* sp.) and leaf miner (*Liriomyza cicerina*) are also important in some chickpea growing areas.

2.3 Genetic variability and genetic parameters

Burton (1952) has given the use of heritability estimates with genotypic coefficients of variation to give precise estimates of genetic advance. Johnson *et al.* (1955) have suggested that in a selection programme heritability values as well as estimates of genetic advance are more useful than heritability alone.

According to Hanson *et al.* (1956) heritability and genetic advance are the two complementary concepts. The heritability values may be used to estimate the expected genetic advance through selection. The heritability enables the plant breeder to base his selection on phenotypic performance for improvement of character.

Maloo and Sharma (1987) analyzed 21 diverse varieties of gram for estimation of variability parameters. The estimates of genotypic coefficients of variation ranged from 1.58 (for days to maturity) to 40.26 (for grain yield/plant). High genetic advance as per cent of mean coupled with high heritability were recorded for seed yield.

Jivani and Yadavendra (1988) estimated high heritability parameters for seed yield/plant and nine yield related traits in 42 genetically diverse genotypes of *Cicer arietinum* and observed that, both coefficients of variation were high for 100-seed weight, plant height, days to flowering and maturity, 100-seed weight and harvest index and high heritability estimates. The greatest genetic gains were expected for 100-seed weight, pods/plant and days to flowering.

Samal and Jagdev (1989) showed high PCV and GCV for seed weight and yield and moderately high for plant height and days to flowering indicating relatively low influence of environment on these characters. High heritability coupled with high genetic advance was observed for seed mass and yield.

Sharma *et al.* (1990) reported high range of variability for harvest index followed by days to maturity and days to flowering. Coefficient of genetic variation was highest for 100-seed weight, while moderate for plant height, harvest index, days to flowering and maturity. High heritability (b.s.) coupled with high genetic advance was observed for 100-seed weight, whereas plant height recorded high heritability with moderate genetic advance.

Pundir *et al.* (1991) reported considerable variation for days to flowering (46 - 71 days), 100-seed mass (5.00 - 60.94 g) and seed yield per plot (204.00 - 3333 kg/ha) in chickpea germplasm.

Sandhu *et al.* (1991) observed a wide range of variability for grain yield, plant height and 100-seed weight in 55 diverse lines of chickpea. They also observed that, grain

yield per plot followed by pods per plant and 100-seed weight had maximum genetic variation. The heritability estimates for different characters ranged from 8.4 % (primary branches) to 86.1 % (seed yield) and plant height and 100-seed weight had considerably high estimates of variability. The genetic advance as per cent of mean was highest (73.44 %) for seed yield per plot.

Jahagirdar *et al.* (1994) reported high genotypic and phenotypic coefficients of variation for number of pods per plant and moderately high values of genetic variability for days to 50 per cent flowering and plant height. High values of heritability coupled with high genetic advance were observed for 100-grain weight and days to 50 per cent flowering.

Mathur and Mathur (1996) studied genotypic and phenotypic coefficients of variations, heritability and expected genetic advance for grain yield and its contributing characters in 34 varieties of chickpea (*Cicer arietinum*) during *rabi* season. They reported that the highest values of genotypic and phenotypic coefficients were obtained for 1000-seed weight. High heritability for all characters was recorded. Seed weight and grain yield per plant had good values of genetic advance with high heritability.

Sable *et al.* (2000) studied 30 genotypes of chickpea characters viz., seed yield per plant and 100-seed weight exhibited high estimates of GCV indicating high degree of variation due to genetic factors. The characters viz., seed yield per plant, 100-seed weight and biological yield per plant exhibited high estimates of genetic advance accompanied with high estimates of heritability.

Gumber *et al.* (2002) studied 30 genotypes of chickpea and observed highest heritability for 100-seed weight (69.4 %). The estimate of phenotypic coefficient of variation (PCV) was highest for seed yield (38.64 %).

Muhammad *et al.* (2004) studied variability, heritability, genetic advance, correlation coefficients and path coefficients for yield and its components in 24 advance lines of chickpea. Grain yield had positive and significant correlation with plant height, pods per plant, 100-seed weight and biological yield. High direct effects were contributed by biological yield and harvest index although the later had negative association with grain yield.

Borate *et al.* (2010) studied population parameters such as range, mean, phenotypic and genotypic variances, PCV and GCV, heritability and genetic advance for 13 agronomic characters in a set of 30 chickpea genotypes. Range of variability was appreciable for days to first flowering, plant height, dry matter and grain yield. PCV showed higher values than GCV for all characters. High heritability coupled with high genetic advance was observed for grain yield.

Sharma and Saini (2010) evaluated that the analysis of variance exhibited significant differences among genotypes for all the characters indicating the presence of genetic variability among the genotypes. The magnitude of the PCV was higher than GCV for all the characters due to the influence of environment on expression of these characters. The highest magnitudes of GCV as well as PCV were recorded for 100-seed weight followed by seed yield per plant, plant height, days to flowering and days to maturity.

Khan *et al.* (2011) observed highly significant differences among the genotypes for days to 50 per cent flowering (90-122 days), days to maturity (163-178 days), plant height (47-94 cm), 100-seed weight (13-39 g), biological yield per plant (14.3-68 g) and grain yield per plant (4.1-28.9 g). The trait grain yield per plant had maximum phenotypic and genotypic coefficient of variation (PCV and GCV) followed by biological yield per plant. Heritability estimates of all the traits were high. Highest heritability was recorded for days to 50 per cent flowering (93) followed by biological yield per plant (89), plant height (88), 100-seed weight (82), grain yield per plant and days to maturity (68). The genotypes were identified as genetically diverse and can be utilized in future chickpea improvement programs.

Akhtar *et al.* (2011) observed higher phenotypic coefficient of variation (PCV) for days to flowering, days to maturity, plant height and seed yield than genotypic coefficient of variation, which means that the expression of these traits is more influenced by environmental effects. Therefore, suggested that the grain yield could be improved by using the 100-seed weight and number of pods per plant as selection criteria in chickpea.

Jha *et al.* (2012) studied thirty chickpea genotypes for seed yield and its components for estimating genetic variability and correlation co-efficient. Analysis of variance for seed yield and its components showed significant differences among the

genotypes for 50% flowering, days to maturity, plant height, 100-seed weight and seed yield/ plant.

Babbar *et al.* (2012) studied 44 promising lines of chickpea under late sown condition. The maximum genotypic coefficient of variation was noticed for total number of seeds per plant and total number of pods per plant. Days to 50 per cent flowering, days to maturity, plant height, 100-seed weight and seed yield per plant showed high heritability coupled with medium genetic advance as percentage of mean.

Gul *et al.* (2013) evaluated 20 chickpea genotypes for genetic potential, heritability, genetic advance and association of yield contributing characters. Analysis of variance revealed significant variation among genotypes for all the characters. Heritability estimates were highest for pods per plants (0.88), test weight (0.82) and seed yield per plant (0.75). Genetic advance was higher for pods per plant (40.14%) and seed yield per plant (38.24%)

Nizama *et al.* (2013) observed high genetic advance as percentage of mean for 100-seed weight and seed yield per plant. Puri *et al.* (2013) studied 42 genotypes of chickpea and variability analysis revealed sufficient variability for most of the characters studied viz. days to first flowering, days to 50 per cent flowering, days to maturity, 100-seed weight, biological yield, harvest index and seed yield per plant. Monpara and Dhameliya (2013) estimated high heritability for flowering periods followed by days to maturity, plant height and seed yield per plant.

Karami and Talebi (2013) studied five genotypes of chickpea in a half diallel cross. High values of narrow-sense heritability were observed for harvest index (67%), 100-seed weight (56%) and plant height (42%).

Singh *et al.* (2013) evaluated 26 genotypes over four environments. High estimates of heritability were observed for 100-seed weight plant height and seed yield.

Jivani *et al.* (2013) conducted field trial to assess the magnitude of genetic variability, heritability, genetic advance and genetic diversity using 105 diverse genotypes of chickpea. High heritability coupled with high genotypic coefficient of variation (GCV) and high genetic advance as per cent of mean (GA %) were observed for seed yield per

plant, biological yield per plant, harvest index and 100-seed weight which indicated that response to selection would be very high for these yield components.

Aarif *et al.* (2014) revealed high phenotypic coefficient of variation for 100-seed weight, followed by seed yield per plant. High heritability observed for 100-seed weight.

Mallu *et al.* (2014) found highest estimates of broad sense heritability for 100-seed weight, followed by days to 50 per cent flowering and plant height. Malik *et al.* (2014) studied the genetic diversity of 113 *desi* chickpea genotypes through descriptive, principal component and cluster analysis and high variances were observed for days to flowering, maturity, plant height, pods per plant, biological yield and harvest index.

Jain *et al.* (2014) studied 30 genotypes of chickpea and high values for phenotypic coefficient of variation (PCV) and genotypic coefficient of variation (GCV) were recorded for grain yield/ plant (g).

Kuldeep *et al.* (2014) evaluated 100 advance breeding lines of chickpea and the analysis of variance revealed significant differences among the genotypes for all the traits indicating presence of sufficient variability among the genotypes for various traits. High estimates of GCV and PCV were observed for seed yield per plant, 100- seed weight, harvest index, number of effective pods per plant, total number of pods per plant and number of secondary branches. High genetic advance as percentage of mean for seed yield per plant, followed by 100-seed weight and harvest index, while moderate genetic advance was observed for plant height and days to flower initiation.

Sharanappa *et al.* (2014) estimated high heritability coupled with high genetic advance for the traits like plant height, biological yield/plant and harvest index.

Singh *et al.* (2014) evaluated 29 genotypes of chickpea in four environments. High phenotypic and genotypic coefficients of variation coupled with high heritability and genetic advance as per cent of mean were reported for seed yield/plant, indicating predominance of additive gene effects in controlling these characters.

Anita *et al.* (2015) carried out studies to estimate genetic variability and heritability for seed yield in 31 genotypes of chickpea grown under rice fallow conditions. Highly significant differences existed among the genotypes tested for all the traits. Traits

such as seed yield per plant (40.20), biological yield (40.10), test weight (36.60) and harvest index (30.60) showed high phenotypic and genotypic coefficients of variation, heritability and genetic advance as percentage of mean.

Babbar *et al.* (2015) estimated genetic variability, heritability, interrelationships and carried out diversity analysis for seed yield and its components in 31 genotypes of chickpea grown under rice fallow conditions. Highly significant differences existed among the genotypes tested for all the traits. Traits such as seed yield per plant, biological yield, 100-seed weight and harvest index showed high phenotypic and genotypic coefficients of variation, heritability and genetic advance as percentage of mean.

Thakur *et al.* (2015) evaluated 38 genotypes for yield and yield contributing traits. High heritability and genetic advance as percentage of mean for 100-seed weight was observed. While, low heritability coupled with low genetic advance was recorded for plant height, days to maturity, biological yield and harvest index indicating high impact of environmental conditions on these characters.

Yadav *et al.* (2015) estimated high heritability coupled with high genetic advance for seed yield/plant in 45 chickpea genotypes. High heritability with moderate genetic advance was observed for plant height. However, low heritability coupled with low to moderate genetic advance has been observed for days to maturity.

Dehal *et al.* (2016) carried out studies by using 25 chickpea genotypes for two different environments. The results found that high heritability coupled with high genetic advance for seed yield per plant (80.08%) and test weight (85.76%) in environment-I. The genotypes showed highly significant differences for all the characters studied in both environments. Environment-I (normal sowing) exhibited its excellent potential for the traits viz., seed yield per plant, biological yield per plant, pods per plant, days to 50 per cent flowering, days to maturity and plant height, whereas environment II (late sowing) exhibited its excellent potential for harvest index and test weight.

Akanksha *et al.* (2016) studied 38 chickpea varieties to find out genetic variability, genetic association and path coefficient analysis for grain yield and its attributing characters. Both genotypic and phenotypic variances were highly significant for all the traits with higher phenotypic coefficient of variation. The low differences

between the phenotypic and genotypic coefficients of variation indicated low environmental influences on the expression of these characters. High heritability coupled with high genetic advance was obtained for harvest index (97.70%), seed yield per plant (86.60%) and days to maturity (92.40%).

2.4 Correlation and path analysis

Kunadia *et al.* (1986) investigated similar magnitudes of phenotypic and genotypic correlation coefficient in *kabuli* chickpea. They further reported seed yield was significantly and positively associated with days to flowering. Association of days to maturity with pods per plant and 100-seed weight were significant but negative. While, correlation between pods per plant and 100-seed weight was positive and significant.

Maloo and Sharma (1987) recorded data on seed yield per plant and seven yield related traits for 21 diverse varieties of chickpea and from path analysis it was observed that pods per plant had the highest direct effect on seed yield followed by 100-seed weight and days to flowering.

Paliwal *et al.* (1987) estimated correlation in seven traits in 36 genotypes and found that seed yield per plant was positively correlated (at phenotypic level) with plant height, days to 50% flowering and days to 95% maturity. Path analysis revealed that 100-seed weight had the highest positive direct effect on yield and days to 95% maturity.

Jivani and Yadavendra (1988) studied correlation and path analysis in 42 genetically diverse chickpea genotypes and reported that the yield was significantly and positively correlated with 100-seed weight ($r = 0.3$) and harvest index ($r = 0.3$). It was suggested that branches and pods per plant, 100-seed weight and harvest index could be used to make indirect selection for improving seed yield.

Zade and Wanjari (1988) estimated correlation among 44 genotypes of chickpea. They reported that yield was significantly and positively correlated with pod weight per plant and 100-seed weight and negatively correlated with crop duration and duration of the reproductive phase. They further suggested that selection for increased pod weight per plant, 100-seed weight and earliness would help in improving the yield.

Wadud and Yaqoob (1989) studied correlation in ten *kabuli* chickpea varieties and found that seed yield was positively correlated with days to maturity, plant height and 100-seed weight. Tagore and Singh (1990) reported that grain yield was positively and significantly associated with 100-seed weight and harvest index.

Yaqoob *et al.* (1990) studied the inter-relationship between grain yield and other important characters in 12 genotypes of chickpea and reported negative correlation between grain yield and days to maturity.

Jain *et al.* (1992) studied multiple correlation and regression in 14 varieties of chickpea and indicated that a combination of two or three variables viz., seed weight, number of days to first flowering and number of days to maturity were better than other combinations of the characters for the improvement of chickpea.

Arora and Kumar (1994) investigated correlation in 40 genotypes of chickpea. They reported positive associations of seed yield per plant with biological yield per plant, plant height and 100-seed weight. They also carried out path coefficient analysis and reported that biological yield per plant, harvest index and 100-seed weight had highest positive direct effect on seed yield per plant.

Sarvalia and Goyal (1994) studied correlation among ten yield components in 76 chickpea genotypes and revealed significant association between yield and 100-seed weight, plant height, days to maturity and days to flowering at genotypic and phenotypic levels.

Khorgade *et al.* (1995) observed positive significant association between seed yield, harvest index and 100-seed weight. They also reported character and path analysis under normal and late sown conditions in chickpea and revealed that biological yield per plant had the maximum positive direct effect on seed yield in both conditions followed by harvest index.

Sandhu and Mangat (1995) estimated correlation and path analysis in 32 diverse genotypes of chickpea and found that yield per plant was significantly and positively correlated with harvest index, whereas negatively associated with plant height and days to

flowering. They also found that harvest index and 100-seed weight had direct effect on yield.

Tripathi *et al.* (1995) evaluated 40 varieties of chickpea for eight quantitative traits and reported high positive association of yield per plant with 100-seed weight.

Mathur and Mathur (1996) found significant positive genotypic correlation of grain yield per plant with 1000-grain weight while negative correlations with plant height and days to flowering in an inter-relationship study using 34 varieties of chickpea.

Khedar and Maloo (1999) evaluated 40 genetically diverse genotypes of chickpea and revealed that seed yield was significantly and positively correlated with 100-seed weight. Path analysis indicated that pods per plant had the highest direct effect on seed yield followed by seeds per pod and 100-seed weight.

Saleem *et al.* (2002) reported that the chickpea genotypes showed highly significant differences for all the characters studied. Seed yield per plant was significant positively correlated with days to flowering and 100-seed weight both at the genotypic and phenotypic levels.

Muhammad *et al.* (2004) studied correlation coefficient and path coefficient for yield and its components in 24 advance lines of chickpea. Grain yield had positive and significant correlation with plant height, 100-seed weight and biological yield. High direct effects were contributed by biological yield and harvest index although the later had negative association with grain yield.

Babbar and Patel (2005) studies on chickpea revealed positive significant association of seed yield with biological yield, pod weight per plant, number of pods per plant and days to maturity.

Meena *et al.* (2006) observed higher estimates of genotypic correlation coefficient than the phenotypic components. The seed yield showed significant positive correlation with 100-seed weight, biomass and plant height at genotypic level and phenotypic level. Significant negative correlations of seed yield were observed with days to maturity at both genotypic and phenotypic levels.

Singh (2007) noticed that the correlation and path analyses are important tools for getting appropriate information regarding inter-relationship among different characters for effective selection programme. In 45 diverse genotypes of chickpea, the genotypic correlations were higher than the corresponding ones. Seed yield had highly significant positive correlation with biological yield per plant and harvest index.

Talebi *et al.* (2007) reported positive and significant relationship between 100-seed weight, plant height, days to maturity, seed yield and harvest index. Negative and significant relationships were recorded for 100-seed weight. Harvest index had the greatest direct effect on seed yield (PC = 0.901). Its indirect effect on seed yield was highly positive through plant height and biomass, but negative and low through days to heading, days to maturity and 100-seed weight. The selection for high seed yield should be based on biomass and harvest index.

Bhavani *et al.* (2008) revealed that the 100 seed-weight, harvest index, days to 50 per cent flowering and days to maturity were the most important attributes which contributed towards higher yield. Therefore, it was emphasized that these components should be given more consideration while selecting for higher yield. The highest direct and positive effect on seed yield was exhibited by days to maturity, days to 50 per cent flowering and harvest index.

Yadava and Singh (2008) reported that harvest index and 100-seed weight showed positive and significant association with seed yield in all the environments.

Zena *et al.* (2008) studied path coefficient analysis in 36 genotypes of chickpea for total 12 quantitative characters including yield. Biological yield had shown highest direct effect on the seed yield which was followed by test weight, yield index, initial flowering period and maturity period, while 50 per cent flowering period has shown negative direct effect on the yield of grain.

Thakur *et al.* (2009) evaluated 53 chickpea genotypes over two seasons to find out association among characters and found that seed yield per plant exhibited stable positive association with biological yield per plant, plant height and harvest index at genotypic and phenotypic levels in individual as well as combined over seasons.

Ali *et al.* (2010) reported that pods per plant, seeds per pod and 100-seed weight were positive and significant at genotypic level but positive and highly significant at phenotypic level with biological yield per plant. Days to flowering was negatively and non-significantly correlated with seed yield per plant at both genotypic and phenotypic levels.

Ozveren and Anlarsal (2010) carried out an experiment to determine selection criteria by using correlation and path coefficient analysis in 22 genotypes of chickpea under Mediterranean conditions. In investigated characters, positive and significant relationships were found among seed yield and harvest index and seed number. The path coefficient analysis revealed that, harvest index had the greatest direct effect on seed yield (0.4206). Both correlation and path analyses indicated that harvest index was the major direct contributor to seed yield.

Akhtar *et al.* (2011) studied the correlation (days taken to 50 per cent flowering, plant height, number of pods per plant, days taken to 90 per cent maturity, 100-seed weight) in 20 advance genotypes of chickpea collected from various sources along with one check variety (Pb-2000) and revealed that significant and positive correlations were found between yield and 100-seed weight, number of pods per plant and plant height.

Ali *et al.* (2011) revealed that days to flowering, days to maturity, primary branches per plant, secondary branches per plant were positively correlated with the grain yield per plant at genotypic and phenotypic levels.

Ahsan (2011) found that number of secondary branches per plant, biomass per plant and number of pods per plant were positively correlated with grain yield per plant at genotypic and phenotypic levels. Genotypic correlation of grain yield per plant with plant height was positive and significant. Genotypic and phenotypic correlations of seed yield per plant with days to maturity were negative and highly significant.

Mishra and Babbar (2011) observed that seed yield per plant was positively and significantly correlated with biological yield per plant, 100-seed weight and harvest index in normal and late sown conditions. Ahmad *et al.* (2012) studied four morphological traits and found that the average coefficient of variation was 56.8% with significant correlation among yield traits.

Jha *et al.* (2012) observed significant and negative genotypic correlation between primary branches per plant and plant height; primary branches per plant and plant width; primary branches per plant and days to 50 per cent flowering; primary branches per plant and days to maturity; plant height and plant width; plant height and pods per plant; seeds per pod and 100-seed weight; days to 50 per cent flowering and 100-seed weight.

Babbar *et al.* (2012) evaluated 44 promising lines of chickpea under late sown condition. Seed yield per plant showed high significant positive correlation with biological yield, plant height and 100-seed weight but, significant negative correlation with days to 50 per cent flowering was also observed.

Kumar *et al.* (2012) observed that there is significant and positive correlation between 100-seed weight, harvest index, seed yield/plant and harvest index at both phenotypic and genotypic level.

Naveed *et al.* (2012) showed that biomass per plant and 100-seed weight were positive and significant at genotypic level but positive and highly significant at phenotypic level.

Naveed *et al.* (2012) carried out correlation and path coefficient analysis on 20 chickpea genotypes and observed that days to flowering and maturity, biomass per plant and 100-seed weight had direct effects on grain yield per plant.

Shrivastava *et al.* (2012) studied that biological yield, harvest index, days to maturity and seeds per pod exhibited maximum direct positive effect on seed yield per plant under rice fallows.

Devasirvatham (2013) found that phenological traits *viz.*, days to first flowering and days to pod maturity showed significant and negative correlation among themselves.

Ramanappa *et al.* (2013) carried out analysis of variability for economically important traits in 179 genotypes of chickpea and found highly significant positive association of grain yield with number of pods per plant, harvest index, number of primary branches, plant height and number of seeds per pod at phenotypic level.

Monpara and Dhamelia (2013) revealed that days to flowering, flowering period and days to maturity had significant positive association among themselves. Pandey *et al.* (2013) reported that grain yield per plant was positively and significantly correlated with days to flowering, plant height, number of branches per plant, number of pods per plant and 100-seed weight.

Aarif *et al.* (2014) observed that seed yield per plant exhibited significant positive association with biological yield per plant and harvest index; and significant negative correlation with days to maturity at genotypic level.

Jadhav *et al.* (2014) studied 45 genotypes of chickpea. Coefficient analysis revealed that seed yield per plant exhibited positive and significant correlation with plant height (0.2736, 0.2957), test weight (0.5892, 0.6193) and harvest index (0.4453, 0.4806) at both genotypic and phenotypic level. The characters biological yield per plant and days to maturity depicted positive but non-significant association. Days to 50 per cent flowering exhibited negative and non-significant association with seed yield per plant at both genotypic and phenotypic level. Thus, these are important traits for the selection in crop improvement program in chickpea.

Mishra and Babbar (2014) observed that positive and significant correlation of seed yield per plant was found with harvest index, seed size and biological yield per plant under timely and late sown conditions.

Malik *et al.* (2014) observed high variances for days to flowering, maturity, plant height, biological yield and harvest index. These traits showed positive significant correlation with yield.

Parhe *et al.* (2014) evaluated 51 diverse genotypes of chickpea for direct and indirect influences of yield components on grain yield and concluded that 100-seed weight, had the highest positive direct effects on grain yield.

Anita *et al.* (2015) estimated correlation for 31 genotypes of chickpea grown under rice fallow conditions. Significant and positive correlations were found between seed yield and biological yield, harvest index and test weight while negative with phenological traits.

Bala *et al.* (2015) studied 20 chickpea genotypes for the nature and extent of genetic variability as well as, direct and indirect effects of yield related traits on seed yield. The path coefficient analysis revealed the presence of positive and high direct effects of biological yield per plant and harvest index on seed yield per plant.

Tilahun *et al.* (2015) assessed genotypic variability, heritability and correlation of 17 kabuli genotypes for yield and yield related traits under rainfed condition and reported that pods per plant, plant biomass, plant height, harvest index and 100-seed weight were positively correlated with grain yield.

Joshi and Yasin (2015) found that biological yield per plant exerted maximum direct effect on seed yield, followed by harvest index. Days to 50 per cent flowering and days to maturity had negative indirect effect on seed yield *via* biological yield and harvest index. Shafique (2016) revealed that maximum positive direct effect toward seed yield was exerted by number of pods per plant with 52.87 %.

Dehal *et al.* (2016) carried out correlation and path analysis studies with 25 chickpea genotypes under two different environments and found seed yield per plant was positively and significantly correlated with pods per plant ($E1=0.767$ and $E2=0.647$), harvest index ($E1=0.767$ and $E2=0.745$), biological yield per plant ($E1=0.612$ and $E2=0.537$) and primary branches per plant ($E1=0.422$ and $E2=0.515$) in both the environments. Path coefficient analysis revealed the high direct effect of biological yield and harvest index on seed yield per plant. The study suggested that selection for high seed yield should be based on biological yield and test weight in chickpea.

Sanjeev (2016) studied correlation and path analysis successively during *rabi* season among 17 diverse genotypes of *desi* chickpea for seed yield and its components. Correlation studies revealed that seed yield per plant were significantly and positively correlated with plant height, test weight and negatively correlated with number of secondary branches and days to 50 per cent flowering. Path analysis indicated that the highest direct and positive effect of plant height (0.46) followed by days to maturity (0.23), and test weight (0.19). The study revealed that test weight and plant height were important characters for improvement of seed yield per plant in chickpea.

Akanksha *et al.* (2016) studied 38 chickpea varieties to find out genetic variability and genetic association of grain yield and its attributing characters. Both GCV and PCV were highly significant for all the traits with higher phenotypic coefficient of variation. The low differences between the PCV and GCV indicated low environmental influences on the expression of these characters. Harvest index, test weight and plant height were the most important characters, which possessed positive association with seed yield per plant. Path coefficient analysis revealed that among the different yield contributing characters harvest index, test weight, plant height, days to maturity and days to 50 per cent flowering influenced seed yield per plant directly.

2.5 Screening of chickpea genotypes for dry root rot resistance

Nene *et al.* (1981) developed pot culture technique for green house screening and blotter paper technique for laboratory screening of chickpea genotypes for resistance to dry root rot. Nene *et al.* (1989) evaluated *Cicer arietinum* genotypes for broad based resistance to root rots caused by *Rhizoctonia bataticola* at 24 locations in 11 countries during 1978-82. Genotypes ICC 2862, ICC9023, ICC 9032, ICC 10803, ICC 11550 and ICC 11551 were resistant to dry root rot.

Oad *et al.* (1995) identified two varieties viz., Sanyasi and C 612, as resistant to *R. bataticola*. Al Shareef *et al.* (2009) observed 35 genotypes showing resistance and other five genotypes showing moderately susceptible reaction out of the 40 chickpea genotypes screened for resistance to wilt and root-rot diseases during 2007-08.

Chaturvedi and Dua (2009) screened 25 resistant cultivars. Cultivars KPG 59, Radhey and K 50 were found to be most promising against dry root rot of chickpea.

Muhammad Saifulla *et al.* (2011b) screened 16 chickpea genotypes for collar rot, dry root rot, wilt and rust diseases at GKVK, Bengaluru during 2010-11. The genotypes HIR 60, BGD 256, A 1, JG 11, BBG 1, BBG 2, JG 62, and L 550 were resistant to moderately resistant to dry root rot. The genotypes HIR 60, BGD 256, A1, BBG 1 and BBG 2 showed multiple disease resistance to collar rot, dry root rot, wilt and rust diseases.

Gupta *et al.* (2012) assessed 170 chickpea genotypes for locating new and better sources of resistance against dry root rot through blotter paper technique and in multiple disease sick fields at JNKVV, Jabalpur during 2007 to 2010. The susceptible cultivar BG 212 was used which exhibited 100 % mortality *i.e.*, rating 9 in 1-9 scale. Out of 170 accessions, 68 genotypes exhibited resistant reaction.

Gowdra Nagamma (2012) screened 165 chickpea genotypes for dry root rot under laboratory condition using blotter paper technique. Thirty-two genotypes showed moderately susceptible reaction, while 97 and 36 genotypes exhibited susceptible and highly susceptible reaction, respectively.

Talekar *et al.* (2017) screened 520 germplasm lines. Four genotypes, viz., PG06102, ICC2867, BG2094 and IC552137 showed resistant reaction to *Rhizoctonia* under laboratory condition.

2.6 Genetics of dry root rot resistance

The literature available on genetics of dry root rot resistance is very meager and thus the literature on other diseases is presented apart from dry root rot.

Rao and Haware (1987) studied the inheritance of resistance to dry root rot of chickpea. Parental, F₁ and F₂ populations of two resistant and two susceptible parents along with 49 F₃ progenies of one of the resistant × susceptible cross were tested for their reaction to dry root rot using blotting paper technique. All the F₁ plants of the resistant × susceptible crosses were resistant, while in F₂ generation, segregation was 3 resistant: 1 susceptible indicating monogenic inheritance, where resistance being dominant over susceptibility.

Talekar (2014) screened a set of 520 chickpea accessions to identify the genetics of dry root rot resistance and study revealed that a high level of resistance in three entries, namely PG 06102, BG 2094 and IC 552137. The pattern of segregation in the F₃ generation of the cross L 550 (susceptible) × PG 06102, the resistance in PG 06102 was under monogenic control and resistance being dominant over susceptibility. Both studies have confirmed the genetics of dry root rot to be monogenic with resistant being dominant over susceptibility.

2.7 Understand the relative comparison of resistant and susceptible RILs for yield along with other agronomical traits and DRR incidence

Ahmad *et al.* (1986) reported that 78.80 % loss in chickpea seed weight/plant which was caused by *Rhizoctonia bataticola* at full podding stage, 48.90 % at pre-harvest stages and 33.90 % reduction in 1000-seed weight. Mohan *et al.* (2005) recorded maximum yield (14.8 q/ha and 14.9 q/ha) when there was low disease incidence (18.10 and 17.30 %) of dry root rot caused by *Macrophomina phaseolina*. The annual yield loss due to this disease alone was 10-20 % (Vishwadhar and Chaudhary, 2001). The DRR is more dominant when the crop is exposed to moisture stress conditions and can cause 50 to 100% yield loss under favourable conditions.

Under hot and dry environmental conditions, many economically important crops are predisposed to the infection and colonisation of *R. bataticola* and caused drastic yield losses on chickpea (Thripathi and Sharma, 1983), soybean (Pearson *et al.*, 1984) and sunflower (Nawaz, 2007).

2.8 QTL mapping for dry root rot disease resistance along with yield and other agronomic traits

2.8.1 DNA markers linked to dry root rot resistance genes

The most crucial factor for marker-assisted breeding is the knowledge of the association between the markers and trait of interest and it is starting point to work for crop improvement. More closely, the markers are associated with the traits, the higher the possibility of success and efficiency. This information can be determined in various ways, linkage-based approach or by linkage disequilibrium (LD) based association mapping, classical mutant analysis, linkage or recombination analysis, bulked segregant analysis, *etc.* In the past, in several crops, genetic mapping based approaches were used to identify QTLs/genes for a trait (Gupta and Varshney, 2004). For conducting marker-trait association by using linkage maps, three widely used methods are: single marker analysis (SMA), simple interval mapping (SIM), and composite interval mapping (CIM) (Tanksley, 1993).

The use of DNA marker systems, such as Random Amplified Polymorphic DNAs (Williams *et al.*, 1990), Amplified Fragment Length Polymorphisms (Vos *et al.*, 1995), Simple Sequence Repeats (Akkaya *et al.*, 1992) and Single nucleotide polymorphism (Jordan and Humphries, 1994) have contributed greatly to the development of genetic linkage maps for many important crop species including chickpea.

Talekar (2014) used a set of 520 chickpea accessions to identify DNA markers linked to genes conferring dry root rot resistance and revealed that screening of 381 SSR primer pairs realized 52 which were informative between L 550 and PG 06102. A bulk segregant analysis approach was applied to the DNAs of L 550 x PG 06102 F₂ progeny showed potential marker/ trait linkage for four SSR primer pairs. A linkage analysis based on 129 F₂ individuals and these four SSR markers showed that two markers (ICCM 0299 and ICCM 0120b) co-segregated with resistance to dry root rot.

Markers that are polymorphic between the pools might be genetically linked to the loci determining the trait (Wang *et al.*, 2014 and Qi *et al.*, 2015). Mapping genes responsible for traits remains one of the most basic goals of genetics (Brauer *et al.*, 2006). Genetic linkage mapping and marker-assisted selection have greatly assisted breeding programmes through the discovery of many important disease resistance genes (Wang *et al.*, 2009).

The marker-assisted selection (MAS) is an efficient strategy to accelerate resistance breeding in chickpea by overcoming the ambiguities of conventional breeding and difficulties in phenotyping for soil borne diseases like dry root rot (Patil *et al.*, 2014). Identification of DNA markers linked to disease resistance is important to initiate marker assisted introgression of genomic region/s for resistance to dry root rot. The availability of sufficient number of microsatellites (Litt and Luty, 1989) or simple sequence repeats (Cregan *et al.*, 1994) in chickpea makes it possible to apply marker assisted selection more effectively and efficiently.

2.8.2 Mapping populations used for QTL mapping

Development of appropriate mapping population is necessary for constructing a genetic linkage map and dissecting complex traits. The first step in producing a mapping population is selecting two genetically diverse parents for one or more traits of interest.

Further, the parents should be genetically divergent enough to exhibit sufficient polymorphism, and on the other hand, they should not be too distant that causes sterility of the progenies and expresses high level of segregation distortion during linkage analysis (Collard *et al.*, 2005). Population sizes used in preliminary genetic mapping studies generally range from 50 to 250 individuals (Mohan *et al.*, 1997). However, larger populations are required for high-resolution mapping (Paterson, 1996). Different types of populations, including progenies from F₂ generation, backcross (BC), recombinant inbred lines (RILs), double haploids (DH) and near isogenic lines (NILs) have been used for genetic mapping in chickpea. F₂ populations are developed by self-pollinating F₁ hybrids derived by crossing two parents, while BC population is produced by crossing F₁ to one of the parents). By repeated backcrossing for at least six generations (BC₆) with the recurrent parent, more than 99% of the genome can be recovered from the recurrent parent. Further, selfing of selected individuals at BC₆F₁ or BC₇F₁ will produce lines that are homozygous for the target gene which are considered to be nearly isogenic with the recipient parent (NILs). NILs are mainly generated for fine mapping of a QTL/genomic region of interest. DH populations are generally developed by chromosome doubling of haploids developed through anther culture (pollen or microspore culture) of F₁ plants.

Recombinant inbred lines (RILs) are developed following single seed descent (SSD) advancement of F₂ plants by six or more generations and then developing single plant progenies. This process leads to lines that contain a different combination of linkage blocks from the original parents. Seed from RILs is predominantly homogeneous and abundant, so the seed can be shared across any lab interested in adding markers to an existing linkage map previously constructed with the RILs. Moreover, RILs can be grown in replicated trials at several locations and/or over several years making them ideal for QTL mapping (Paterson, 1996 and Collard *et al.*, 2005).

2.8.3 Genetic mapping

Genetic mapping is a procedure of locating the molecular markers or gene loci/QTLs in order, indicating the relative distances among them and assigning to their linkage groups on the basis of their recombination values from all pairwise combinations. Genetic mapping is based on the principle that genes (markers or loci) segregate via chromosome recombination (called crossing-over) during meiosis (*i.e.*, sexual

reproduction), thus allowing their analysis in the progeny (Paterson, 1996). In a segregating population, the frequency of recombinant genotypes can be used to calculate recombination fractions, which may be used to infer the genetic distance between markers (Collard *et al.*, 2005). By analyzing the segregation of markers, the relative order and distances between markers can be determined. Markers that have a recombination frequency of 50% are described as 'unlinked' and assumed to be located far apart on the same chromosome or on different chromosomes (Hartl and Jones, 2001). The recombination fractions are converted into map distances (cM) by using mapping functions. Genetic linkage map construction requires following steps: (1) Development of appropriate mapping population, (2) Identification of polymorphic markers between parental genotypes and (3) Linkage analysis of markers. Linkage between markers is usually calculated using odds ratios (*i.e.*, the ratio of linkage versus no linkage). This ratio is more conveniently expressed as the logarithm of the ratio and is called a logarithm of odds (LOD) value or LOD score (Risch, 1992). LOD >3 are typically used to construct linkage maps. A LOD value of 3 between two markers indicates that linkage is 1000 times more likely (*i.e.*, 1000:1) than no linkage (null hypothesis). LOD values may be lowered in order to detect a greater level of linkage or to place additional markers within maps constructed at higher LOD values. Commonly used software programs include Mapmaker/EXP (Lander *et al.*, 1987), MapManager QTX (Manly *et al.*, 2001) and GMendel (<http://cropandsoil.oregonstate.edu/Gmendel>). JoinMap is another commonly used program which is generally used for constructing and combining the genetic maps developed for different mapping populations (Stam, 1993).

The beginning of the linkage map development in chickpea was based on morphological and isozyme loci. However, their small numbers and the fact that expression of these markers is often influenced by the environment, makes them unsuitable for routine use. Further, limited polymorphism exhibited by cultivated chickpea for the molecular markers developed in the early phase forced researchers to use interspecific crosses in linkage mapping of chickpea. The first linkage map of chickpea was developed by Gaur and Slinkard (1990a and 1990b) using isozyme markers in the inter-specific crosses of *C. arietinum* with *C. reticulatum* and *C. echinospermum*. Later, DNA-based markers such as RFLP and RAPD (Simon and Muehlbauer, 1997), STMS and amplified fragment length polymorphism (AFLP) (Winter *et al.*, 1999),

morphological isozyme, inter simple sequence repeat (ISSR) and RAPD loci (Santra *et al.*, 2000) and STMS markers (Tekeoglu *et al.*, 2002; Flandez-Galvez *et al.*, 2003; Udupa and Baum, 2003; Cho *et al.*, 2004 and Taran *et al.*, 2007) were used.

The interspecific RIL population of the cross *C. arietinum* (ICC 4958) × *C. reticulatum* (PI 489777) has been considered as the reference mapping population and extensively used for genome mapping (Nayak *et al.*, 2010; Gujaria *et al.*, 2011; Thudi *et al.*, 2011; Choudhary *et al.*, 2012 and Hiremath *et al.*, 2012). The first integrated genetic map based on this population comprised 37 inter simple sequence repeats (ISSRs), 70 amplified fragment length polymorphisms (AFLPs), 118 sequence-tagged microsatellite sites (STMSs), 96 DNA amplification finger printings (DAFs), 17 RAPDs, 3 cDNAs, 8 isozymes and 2 sequence characterized amplified regions (SCARs) covering a total distance of 2077.9 cM (Winter *et al.*, 2000). An advanced genetic map with 521 markers including simple sequence repeats (SSRs) and single nucleotide polymorphisms (SNPs) with an inter marker distance of 4.99 cM spanning 2602.1 cM was developed from the above population (Nayak *et al.*, 2010). In this map, they integrated 71 SNP loci based on gene-specific primers developed by Choi *et al.* (2004). This effort demonstrated the power of a comparative genomics approach (Medicago and chickpea) to identify molecular tools.

Later, this map was integrated with BES-SSRs, DArT and gene-based markers by Thudi *et al.* (2011), which comprised of 1291 loci on eight linkage groups (LGs). In this map, highest number of markers per LG was on LG 3 (218) and the lowest was on LG 8 (68) with an average inter-marker distance of 0.65 cM. Later, an advanced gene-rich map of chickpea comprising of 406 loci (including 177 gene-based markers) spanning 1,497.7cM genetic distance has been developed for this reference population (Choudhary *et al.*, 2012). Recently, Hiremath *et al.* (2012) developed large-scale KASPar assays for SNP genotyping and developed a genetic map comprising 1328 marker loci, including novel 625 CKAMs (Chickpea Kaspar Assay Markers), 314 TOG-SNPs and 389 published marker loci for this reference population. Gujaria *et al.* (2011) has developed a transcript map that comprised of 300 loci (including 126 genic molecular markers [GMMs]) and spans a genomic region of 766.56 cM.

Cho *et al.* (2002) developed the first intra-specific map of cultivated chickpea from ICCV 2 × JG 62 RILs. This map was developed using 3 ISSRs, 20 RAPDs, 55 STMSs along with two phenotypic markers comprised of 14 linkage groups and spanning 297.5 cM. This map was used to map genes for important morphological traits along with the double podded trait. Tekeoglu *et al.* (2002) developed a chickpea map from 65 STMS primer pairs and a population size of 142 RILs from an interspecific cross between FLIP84-92C (*Cicer arietinum*) and PI599072 (*Cicer reticulatum*). They also integrated this map with marker data from Santra *et al.* (2000) and reported a total genetic map with 167 markers on nine linkage groups covering 1,174.5 cM with an average distance of 7.0 cM between markers.

The genetic map reported by Flandez-Galvez *et al.* (2003) was based on 66 markers, including 51 SSRs and a population of 85 F₂ plants from an intraspecific cross between *desi* cultivars ICC12004 and Lasseter. Udupa and Baum (2003) generated a map from 52 SSRs and a population size of 97 RILs from an intraspecific cross (*kabuli* type chickpea) between ILC 1272 and ILC 3279. The genetic map reported by Cho *et al.* (2004) was generated from 53 STMS primer pairs based on the population of RILs from a cross between PI359075(1) and FLIP84-92C(2). Another map developed based on a “*kabuli* × *desi*” cross included a total of 134 molecular markers (3 ISSR, 13 STMSs and 118 RAPDs) mapped to 10 linkage groups (Cobos *et al.*, 2005). This map spanned a genomic region of 534.5 cM with an average marker interval of 8.1 cM. Later, Radhika *et al.* (2007) developed an integrated intraspecific map spanning a region of 739.6 cM, including 230 markers at an average distance of 3.2 cM between markers. The genetic map published by Taran *et al.* (2007) was generated from 135 primer pairs including 134 SSRs and was based on a population of 186 F₂ plants from an intraspecific cross of *desi* cultivar ICCV 96029 and *kabuli* cultivar CDC Frontier. Markers reported in this map were assigned to 8 linkage groups with a combined linkage distance of 1,285 cM with average linkage distance of 8.9 cM between primer pairs. Another map developed recently by Anuradha *et al.* (2011) included 144 markers assigned to 11 linkage groups, spanning 442.8 cM, with an average marker interval of 3.3 cM. Consensus genetic maps using both interspecific and intraspecific populations were also developed. A consensus map based on five interspecific (*C. arietinum* × *C. reticulatum*) and five intraspecific (*Desi* × *Kabuli* types) populations was developed (Millan *et al.*, 2010). It integrated 555

marker loci including RAPDs (251), STMSs (149), AFLPs (47), 33 cross-genome markers, 28 gene-specific markers, 10 isozyme markers, 10 inter-simple sequence repeats (ISSRs) and 7 RGA loci.

Recently, Varshney *et al.* (2014) developed an intraspecific genetic map of chickpea from ICC 4958 × ICC 1882 and ICC 283 × ICC 8261 RIL populations. The genetic maps comprised of 241 and 168 SSR loci with genome coverage of 621.51 cM and 533.06 cM, respectively. A consensus map developed from these two populations integrated 352 SSR loci covering map distance of 771.39 cM. Common markers in these and future maps with SSR primer pairs could lead to the development of a high density genetic map of chickpea to identify tightly linked flanking markers for genes of interest, which ultimately help in marker-assisted selection (MAS) and positional cloning of agronomically important genes.

2.8.5 Approaches for QTL mapping

The main principle of QTL detection is to obtain association between the trait value under consideration and the genotype of every marker. A significant correlation test means that at least one QTL may exist near the marker locus. To be able to detect QTL using the marker loci, linkage disequilibrium must exist between alleles at the marker loci and alleles of the linked QTL (Tanksley, 1993). Tests for QTL/trait association are often performed by the following approaches. The single marker approach, also referred to as the single factor analysis of variance (SF-ANOVA) or single point analysis (Edwards *et al.*, 1987). In this method, SF-ANOVA is done for each marker locus independent of information from other loci. F-tests provide evidence whether differences between marker locus genotype classes are significant or not. Although this approach is computationally simple, it suffers from major limitations: (i) the likelihood of QTL detection significantly decreases as the distance between the marker and QTL increases (ii) the method cannot determine whether the markers are associated with one or more QTLs and (iii) the effects of QTLs are likely to be underestimated because they are confounded with recombination frequencies. SIM (Simple Interval Mapping) was first proposed by Lander and Botstein (1989). This method takes full advantage of a linkage map and evaluates the target association between the trait values and the genotype of a hypothetical QTL (target QTL) at multiple analysis points between pair of adjacent marker loci (the target interval).

Presence of a putative QTL is estimated if the log of odds ratio (LOD) exceeds a critical threshold.

However, when multiple QTLs are segregating in a cross (which is usually the case), SIM fails to take into account genetic variance caused by other QTLs. CIM (Composite Interval Mapping, Zeng, 1994) and MQM (multiple-QTL model or marker-QTL-marker analysis) developed by Jansen and Stam (1994) combine interval mapping for a single QTL in a given interval with multiple regression analysis on marker associated with other QTL. The advantages of CIM are as follows; (i) mapping of multiple QTLs can be accomplished by the search in one dimension (ii) by using linked markers as cofactors, the test is not affected by QTL outside region, thereby increasing the precision of QTL mapping and (iii) by eliminating much of the genetic variance by other QTL, the residual variance is reduced thereby increasing the power of detection of QTL. CIM is more precise and effective at mapping QTLs as compared to single-point analysis (SMA) and SIM, especially when linked QTLs are involved.

2.8.5.1 Days to flowering

Earlier, Cho *et al.* (2002) detected a single QTL for flowering time on LG3 between the markers TS57 and TA127. However, Lichtenzweig *et al.* (2006) in the cross involving Hadas and ICC 5810 reported three QTLs on LG1, LG2 and LG8 for flowering time.

Previously, Cobos *et al.* (2007) reported a major QTL for days to 50 per cent flowering (QTLDF1; 20% PVE) on LG4 closely linked to the marker GAA47. Whereas, Cobos *et al.* (2009) and Aryamanesh *et al.* (2010) mapped a QTL for flowering time on LG3 closely linked to marker TA142. The inheritance of flowering time in chickpea is controlled by two QTLs with epistatic interaction (Hegde, 2010). Recently, Rehman *et al.* (2011) reported four QTLs for flowering time on LG1, LG3, LG4 and LG8. Jamalabadi *et al.* (2013) also reported a QTL for days to flowering on LG3 closely linked to the marker TA117 using a RIL population from a cross involving ICCV 2 as one of the parents. Bajaj *et al.* (2015) identified significant QTL associated with seed weight on linkage groups 1, 2, 3, 4 and 6.

Upadhyaya *et al.* (2015) described a QTL on CaLG4 as *Efl1*, implying that it is synonymous with Ca_14192/ XM_012715013, a homolog of Arabidopsis PIE1. Daba *et al.* (2016) identified additional major QTLs for days to flowering on LG5 and LG8 that are consistent across years and sites. Recently, Daba *et al.* (2016) also mapped a minor QTL on LG3 in the same cross. Major QTLs corresponding to flowering time genes *efl-1* from ICCV 96029, *efl-3* from BGD 132 and *efl-4* from ICC 16641 were mapped on CaLG04, CaLG08 and CaLG06 reported by Mallikarjuna *et al.* (2017).

2.8.5.2 Days to maturity

Pushpavalli *et al.* (2015) reported four major QTLs (up to 67% PVE) for DF which were identified in the control treatment but not in the salinity treatment during 2010. However, in 2011, three QTLs (up to 65% PVE) were identified in the salinity treatment. Four stable QTLs for DM were detected in control treatment, two each in CaLG05 (with flanking markers CaM0463-ICCM272) and in CaLG08 (CKAM1903-CKAM0343).

2.8.5.3 Plant height (cm)

Aryamanesh *et al.* (2010) found a QTL for plant height which was mapped on LG3. Varshney (2014) reported five QTLs for plant height (CaLG03, CaLG04, CaLG05, CaLG06 and CaLG08). However, both populations shared at least one QTL for plant height on CaLG08 that contributed 14.73 % PVE (ICCRIL03) to 31.32 % PVE (ICCRIL04).

Kujur *et al.* (2016) used SNPs on an ultra-high resolution intra-specific chickpea genetic linkage map (dwarf/semi-dwarf *desi* cv. ICC12299 x tall *kabuli* cv. ICC8261), identified six major genomic regions underlying six robust QTLs (11.5–21.3% PVE with 35.8% combined PVE) governing plant height traits in chickpea were mapped within <0.5 cM average marker intervals on six chromosomes.

2.8.5.4 Grain yield per plot (g)

Pushpavalli *et al.* (2015) identified four yield QTLs (three major and one minor QTL) across two years. In 2010, in the salinity treatment one major QTL was identified on CaLG07 and explained 17% of the PVE. In 2011, one major QTL in the salinity

treatment that explained 12% PVE was also identified on CaLG05, while one major QTL (16% PVE) and one minor QTL (8% PVE) were identified on each of CaLG05 and CaLG07 in the control treatment. The two major QTLs identified in the control and salinity treatments during 2011 were located at the same position on CaLG05 with flanking markers, CaM0463 and ICCM272.

2.8.5.5 Hundred seed weight (g)

For 100-seed weight, two genomic regions on CaLG01 (1.08 Mb) and CaLG04 (2.70 Mb) were identified. It is interesting to note that several studies reported the presence of QTL for 100-seed weight on CaLG04 in chickpea (Abbo *et al.*, 2005; Cho *et al.*, 2002 and Cobos *et al.*, 2007), showing the importance of this linkage group in yield trait improvement for chickpea.

Abbo *et al.* (2005) reported most significant QTL for seed weight was located on LG 4B flanked by markers GA24 and STMS11 on one side and GA2 on the other. Two other, less significant seed weight QTLs were detected on LG 1B (QTL2) and 2B (QTL3).

Radhika *et al.* (2007) identified two major QTLs for chickpea seed weight at a LOD > 3.0 on LG 1 and LG 4 in a different genetic background. Alternatively, Cobos *et al.* (2007) reported two QTLs for chickpea seed size located on LG 4 and LG 8 that explained up to 30.4% of the phenotypic variation and Cho *et al.* (2002) found two putative QTLs for 100-seed weight located on LG 4 and LG 9.

Hossain *et al.* (2010) studied two RIL populations derived from intraspecific crosses of a *kabuli* type (S95362; light cream colour) crossed to two *desi* types (Howzat and ICC 3996; medium tan and dark tan colour, respectively) across two environments. Two major quantitative trait loci (QTL) were identified, one on LG 4 (QTL1) and one on LG 1 (QTL2) with 37% of the phenotypic variance for 100-seed weight across the two environments.

In the case of 100 SDW, Varshney *et al.* (2014a) reported two major QTLs one each on CaLG01 and CaLG04 that explained 10.31% and 58.20% of phenotypic variation, respectively. Based on the physical position of flanking markers, the QTLs for

100 SDW spanned 6.57 Mb (2.93–9.51 Mb) on CaLG01 and 6.75 Mb (10.07–16.83 Mb) on CaLG04, while Jaganathan *et al.* (2015) reported two major QTLs one each on CaLG01 and CaLG04 that explained 16.23% and 60.41% of phenotypic variation, respectively. Das *et al.* (2015) found major genomic region harbouring a robust 100-seed weight QTL (CaqSW1.1) on chromosome 1 using an intra-specific *desi* chickpea mapping population.

Verma *et al.* (2015) reported seven QTLs for seed weight mapped on LGs 1, 2, 5, 6 and 7 with the LOD thresholds ranging between 3.02 (*qSW-1*) to 5.52 (*qSW-3*). The QTLs explained phenotypic variance in the range of 10.07% (*qSW-7*) to 22.31% (*qSW-5*). Singh *et al.* (2015) reported genome wide SNP profiling of extreme phenotypic bulks from the ICC 49589, ICC 1882 population identified two significant genomic regions, one on CaLG01 (1.08 Mb) and another on CaLG04 (2.7 Mb) linkage groups for 100 SDW.

Pushpavalli *et al.* (2015) identified four major QTLs for 100-seed weight, one each per treatment and per year, were identified on CaLG05. Three of the four QTLs for 100-seed weight were identified at the same locus of CaLG05 (CaM0463-ICCM272) and explained PVE up to 40%. Two stable QTLs for 100-seed weight under control treatment were identified. The fourth QTL was also identified on CaLG05, but at a different position which explained 17% of the PVE.

2.8.5.6 Harvest index (%)

Pushpavalli *et al.* (2015) identified nine QTLs for HI across years and treatments. In 2010, in the salinity treatment a minor QTL (6% PVE) for HI was identified on CaLG04d while in the control treatment two major QTLs were identified for HI, one each on CaLG05 (46% PVE) and CaLG08 (10% PVE). In 2011, in the salinity and control treatment, three major QTLs per treatment for HI explaining PVE of 30-49% and 32 to 56%, one each on CaLG05, CaLG04d and CaLG08 were identified. Four stable QTLs were identified for HI under control treatment.

Rehman (2009) reported two QTLs for grain yield under drought on LG1 and LG3, five QTLs for harvest index (HI) on LG1, LG3, LG4 and LG8. Four QTLs were identified for days to flowering on LG1, LG3, LG4 and LG6. QTL on LG1 and LG3 were

also associated with days to maturity, reproductive period and plant height which indicated their possible pleiotropic nature. The QTL on LG1 explained 13% of the total phenotypic variability for days to flowering with a LOD score of 6.2. This QTL also explained 13% of the phenotypic variability for days to maturity (LOD score = 7.3), 7% of the phenotypic variability for reproductive period (LOD score = 3.9) and 21% of the total phenotypic variability for plant height (LOD score = 8.5). Similarly, a QTL on LG3 explained 28% of the total phenotypic variability for days to flowering with a LOD score of 9.7. This QTL explained 49% and 26% of the total variability with LOD scores of 16.7 and 5.9 for days to maturity. Two QTLs, on LG6 and LG8, were unique for plant height explaining a total of 28% of the phenotypic variability.

Jamalabadi *et al.* (2013) identified potential markers linked to days-to-flowering, 100-seed weight and plant height in a chickpea intraspecific $F_{2:3}$ population derived from the cross ILC3279 $\times$ ICCV2. A closely linked marker (TA117) on linkage group LG3 was identified for the days to 50 per cent flowering trait, explaining 33% of the variation. In relation to plant height, a QTL was located in LG3, close to the Ts5 marker that explained 29% of phenotypic variation. A QTL for 100-seed weight located in LG4, close to TA176, explained 51% of variation. Kale *et al.* (2015) mapped QTLs for 100-seed weight and plant height on CaLG04, while a QTL for DF was reported to be present on CaLG08.

Karami *et al.* (2015) studied for quantitative traits of seed weight, plant height and days to flowering in a chickpea intraspecific $F_{3:4}$ lines population derived from a ILC3279 $\times$ ILC588 cross. Two QTLs for days to flowering were detected on LG2 and LG3 accounted for 58% of the total phenotypic variation. A QTL for plant height was located in LG3 explaining around 42% of the variation shown to be under a major gene control. For 100-seed weight, a QTL located in LG4 explained around 37% of the phenotypic variations.

Shefali *et al.* (2015) detected two QTLs each for seed weight, qSW- 1 and qSW- 2 (explaining 11.54% and 19.24 % of phenotypic variance, respectively) and plant height, qPH-1 and qPH- 2 (explaining 13.98% and 12.17% of phenotypic variance, respectively).

Material and Methods

III. MATERIAL AND METHODS

The present study on “Molecular mapping of dry root rot resistance gene(s) in chickpea (*Cicer arietinum* L.)” was carried out at International Crops Research Institute for the Semi-Arid Tropics (ICRISAT), Patancheru, Telangana, India (located at 17.51°N, 78.26°E, 545 m above sea level and 600 km away from the sea). The weather and climate of semi-arid tropic region is characterized by highly variable, low to medium rainfall, poor soil and further characterized by lack of irrigation, tropical wet and dry climate bordering on a hot semi-arid climate. In general, the historical average annual rainfall in the semi-arid tropics is below 700 mm. The field experiments for phenotyping of the mapping populations were conducted during post-rainy season (October to February) of 2015-2016 and 2016-17 at Patancheru.

The details of material used and the methods adopted in the present study are described under the following headings.

3.1 Selection of parents and development of mapping populations

The experimental material for the present study consists of the female parent BG 212, susceptible to DRR and the male parent ICCV 08305, moderately resistant to DRR. Recombinant inbred lines (RILs) obtained by hybridization of contrasting parents, followed by single seed descent method from F₂ onwards are considered to be the best among mapping populations. Chickpea breeding unit at ICRISAT developed 182 F₉ RIL population from the cross BG 212 x ICCV 08305 for the present study.

3.2 Experimental details and design

The experiments for evaluation of RILs for various agronomic traits were carried out in an alpha lattice design with three replications in the black precision (BP-8) field at ICRISAT, Patancheru. The field used for the phenotyping was solarized using polythene mulch during the preceding summer to sanitize the field, especially to get rid of soil-borne diseases. Seeds were treated with a mixture of 2 g of Thiram and 1 g of Carbendazim per kilogram of seeds and dibbled when sufficient moisture was available in the soil. The seeds were sown at a wider spacing of 60 cm x 10 cm with single seed per hill in 4m-long rows (considering each plot consists of 2m in a row). Diammonium

phosphate fertilizer@ 100 kg/ha was applied to provide a basal dose of 18 kg N and 46 kg P_2O_5 ha⁻¹ to raise healthy crop. Plot was maintained weed-free by regular manual weeding. One supplementary irrigation was given at 30 DAS to overcome moisture stress conditions. Control of pod borer (by spraying of 20 mL/ha Indoxacarb in 300 L water) was taken up when the pest incidence reached the economic threshold level. All other crop management practices were carried out to ensure good crop establishment.

3.3 Collection of data

Observation was recorded on all the 182 RILs along with the parents on the following traits.

3.3.1 Days to 50 per cent flowering

Number of days counted from the date of sowing till the day on which 50 per cent of the plants open their first flower bud was recorded.

3.3.2 Flower color

Colour of the freshly opened flower i.e. pink or white was recorded on individual plant basis.

3.3.3 Days to maturity

Number of days after sowing when more than 90% of pods of the plant have changed from green to yellow.

3.3.4 Plant height (cm)

The plant height was measured from ground level to the growing tip at maturity and recorded in centi meters, where the number of plants present in the two meter row was considered as plot.

3.3.5 Seed yield per plot (g)

The harvested seeds from each plant of two meter row was considered as plot, were properly dried and weighed in grams to obtain seed yield per plot.

3.3.6 Plant biomass per plot (g)/ Biological yield per plot (g)

The above ground plant weight of the fully dried plant was recorded as biological yield per plot. Here, two-meter row considered as plot.

3.3.7 Hundred seed weight (g)

One hundred seeds were counted from the selected plants and their weight was recorded in grams and termed as “test weight”.

3.3.8 Harvest index (%)

It was calculated by the following formula:

$$\text{Harvest Index} = \frac{\text{Grain yield per plant (g)}}{\text{Biological yield per plant (g)}} \times 100$$

3.4 Screening of chickpea genotypes for resistance to DRR disease

3.4.1 Paper towel method

3.4.1.1 Screening of genotypes in laboratory condition

The parents and 182 RILs were phenotyped for their resistance to DRR (caused by *Rhizoctonia bataticola*) by paper towel technique (Pande *et al.*, 2012) with two replications

Raising of seedlings

RILs were grown in separate polythene bags in a greenhouse maintained at 25±1°C for 7 days. Plastic bags were filled up to 2/3 of its volume with sterilized river sand. Seeds of each RIL were surface sterilized using 2% sodium hypochlorite for 2 minutes, rinsed in sterile water for 2-3 minutes in order to wash off sodium hypochlorite, sown (30 seeds per RIL) in plastic bags and allowed to grow for 7 days.

Inoculum preparation

1. Pure culture of *R. bataticola* (Rb 6) (on PDA) isolated from DRR infected chickpea plants was obtained from Legume Pathology lab. Potato Dextrose Broth (PDB) was prepared, placed 100 ml of broth into a 250 ml flask and prepared as many flasks as needed. Autoclave was done at 15 lb for 20 minutes. One flask of inoculum was needed to test 10 lines at once.
2. Sterilized PDB medium with the fungus was inoculated and incubated for seven days at 25°C.
3. Mycelial mats were removed from the flasks after seven days of incubation. Two mycelial mats were added to 100 ml of sterile distilled water and macerated in a blender for 1 minute (operate the blender intermittently). Placed this inoculum in a beaker of a suitable size.
3. All seedlings of a test line were held in hand and dipped the roots into the inoculum for about 30 seconds. Excess inoculum was removed by touching the edge of the beaker.
4. Ten seedlings of the test line were placed side by side on a blotter paper (size 45 × 25 cm with one fold; any color; thin) so that only the cotyledons and roots are covered, and the green tops of the seedlings remain outside the blotter paper after it is folded.
5. The blotter paper was folded and moistened adequately but not excessively. One folded blotter paper had seedlings of one test line only.
6. Un-inoculated and inoculated seedlings of susceptible control (BG 212) were kept separately with each batch of test seedlings. The folded blotters were kept, one on top of the other, in heaps of 10 in a tray.
7. The trays were placed in an incubator at 35°C for eight days. Twelve hours of artificial light was provided and the blotters were moistened everyday adequately.
8. At the end of the incubation period (8 days), the seedlings were examined for the extent of root damage and disease scoring was done on 1-9 rating scale (Pande *et al.*, 2012).

Table 1: Scoring scale for chickpea DRR disease

Scale	Category	Symptom
1	Resistant	No infection on roots
3	Moderately resistant	Very few small lesions on roots
5	Moderately susceptible	Lesions on roots clear but small, new roots free from infection
7	Susceptible	Lesions on roots many, new roots generally free from lesions
9	Highly susceptible	Roots infected and completely discolored

3.5 Biometrical analysis

The data was analyzed using computerized statistical packages like MS Excel, Genstat 14th edition, SAS and Windostat.

3.5.1 Analysis of variance (ANOVA):**Table 2: Skeleton of analysis of variance (ANOVA) for alpha lattice design**

Source of variation	Degrees of freedom
Environment	e-1
Replication	e (r-1)
Block	er (b-1)
Genotype	g-1
Genotype × Environment	(g-1)(e-1)
Residual	e (rt-rb-t+1)
Total	ert-1

Where,

r = Number of replications,

k2 = Number of treatments (genotypes)

The adjusted variable mean differences were tested for significance as follows

Source of variation	Degree of freedom	Sum of squares	Mean squares	F ratio
Genotypes (adjusted)	k2-1	SSg	MSg	MSg/MSe
Error (intra block)	(k-1) (rk-k-1)	SSE	MSe	

The computed F value was compared with the table F value at (k2-1) and (k-1) (rk-k-1) degrees of freedom for 5% and 1% level of significance. To test significance of the differences between any two adjusted genotypic means, the standard error of mean was computed using the formula

$$S.E = [2MSe/r (1+2ku/k+1)]^{0.5}$$

Where,

S.E = Standard error of mean

MSe = Mean sum of squares for error (intra block)

r = Number of replications

k = Number of genotypes in each sub-block

u = Weightage factor computed

3.5.2 Mean

On the basis of individual plant observations, the population mean for each character was computed as follows.

$$\bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i$$

Where,

$\bar{Y}$ = Population mean

Y_i = Individual value

n = Number of observations

3.5.3 Range

The minimum and maximum values on the basis of individual plant observations were used to indicate the range for a given character.

3.5.4 Variance

Variance was computed for each character as follows:

$$\text{Variance } (\sigma^2) = \frac{1}{n-1} \sum [(Y_i - \bar{Y})^2]$$

Where,

Y_i = Individual value

$\bar{Y}$ = Population mean

n = Number of observations

3.5.5 Standard deviation (SD)

$$\sqrt{\frac{1}{n-1} \sum [(Y_i - \bar{Y})^2]}$$

Where,

Y_i = Individual value

$\bar{Y}$ = Population mean

n = Number of observations

3.5.6 Standard error (SE)

$$SE = \frac{SD}{\sqrt{n}}$$

Where,

SD= Standard deviation

n=Number of observations

3.6 Estimation of genetic variability parameters

3.6.1 Phenotypic and genotypic variances

Phenotypic and genotypic components of variance were estimated by applying the formula suggested by Cochran and Cox (1957).

$$\text{Genotypic Variance } (\sigma^2_g) = \frac{\text{MSS due to genotypes (adj)} - \text{MSS due to error (intra block)}}{R}$$
$$\text{Phenotypic variance } (\sigma^2_p) = \sigma^2_g + \sigma^2_e \text{ (MSS due to error)}$$

3.6.2 Phenotypic and genotypic co-efficient of variation

The co-efficient of variability both at phenotypic and genotypic levels for all the characters were computed by applying the formula suggested by Burton and De Vane (1953).

3.6.3 Phenotypic coefficient of variation (PCV %)

$$PCV (\%) = \frac{\text{Phenotypic standard deviation}}{\bar{X}} \times 100$$

3.6.4 Genotypic coefficient of variation (GCV %)

$$GCV (\%) = \frac{\text{Genotypic standard deviation}}{\bar{X}} \times 100$$

Where,

$\bar{X}$ = General mean

GCV and PCV were classified as 0-10 %: Low, 10-20 %: Moderate and above 20%: High as suggested by Siva subramanian and Menon (1979).

3.6.5 Heritability (h^2)

Heritability in broad sense for all the characters was computed by the formula as suggested by Hanson *et al.*, (1956).

$$h^2 = \frac{\sigma^2_g}{\sigma^2_p} \times 100$$

Where,

h^2 = Broad sense heritability

σ^2_g = Genotypic variance

σ^2_p = Phenotypic variance

Heritability was classified as suggested by Robinson *et al.* (1949) and described as 0-30 %: Low, 30-60 %: Moderate, 60 % and above as High.

3.6.6 Genetic advance

Genetic advance for each character was predicted by the formula given by Johnson *et al.* (1955).

$$GA = h^2 \times \sigma_p \times K$$

The expected GA as per cent of mean (GAM) was estimated as given below

Expected genetic advance as a per cent of mean = $\frac{GA}{\bar{X}} \times 100$

Where,

h^2 = Heritability (Broad sense)

σ_p = Phenotypic standard deviation

K = Standard selection differential (2.06)/ Selection differential at 5% selection intensity=2.06 (constant value)

GA = Genetic advance

$\bar{X}$ = General mean

The genetic advance as per cent of mean was categorized into Low (0-10%); Moderate (10-20%); High (20 % and above) as suggested by Johnson *et al.* (1955).

3.6.7 Correlation co-efficient analysis

To determine the degree of association of characters with yield and also among the yield components, the correlation coefficients were calculated. Both genotypic and phenotypic correlation coefficients between two characters were determined by using variance and covariance components as suggested by Al-Jibouri *et al.* (1958)

$$r_g(xy) = \frac{\text{Cov}_g(xy)}{\sigma_{2g}(x) \times \sigma_{2g}(y)} \qquad r_p(xy) = \frac{\text{Cov}_p(xy)}{\sigma_{2p}(x) \times \sigma_{2p}(y)}$$

Where,

$r_g(xy)$ = Genotypic correlation coefficient

$r_p(xy)$ = Phenotypic correlation coefficient

$\text{Cov}_g(xy)$ = Genotypic co-variances of xy

$\text{Cov}_p(xy)$ = Phenotypic co-variances of xy

The calculated value of 'r' was compared with table value at n-2 degree of freedom at 5% and 1 % level of significance.

3.6.8 Chi- square test

The observations recorded on reaction of F₉s for DRR disease were subjected to chi-square test based on expected ratios. The goodness of fit between expected and observed segregation ratios were worked out. The Chi-square was calculated by using the following formula.

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

Where,

O_i = Observed frequency

E_i = Expected frequency

The significance of Chi- square value was tested against table value with (n-1) degrees of freedom, where n is the total number of segregating classes (Stansfield, 1986). A p-value of 0.05 is used as cut-off between significant and not-significant results. If the p-value is less than or equal to 0.05 probability level, then the null hypothesis is rejected. Otherwise, the null hypothesis is accepted.

3.6.9 t-Test

To screen the RILs against DRR disease, sick plot was not available. So, the screening was conducted two times under controlled environment conditions by using paper towel method, while the crop was grown in solarized field for evaluating yield other agronomic characters. Here, t-test was used to find out the relative comparison between susceptible and resistant RILs for yield and other agronomic traits including disease incidence.

3.7 Construction of genetic linkage map

3.7.1 Sample preparation and DNA extraction

Young leaf samples were collected from 15 days old seedlings of parents and all the 182 RILs. Genomic DNA was extracted by following the mini-prep CTAB method as suggested Mace *et al.* (2003).

3.7.2 Buffers and solutions prepared for DNA extraction

The chemicals used for preparation of DNA extraction buffer are listed as follows.

- i. Tris chloride (100mM)
- ii. EDTA (25mM)
- iii. NaCl (1.5M)
- iv. CTAB (3 %)
- v. β -mercaptoethanol (0.2%)
- vi. PVP (1 %)
- vii. Chloroform: Isoamylalcohol (24:1)

3.7.3 DNA extraction protocol

CTAB method (Dipankar Chakraborti *et al.*, 2006) was followed for extracting DNA from all the samples with minor modifications and the protocol is as follows.

1. 0.4 g of collected plant material was ground into fine powder in liquid nitrogen. One ml of extraction buffer was added to the ground material and transferred to 2 ml polypropylene tube.
2. The mixture was shaken gently and the tubes were placed in hot water bath at 60°C for 30 minutes (with intermittent shaking).
3. Upper clear aqueous layer was carefully transferred to a fresh labeled 1.5 ml tube. Equal volume of chloroform: Isoamyl alcohol (24:1) was added to this and mixed by gentle inversion for about 5 min and spun at 7200 rpm for 5 min at 25-30°C.
4. The upper clear aqueous layer was transferred to a fresh labeled tube and equal volume of 100% chilled ethanol was added to precipitate the nucleic acid and allowed the mixture to stand at room temperature for 30 min and samples were centrifuged at 12000 rpm for 12 minutes.

5. The supernatant was discarded and the pellet was washed with 70% ethanol and centrifuged at 13000 rpm for 5 minutes. The pellet was vacuum dried for 15 minutes.
6. 50-100 µl of T₁₀E₁ buffer was added to the pellet and the pellet was dissolved.
7. Finally the quality and quantity of DNA was determined through agarose gel electrophoresis using 0.8% agarose gel.

3.7.4 Analysis method:

DNA was isolated from 185 RILs (F₉) along with 2 parents. The cost effective SNP genotyping platform, comprising of 50,590 high quality non-redundant SNPs tiled on to Affymetrix® Axiom® genotyping array was used for genotyping the RIL population. Sequence alignment and SNP calling identified 50,590 SNPs which contained 16,211 (32.04%) SNPs poly high resolution SNPs. Upon processing and filtering of SNPs, all SNP markers monomorphic across the parental genotypes were removed. All SNP markers with more than 10 % missing SNP calls were removed before further analysis. As a result, a total of 13,110 SNP markers were used for further analysis. With the breeding scheme of the RIL population, only three possible genotypes may exist at polymorphic loci with bi-allelic SNP calling. The 2, 0, -1 coding numbers were then used to code for different alleles in the genotype data. SNP call for each marker across the population was coded as 2 if the allele was the same as BG 212. A code of 0 was given to the alternative allele from ICCV 08305. The missing data SNP calls were coded as -1.

3.7.5 Genetic map construction

A total of 13,110 SNPs were found highly polymorphic among the parental genotypes of DRR mapping population. The high quality genotyping data containing 13,110 SNPs on 185 RIL population was used for genetic map construction using Inclusive Composite Interval Mapping (ICIM) v. 4.1 software. Chi-square test was performed ($p < 0.05$) to test the segregation distortion for each marker. Marker grouping was done with LOD score of 5 and recombination frequency of 0.40. Final marker order on each linkage group was determined by the RECORD algorithm. Distance was calculated using Kosambi's mapping function and the linkage map was generated with map option in ICIM software.

3.7.6 QTL analysis

The linkage map data and phenotyping data was used for QTL analysis using ICIM through the QTL-ICiMapping software version 4.1 (Wang *et al.*, 2014). ICIM-Add mapping performs first a stepwise regression to identify the most significant markers and marker-pair multiplications at 0.001 probability level and the scanning step of 1 cM. Then a one-dimensional scanning or interval mapping was conducted to identify additive QTL. The threshold levels to declare significant QTL were determined by performing 1000 permutations by maintaining the chromosome-wise type I error rate of 0.05. The LOD score peaks were used to estimate the most likely position of QTL on the linkage map. The amount of variation explained by marker was determined using the coefficient of determination (R^2) value.

Experimental Results

IV. RESULTS

The results obtained from the present study on “Molecular mapping of dry root rot resistance genes in chickpea (*Cicer arietinum* L.)” are presented under the following subheadings.

4.1 Study of genetic variability parameters for yield and yield contributing traits.

4.2 Correlation between yield and its related traits.

4.3 Path analysis for yield and its related traits.

4.4 Screening of chickpea RILs for DRR resistance.

4.5 Genetics of DRR resistance.

4.6 Understand the relative comparison between resistant and susceptible RILs for yield and yield contributing traits including DRR disease incidence.

4.7 Genetic map construction.

4.8 QTL analysis for DRR resistance and yield attributing traits in RILs of chickpea.

4.1 Genetic variability parameters for yield and yield contributing traits in RILs of chickpea evaluated during *rabi* 2015-16.

The 182 RILs were used to study the extent of genetic variability parameters for seed yield and yield contributing traits and the results are presented below.

4.1.1 Analysis of variance

Analysis of variance was carried out for grain yield and its attributing traits in 182 RILs along with parents. The variance due to known and unknown causes was worked out using the method suggested by Choudary and Prasad (1967) and Lush (1949) and results are presented below.

Table 3. Analysis of variance for yield and yield contributing traits in RILs of *rabi* chickpea (2015-16)

Source of variation	Degree of freedom	Days to 50 per cent flowering	Days to maturity	Plant height (cm)	Plant biomass per plot (g)	Grain yield per plot (g)	Hundred seed weight (g)	Harvest index (%)
Replication	2	3.72	3.77	20.76	2329.70	3403.6	18.27	280.01
Block(Replication)	33	106.90**	116.68**	34.26**	10710.3**	2547.2**	23.29**	48.29**
Entry	183	258.04**	369.45**	60.41**	22463.7**	5907.2**	86.87**	82.54**
Error	333	3.53	6.81	10.14	851.80	188.5	3.08	16.34
Mean		50.64	100.45	39.66	340.37	167.38	19.99	49.43
Range		35.33-74.67	82-128.67	26.89-54.89	146.67-600	68.33-292.33	11.84-35.34	33.52-59.87
C.V(%)		3.71	2.6	8.03	8.57	8.2	8.78	8.17

Table 4. Analysis of variance for yield and yield contributing traits in RILs of *rabi* chickpea (2016-17)

Source of variation	Degree of freedom	Days to 50 per cent flowering	Days to maturity	Plant height (cm)	Plant biomass per plot(g)	Grain yield per plot (g)	Hundred seed weight (g)	Harvest index (%)
Replication	2	1.616	51.53	520.01	5685	128.70	84.66	33.93
Block(Replication)	27	110.78	24.41**	33.84*	8767**	3030.9**	42.95**	67.99**
Entry	183	294.353	26.86**	127.01**	19887**	6518.7**	78.93**	142.07**
Error	339	3.151	4.48	19.92	1248	287.30	3.09	21.85
Mean		44.41	96.15	46.49	409.84	213.41	19.69	52.49
Range		32-64	91-119	29.67-66.44	126.67-635.00	60-381.67	11.03-37.73	30.32-63.74
C.V(%)		4	2.2	9.6	8.62	7.94	8.93	8.90

Table 5. Pooled anova for yield and yield contributing traits in RILs of both *rabi* chickpea (2015 and *rabi* 2016)

Source of variance	Degrees of freedom	Days to 50 per cent flowering	Days to maturity	Plant height (cm)	Plant biomass per plot(g)	Grain yield per plot (g)	Hundred seed weight (g)	Harvest index (%)
Season	1	10377.3	4953.29	12794.86	1313704	582028.2	27.97	2560.66
Rep(season	4	3.21	30.05	283.78	4217.09	1868.19	49.38	153.82
Block(season*REP)	60	4.97	6.93	11.73	1195.87	364.76	6.38	16.3
Treatment	183	475.23**	249.66**	143.75**	30227.86**	8515.53**	152.7**	127.6**
season* Treatment	183	76.81**	145.12**	39.09**	12439.43**	4008.05**	11.87**	97.19**

4.1.1.1 Analysis of variance for RILs of chickpea evaluated during *rabi* 2015-16.

As evident from the Table 2, the RILs exhibited highly significant differences for all the characters days to 50 percent flowering, days to maturity, plant height, plant biomass per plot, grain yield per plot, 100 seed weight and harvest index at both levels (Table 3).

4.1.1.2 Analysis of variance for RILs of chickpea evaluated during *rabi* 2016-17

The RILs exhibited highly significant differences for all the characters which are under study at both levels (Table 4).

4.1.1.3 Pooled analysis of variance for both seasons *rabi* 2015-16 and 2016-17

The RILs showed highly significant differences for all characters days to 50 percent flowering, days to maturity, plant height, plant biomass per plot, grain yield per plot, 100 seed weight and harvest index at both levels and in both seasons (Table 5).

4.1.2 Genetic variability parameters

The variability parameters such as mean, range, genotypic coefficient of variation (GCV), phenotypic co-efficient of variation (PCV), heritability in broad sense (h^2_{bs}) and genetic advance as per cent of mean (GAM) were computed for all the characters during *rabi* 2015 and 2016.

4.1.2.1 Genetic variability studies in RILs of chickpea evaluated during *rabi* 2015-16

Genetic variability parameters were studied and the results are presented below (Table 6; Figure 1 and 2).

4.1.2.1.1 Days to 50 per cent flowering

The average days to 50 per cent flowering was 51 days and mean values ranged from 35-75 days. The estimates of phenotypic and genotypic coefficient of variation were moderate about 18.31 and 18.19 per cent respectively, with high heritability of 98.63 per cent coupled with high genetic advance as per cent of mean (37.21) recorded for this trait.

4.1.2.1.2 Days to maturity

The overall mean of days to maturity in RIL population recorded was 100 days and mean values ranged from 82-129 days. The magnitude of PCV (11.05) and GCV (10.95) was moderate. The heritability was 98.16%, while genetic advance as per cent mean was 22.34.

4.1.2.1.3 Plant height (cm)

The average plant height of RILs was 39.66 cm and mean value ranged from 26.89 cm to 54.89cm. The PCV (11.32%) and GCV (10.32%) estimates were moderate with high heritability (83.21%) and moderate genetic advance as per cent of mean (19.40) for this trait.

4.1.2.1.4 Plant biomass per plot (g)

Plant biomass per plot showed a wide range varying from 146.67g to 600g with a mean of 340.37g and showed high PCV (23.43%) and GCV (22.75%). The estimates of heritability (94.32%) and genetic advance as per cent of mean (45.51) were high for this trait.

4.1.2.1.5 Grain yield per plot (g)

The overall mean of grain yield per plot was 167.38 g and the mean value ranged from 68.33g to 292.33g. High PCV (26.51) and GCV (26.08) were recorded for this trait. Grain yield per plant also exhibited high heritability (96.81 %) and high genetic advance as per cent of mean (52.90).

4.1.2.1.6 Hundred seed weight (g)

The average 100 seed weight was 19.99g and mean value ranged from 11.84g to 35.34g. The PCV and GCV estimates were found to be high for this trait (26.92% and 296.44%, respectively). The heritability was high (96.45%) with high genetic advance as per cent of mean (53.48).

Table 6. Genetic variability, heritability and genetic advance for seed yield and its components in RILs of chickpea (2015-2016)

Trait Genetic Parameter	Days to 50 per cent flowering	Days to maturity	Plant height (cm)	Plant biomass per plot(g)	Grain yield per plot (g)	Hundred seed weight (g)	Harvest index (%)
GCV	18.19	10.95	10.32	22.75	26.08	26.44	14.21
PCV	18.31	11.05	11.32	23.43	26.51	26.92	15.84
Hbs (%)	98.63	98.16	83.21	94.32	96.81	96.45	80.5
GA	18.84	22.44	7.69	179.40	88.5	10.69	15
GAM	37.21	22.34	19.40	45.51	52.9	53.48	26.3

Table 7. Genetic variability, heritability and genetic advance for seed yield and its components in RILs of chickpea (2016-2017)

Trait Genetic parameter	Days to 50 per cent flowering	Days to maturity	Plant height (cm)	Plant biomass per plot(g)	Grain yield per plot (g)	Hundred seed weight (g)	Harvest index (%)
GCV	21.09	12.85	2.84	23.25	21.12	25.53	18.26
PCV	21.50	14.00	3.11	23.84	21.80	26.05	19.58
Hbs (%)	96.27	84.32	83.31	95.08	93.89	96.08	86.97
GA	18.89	11.30	5.14	184.22	92.96	10.15	20.2
GAM	42.63	24.31	5.34	46.69	42.16	51.55	35.1

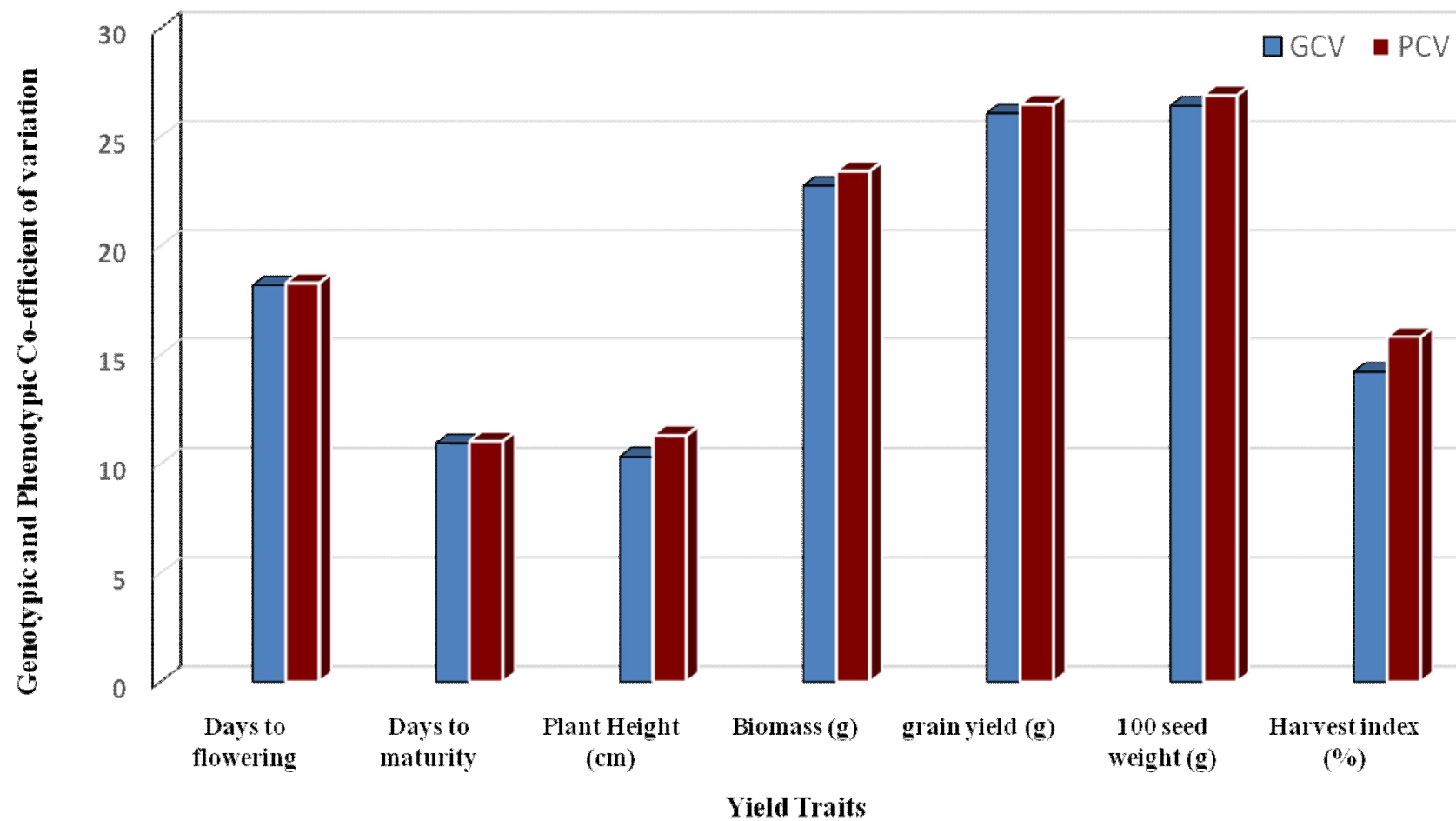


Fig. 1. Genotypic and phenotypic co-efficient of variation for yield and yield related traits in chickpea (rabi 2015-16)

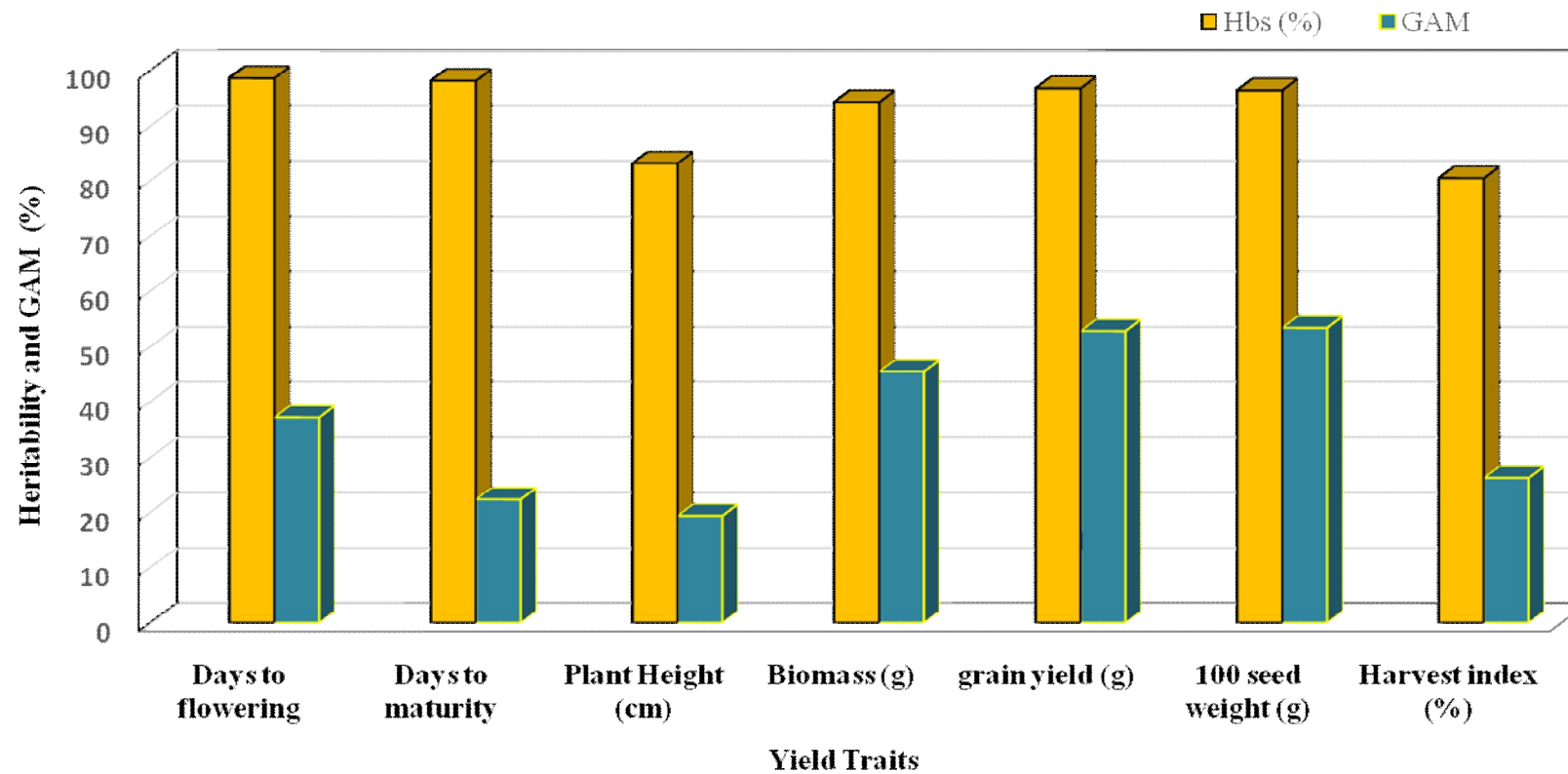


Fig. 2. Heritability and GAM (%) for seed yield and yield related traits in chickpea (*rabi* 2015-16).

4.1.2.1.7 Harvest index (%)

The overall mean of harvest index was 49.43%, while the mean value ranged from 33.52% to 59.87%. The estimates of PCV (15.84%), GCV (14.21%), heritability (80.20%) and genetic advance as per cent mean (26.30) were high for this trait.

4.1.2.2 Genetic variability studies in RILs of chickpea evaluated during *rabi* 2016-17

Genetic variability parameters were studied and the results are presented below Table 7; Figure 3 and 4.

4.1.2.2.1 Days to 50 per cent flowering

Days to 50 per cent flowering showed a wide range of variation from 32 days to 64 days with a mean of 44 days. The character showed high PCV (21.50 %) and GCV (21.09 %). The estimates of heritability (96.27 %) and genetic advance as per cent of mean (42.63) were high.

4.1.2.2.2 Days to maturity

The overall mean of days to maturity recorded in RIL population was 96 days and the mean values ranged from 91 to 119 days. The magnitude of PCV (14.00 %) and GCV (12.85 %) was moderate. The heritability (84.32%) and genetic advance as per cent mean (24.31) were high for this character.

4.1.2.2.3 Plant height (cm)

The average plant height was 46.49cm and the mean value ranged from 29.67cm to 66.44cm. The PCV and GCV estimates were found to be high for this trait (14.00% and 12.85%, respectively). The heritability was high (83.31%) with low genetic advance as per cent of mean (5.34).

4.1.2.2.4 Plant biomass per plot (g)

The overall mean of plant biomass per plot in RIL population was recorded as 409.84g and mean values ranged from 126.67g to 635g. The magnitude of PCV and GCV was high (23.84% and 23.35%, respectively). The heritability was 95.08% with high genetic advance as per cent mean (46.69).

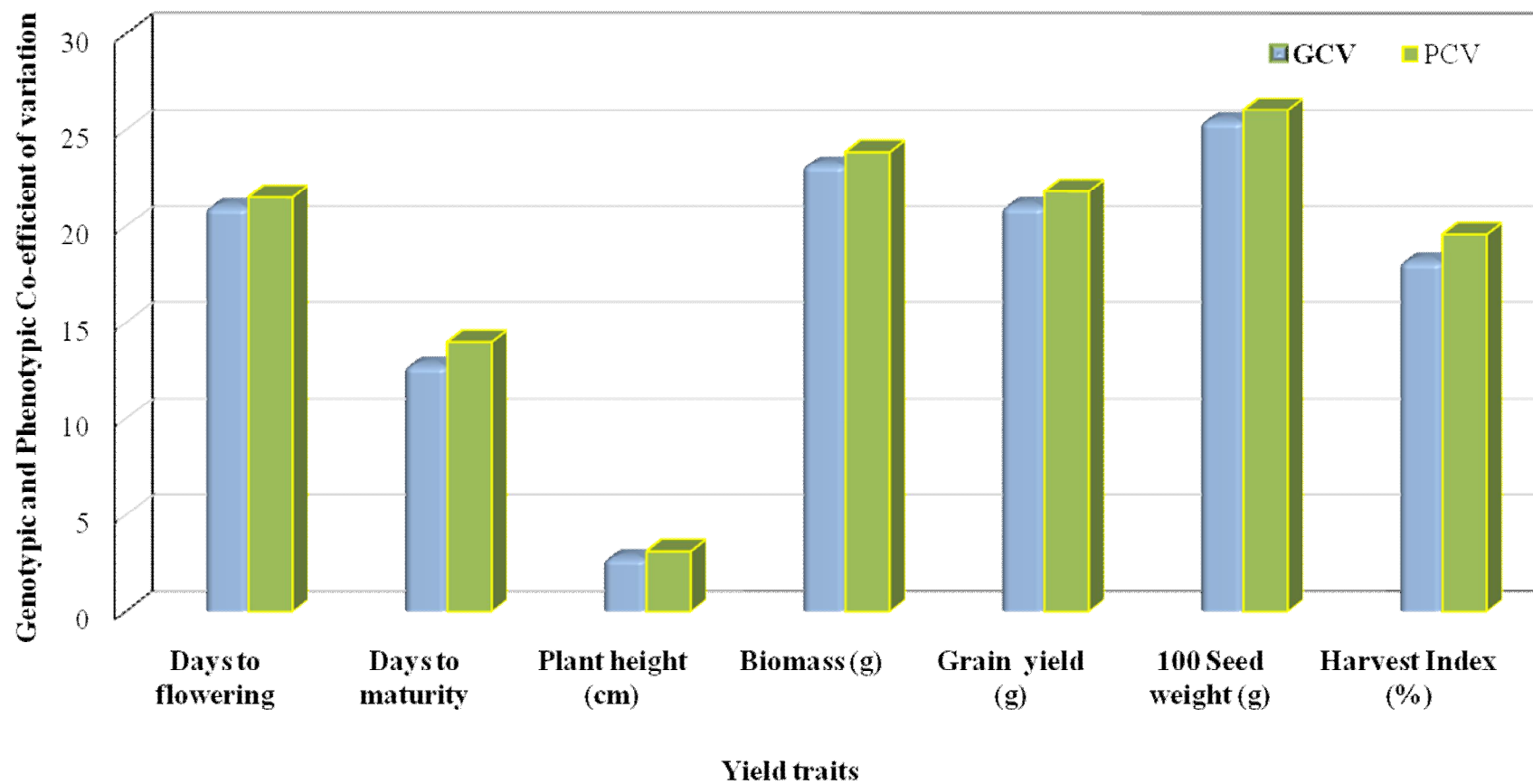


Fig. 3. Genotypic and phenotypic co-efficient of variation for yield and yield related traits in chickpea (rabi 2016-17)

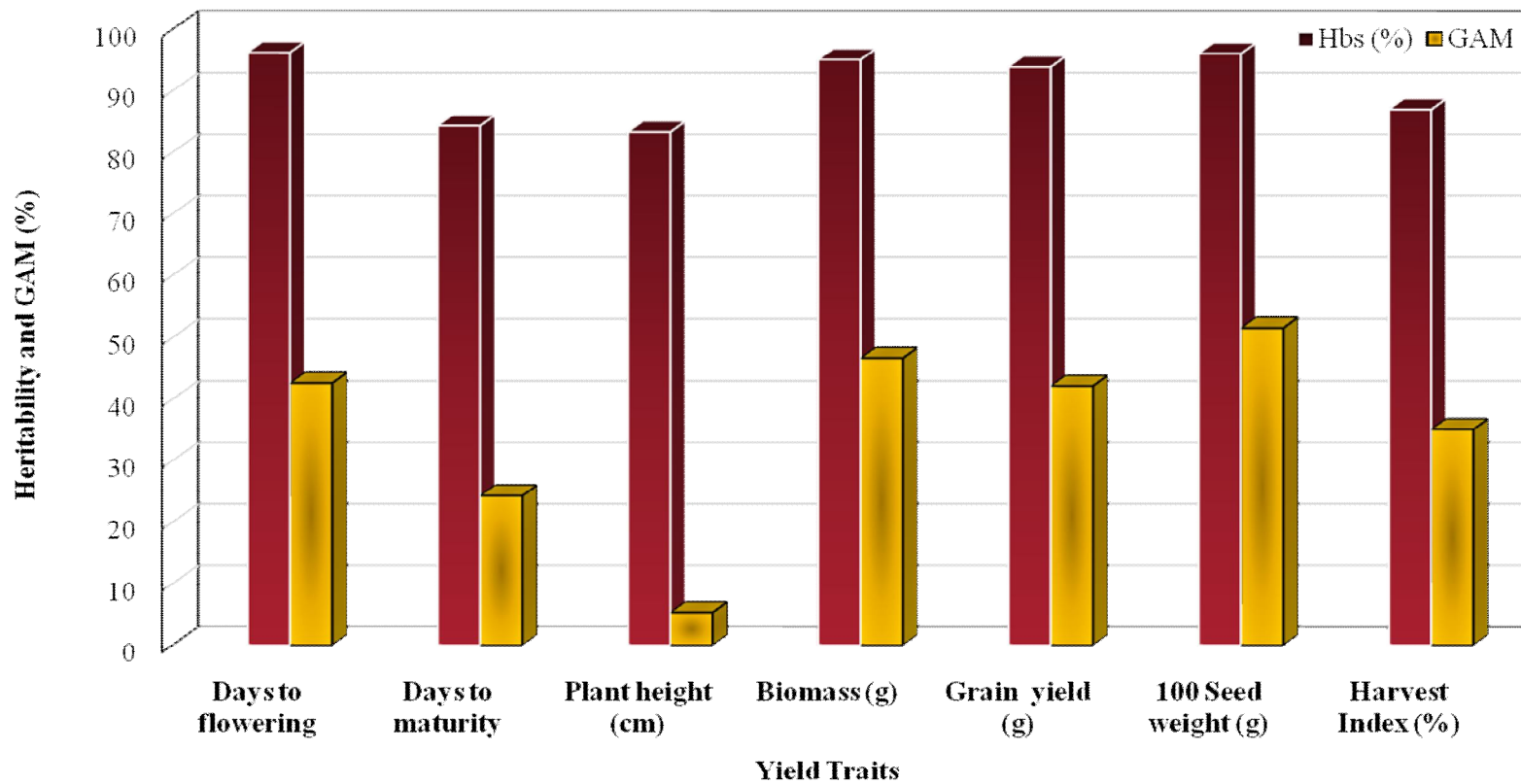


Fig. 4. Heritability and GAM (%) for seed yield and its components in chickpea (*rabi* 2016-2017)

4.1.2.2.5 Grain yield per plot (g)

Grain yield showed a wide range, which is varying from 60 g to 381.67g with a mean of 213.41g. It showed high PCV (21.80%) and GCV (21.12%). The estimates of heritability were high with (93.89%) and genetic advance as per cent of mean (42.16%) was also high for this character.

4.1.2.2.6 Hundred seed weight (g)

The average 100 seed weight of RILs was 19.69 g and mean value ranged from 11.03g to 37.73g. The PCV (26.05%) and GCV (25.53%) estimates were high. Heritability was high (96.08%) with high genetic advance as per cent of mean (51.55).

4.1.2.2.7 Harvest index (%)

The overall mean of harvest index in RIL population recorded was 52.49 % and mean values ranged from 30.32% to 63.74%. The magnitude of PCV and GCV was moderate with 19.58 and 18.26%, respectively. The heritability was 86.97% and high genetic advance as per cent mean (35.10) for this character.

4.1.3 Mean performance of chickpea RILs for yield and yield contributing traits.

Mean performance of RILs of chickpea for yield and yield contributing traits during *rabi* (2015-16) and (2016-17)

4.1.3.1 Mean performance of RILs of chickpea for yield and yield contributing traits during *rabi* (2015-16)

Mean performance of RILs of chickpea for yield and yield contributing traits are presented below Table 8.

4.1.3.1.1 Days to 50 per cent flowering.

Early flowering was recorded in RILs viz., ICCRIL14-60 and 160 (35days) followed by 2,49 and 133 (36 days) which can be used for developing early duration varieties. The late flowering RILs were ICCRIL14-78 (75 days) followed by 95, 110, 161 and 165 (69 days); 55 (68 days); 91 and 112 (67 days); and 08 (66 days). The female parent BG 212 (*desi*) recorded 56 days for days to 50% flowering, while the male parent ICCV (*kabuli*) flowered in 37 days.

4.1.3.1.2 Days to maturity

Early maturity is an advantageous trait in chickpea to avoid terminal drought which frequently occurs in rainfed areas. ICCRILs 133 (82 days) followed by 25 and 143 (83 days) were identified for early maturity. The ICCRILs 120 (129 days) followed by 54 (125 days); 55, 74, 142 and 136 (121 days); 91 (120 days); 11, 112, 95, 75 and 162 (119 days), 78, 110 and 133 (118 days) matured late.

4.1.3.1.3 Plant height (cm)

Tall plant is considered as favourable trait for mechanical harvesting. Among 182 RILs, ICCRILs 145 (54.89 cm) was the tallest followed by 85 (53.56 cm), 80 (51.11 cm), 110 (50.44 cm) and 162 (50.11 cm). The ICCRIL14-47 (26.89cm) was found as a dwarf type among the RILs. The plant height of BG212 was recorded as 44.44cm and that of ICCV 08305 as 39.67cm.

4.1.3.1.4 Plant biomass per plot (g)

Out of 182 RILs, the highest plant biomass as per plot was found in ICCRIL14-119 (600 g) followed by 85 (580g), 48 (558.33g), 82 (536g) and 43 (527.33g). The lowest plant biomass was observed in ICCRIL14-107 (146.67g). The plant biomass per plot for BG 212 was recorded as 379g and that for ICCV 08305 as 312.33 g.

4.1.3.1.5 Grain yield per plot (g)

Among the 182 RILs, ICCRIL14-151 was found as highest grain yielder with (292.33g) followed by 152 (281.33g), 85 (276g), 129 (275g) and 119 (267.33g). The lowest grain yielder was ICCRIL14-133 (68.33g). The grain yield of female parent BG 212 was 167.38 and male parent ICCV 08305 was 166 g.

4.1.3.1.6 Hundred seed weight (g)

The highest 100 seed weight in RILs was observed in ICCRIL14-013 (35.34g) followed by 87 (35.19g), 154 (33.62g), 21 (32.93g) and 35 (31.94g). The lowest was observed in RIL 20 (10.90 g). The 100 seed weight of female parent BG212 was 11.84g, while it was 35.30 g in male parent ICCV 08305.

Table 8. Mean performance of RILs for seed yield and its components during *rabi* chickpea (2015-16)

Trait	Days to 50 per cent flowering	Days to maturity	Plant height (cm)	Plant biomass per plot (g)	Grain yield per plot (g)	Hundred seed weight (g)	Harvest index (%)
Highest 1	ICCRIL14-0078 (74.67)	ICCRIL14-0120 (128.67)	ICCRIL14-0145 (54.89)	ICCRIL14-0119 (600.00)	ICCRIL14-0151 (292.33)	ICCRIL14-0013 (35.34)	ICCRIL14-0152 (59.87)
2	ICCRIL14-0095 & 0110 (69.00)	ICCRIL14-0054 (125.00)	ICCRIL14-0085 (53.56)	ICCRIL14-0085 (580.00)	ICCRIL14-0152 (281.33)	ICCRIL14-0087 (35.19)	ICCRIL14-0047 (59.04)
3	ICCRIL14-0161 (68.67)	ICCRIL14-0074 & 0142 (121.33)	ICCRIL14-0080 (51.11)	ICCRIL14-0048 (558.33)	ICCRIL14-0085 (276.00)	ICCRIL14-0154 (33.62)	ICCRIL14-0176 (58.92)
4	ICCRIL14-0165 (68.33)	ICCRIL14-0136 (121.00)	ICCRIL14-0110 (50.44)	ICCRIL14-0082 (536.00)	ICCRIL14-0129 (275.00)	ICCRIL14-0021 (32.93)	ICCRIL14-0147 (58.34)
5	ICCRIL14-0055 (67.33)	ICCRIL14-0055 (120.67)	ICCRIL14-0162 (50.11)	ICCRIL14-0043 (527.33)	ICCRIL14-0119 (267.33)	ICCRIL14-0035 (31.94)	ICCRIL14-0129 (58.29)
Lowest	ICCRIL14-0160 (35.33)	ICCRIL14-0133 (82.00)	ICCRIL14-0047 (26.89)	ICCRIL14-0107 (146.67)	ICCRIL14-0133 (68.33)	ICCRIL14-0020 (10.90)	ICCRIL14-0132 (33.52)
Mean	50.63	100.42	39.65	167.38	340.37	19.99	49.48
BG212	60.33	105.33	44.44	166.00	379.00	11.84	44.44
ICCV 08305	48.00	85.00	39.67	131.67	312.33	35.30	42.41

4.1.3.1.7 Harvest index (%)

Among the 182 RILs, the highest harvest index was observed in ICCRIL14-152 (59.87%) followed by 47 (59.04%), 176 (58.92%), 0147 (58.34%) and 129 (58.29%). The lowest was observed in ICCRIL14-132 (33.52%). The harvest index of parent BG212 was 44.44% and ICCV 08305 was of 42.41%.

4.1.3.2 Mean performance of chickpea RILs for yield and yield contributing traits during *rabi* (2016-17).

Mean performance of RILs for yield and its components are presented below in Table 9.

4.1.3.2.1 Days to 50 per cent flowering

The RILs ICCRIL14-62 (32 days) followed by 2, 49, 105, 93 and 133 (33 days) were the early flowering lines. The late flowering lines were ICCRIL14-134 (62 days), 175 (61days), 135 (60 days), 79 (59 days) and 8 (58 days). The days to 50 per cent flowering was recorded as 56 days in BG 212 and 37 days in ICCV 08305.

4.1.3.2.2 Days to maturity

The RIL ICCRIL14-32 matured in 91 days followed by 105, 115 and 154 (92 days) which were identified as early maturing lines. ICCRIL14-144 was found as late maturing type (119 days) followed by 91(108 days); 55(103 days); 179 (102 days); and 78, 81, 95 and 120 (101 days). The female parent BG 212 matured in 97 days and male parent ICCV 08305 matured in 95 days.

4.1.3.2.3 Plant height (cm)

Out of 182 RILs, the highest plant height was observed in ICCRIL14-80 (66.44cm) followed by 110, 120, 132 and 163 with 65.44 cm, 60.89 cm, 60.56cm and 59.56cm, respectively. The lowest plant height was observed in ICCRIL14-136 (29.67cm). The female parent BG212 recorded a plant height 47.33cm, while the male parent ICCV 08505 recorded 46.22cm.

4.1.3.2.4 Plant biomass per plot (g)

The highest plant biomass per plot was observed in ICCRIL14-56 (635g) followed by 69 (610g), 106 (603.33g), 118 (578.33g) and 172 (576.67g). The lowest plant biomass per plot was observed in ICCRIL14-136 (126.67g). The plant biomass per plot of female parent BG212 was 470g and for the male parent ICCV 08305, it was 535g.

4.1.3.2.5 Grain yield per plot (g)

Among the 182 RILs, the highest grain yielding line observed was ICCRIL14-69 (381.67g) followed by 58, 56, 34 and 89 with grain yield of 336.67g, 331.67g, 323.33g and 313.33g, respectively.

4.1.3.2.6 Hundred seed weight (g)

The highest 100 seed weight was observed in ICCRIL14-13 (37.73g) followed by 21 (34.03g), 154 (32.69g), 78 (30.50g) and 65 (30.11g). The lowest 100 seed weight was observed in ICCRIL14-136 (11.03g). The hundred seed weight of female parent BG212 was 12.51 g, whereas for the male parent ICCV 08305, it was 34.72g.

4.1.3.2.7 Harvest index (%)

Among 182 RILs, ICCRIL14-33 recorded the highest harvest index (63.74%) followed by 141 (63.67%), 147 (63.10%), 145 (62.76%) and 69 (62.60%). The lowest harvest index was observed in ICCRIL14-162 (30.32%). The harvest index was recorded as 52.60% for the female parent BG212 and 35.77% for the male parent ICCV 08305.

4.2 Association studies for yield and yield contributing traits in chickpea RILs in *rabi* (2015-16).

Association studies results are presented below for both the seasons *viz.*, *rabi* 2015-16 and *rabi* 2016-17.

4.2.1 Association studies for yield and yield contributing traits in chickpea RILs in *rabi* (2015-16).

Table 9. Mean performance of RILs for seed yield and its components during *rabi* chickpea (2016-17)

Trait	Days to 50 per cent flowering	Days to maturity	Plant height (cm)	Plant biomass per plot (g)	Grain yield per plot (g)	Hundred seed weight (g)	Harvest index (%)
Highest 1	ICCRIL14-0134 (62)	ICCRIL14-0144 (119)	ICCRIL14-0080 (66.44)	ICCRIL14-0056 (635.00)	ICCRIL14-0069 (381.67)	ICCRIL14-0013 (37.73)	ICCRIL14-0033 (63.74)
2	ICCRIL14-0175 (61)	ICCRIL14-0091 (108)	ICCRIL14-0110 (65.44)	ICCRIL14-0069 (610.00)	ICCRIL14-0058 (336.67)	ICCRIL14-0021 (34.03)	ICCRIL14-0141 (63.67)
3	ICCRIL14-0135 (60)	ICCRIL14-0055 (103)	ICCRIL14-0120 (60.89)	ICCRIL14-0106 (603.33)	ICCRIL14-0056 (331.67)	ICCRIL14-0154 (32.69)	ICCRIL14-0147 (63.10)
4	ICCRIL14-0079 (59)	ICCRIL14-0110, 0179 (102)	ICCRIL14-0132 (60.56)	ICCRIL14-0118 (578.33)	ICCRIL14-0034 (323.33)	ICCRIL14-0078 (30.50)	ICCRIL14-0145 (62.76)
5	ICCRIL14-0008 (58)	ICCRIL14- 0095,0120,0078,0081 (101)	ICCRIL14-0163 (59.56)	ICCRIL14-0172 (576.67)	ICCRIL14-0089 (313.33)	ICCRIL14-0065 (30.11)	ICCRIL14-0069 (62.60)
Lowest	ICCRIL14-0062 (32)	ICCRIL14-0032 (91)	ICCRIL14-0136 (29.67)	ICCRIL14-0136 (126.67)	ICCRIL14-0136 (60.00)	ICCRIL14-0136 (11.03)	ICCRIL14-0162 (30.32)
Mean	44.41	96.15	46.49	408.83	213.38	19.65	52.60
BG212	56	97	47.33	470.00	168.33	12.51	35.77
ICCV 08305	37	95	46.22	535.00	265.00	34.72	49.53

The phenotypic and genotypic correlation coefficients were determined to know the nature of relationship existing among RIL population. The results of genotypic and phenotypic correlation are presented in Tables 10 and 11

4.2.1.1 Day to 50 per cent flowering

Days to 50 percent flowering exhibited highly significant positive association at genotypic and phenotypic levels with days to maturity (0.8535**G, 0.8259**P), plant height (0.5937**G, 0.4581**P), plant biomass per plot (0.6761**G, 0.6280**P) and grain yield per plot (0.6077**G, 0.5633**P) but significant negative association with harvest index at both levels (-0.1138**G, -0.0950 **P).

4.2.1.2 Days to maturity

At both levels, days to maturity had highly significant association with plant height (0.6004**G, 0.4564**P), plant biomass per plot (0.5806**G, 0.5361**P) and grain yield per plot (0.4968**G, 0.4609**P) but highly significant negative association was noticed with harvest index (-0.1533**G, -0.1186**P) and 100 seed weight (-0.099*G, -0.0853 *P) at both levels.

4.2.1.3 Plant Height (cm)

Plant height exhibited highly significant association with plant biomass per plot (0.4525**G, 0.3408**P), 100 seed weight (0.2142**G, 0.1579**P) and grain yield per plot (0.2837**G, 0.2214**P) but significant negative association with harvest index (-0.3637**G, -0.2167**P) at both levels.

4.2.1.4 Plant biomass per plot (g)

Plant biomass per plot had highly significant association with grain yield per plot (0.9207**G, 0.8832**P) and had significant negative association with harvest index (-0.0950**G, -0.1736**P) at both genotypic and phenotypic levels.

4.2.1.5 Hundred seed weight (g)

Hundred seed weight had highly significant negative association with grain yield per plot (-0.0660**G, -0.0648**P) at both levels.

Table 10. Genotypic correlation for seed yield and yield attributing traits in RILs of *rabi* chickpea (2015-16)

Trait	Days to 50 per cent flowering	Days to maturity	Plant height (cm)	Plant biomass per plot (g)	Hundred seed weight (g)	Harvest index (%)	Grain yield Per plot (g)
Days to 50 per cent flowering	1	0.8535**	0.5937**	0.6761**	-0.0447	-0.1138*	0.6077**
Days to maturity		1	0.6004**	0.5806**	-0.099*	-0.1533**	0.4968**
Plant height (cm)			1	0.4525**	0.2142**	-0.3637**	0.2837**
Plant biomass per plot (g)				1	-0.0573	-0.095**	0.9207**
Hundred seed weight (g)					1	-0.0179	-0.066**
Harvest index (%)						1	0.2889**

Table 11. Phenotypic correlation for seed yield and its components in RILs of *rabi* chickpea (2015-16)

Trait	Days to 50 per cent flowering	Days to maturity	Plant height (cm)	Plant biomass per plot (g)	Hundred seed weight (g)	Harvest index (%)	Grain yield per plot (g)
Days to 50 per cent flowering	1.0000	0.8259 **	0.4581 **	0.6280 **	-0.0369	-0.0950 *	0.5633**
Days to maturity		1.0000	0.4564 **	0.5361 **	-0.0853 *	-0.1186 **	0.4609**
Plant height (cm)			1.0000	0.3408 **	0.1579 **	-0.2167 **	0.2214**
Plant biomass per plot (g)				1.0000	-0.0540	-0.1736 **	0.8832**
Hundred seed weight (g)					1.0000	-0.0219	-0.0648**
Harvest index (%)						1.0000	0.2902**

4.2.1.6 Harvest index (%)

Harvest index had highly significant positive association with grain yield per plot (0.2889**G, 0.2902**P) at both genotypic and phenotypic levels.

4.2.1.7 Grain yield per plot (g)

Grain yield per plot had highly significant positive association with days to 50 per cent flowering (0.6077**G, 0.5633**P), days to maturity (0.4968**G, 0.4609**P), plant height (0.2837**G, 0.2214**P), plant biomass per plot (0.9207**G, 0.8832**P), harvest index (0.2889**G, 0.2902**P) but had highly significant negative association with 100 seed weight (-0.0660**G, -0.0648**P) at both levels.

4.2.2 Association studies for yield and its attributing traits in chickpea RILs in *rabi* (2016-17)

The results of genotypic and phenotypic correlation are presented in Tables 12 and 13.

4.2.2.1 Days to 50 per cent flowering

In RILs, days to 50 per cent flowering had highly significant association with days to maturity (0.6594**G, 0.5228**P), plant height (0.3437**G, 0.2820**P), plant biomass per plot (0.4335**G, 0.3936**P) and grain yield per plot (0.2701**G, 0.2502**G) but significant negative association was noticed with 100 seed weight (-0.0992*G, -0.0911*P) and harvest index (-0.2249**G, -0.1795**P) at both levels.

4.2.2.2 Days to maturity

Days to maturity exhibited highly significant positive association with plant height (0.3643**G, 0.2529**P), plant biomass per plot (0.4183**G, 0.3113**P) and grain yield per plot (0.2234**G, 0.1854**P) but significant negative association was observed with harvest index (-0.2640**G, -0.1445**P) at both levels.

4.2.2.3 Plant height (cm)

At both genotypic and phenotypic levels, plant height had highly significant positive association with plant biomass per plot (0.5483**G, 0.4088**P), 100 seed

weight (0.4192**G, 0.3227**P) and grain yield per plot (0.3199**G, 0.2461**P) but significant negative association with harvest index (-0.3067**G, -0.1993**P).

4.2.2.4 Plant biomass per plot (g)

Plant biomass per plot exhibited highly significant positive association with 100 seed weight (0.2103**G, 0.1818**P) and grain yield per plot (0.8151**G, 0.7580**P) but significant negative association with harvest index (-0.1514**G, -0.2409**P) at both genotypic and phenotypic levels, respectively.

4.2.2.5 Hundred seed weight (g)

In RILs, 100 seed weight had significant positive association with grain yield per plot (0.2895**G, 0.2572**P) at both levels.

4.2.2.6 Harvest index (%)

At both genotypic and phenotypic levels, harvest index exhibited highly significant positive association with grain yield per plot (0.4385**G, 0.4356**P).

4.2.2.7 Grain yield per plot (g)

Grain yield per plot had highly significant positive association with days to 50 per cent flowering (0.2701**G, 0.2502**P), days to maturity (0.2234**G, 0.1854**P), plant height (0.3199**G, 0.2461**P), plant biomass per plot (0.8151**G, 0.7580**P), 100 seed weight (0.2895**G, 0.2572**P) and harvest index (0.4385**G, 0.4356**P) at both levels.

4.3.1.1 Day to 50 per cent flowering

Days to 50 per cent flowering had low positive direct effect (0.0249 G, 0.0134 P) on grain yield per plot at both levels. The positive indirect effect was noticed through plant biomass per plot and 100 seed weight at both levels. Whereas, negative indirect effect was recorded *via* days to maturity, plant height and harvest index on grain yield per plot at both level.

Table 12. Genotypic correlation for seed yield and its components in RILs of *rabi* chickpea (2016-17)

Trait	Days to 50 per cent flowering	Days to maturity	Plant height (cm)	Plant biomass per plot (g)	Hundred seed weight (g)	Harvest index (%)	Grain yield per plot (g)
Days to 50 per cent flowering	1	0.6594**	0.3437**	0.4335**	-0.0992*	-0.2249**	0.2701**
Days to maturity		1	0.3643**	0.4183**	-0.0902	-0.264**	0.2234**
Plant height (cm)			1	0.5483**	0.4192**	-0.3067**	0.3199**
Plant biomass per plot (g)				1	0.2103**	-0.1514**	0.8151**
Hundred seed weight (g)					1	0.1773**	0.2895**
Harvest index (%)						1	0.4385**

Table 13. Phenotypic correlation for seed yield and its components in RILs of *rabi* chickpea (2016-17)

Trait	Days to 50 per cent flowering	Days to maturity	Plant height (cm)	Plant biomass per plot(g)	Hundred seed weight (g)	Harvest index (%)	Grain yield per plot (g)
Days to 50 per cent flowering	1.0000	0.5228 **	0.2820 **	0.3936 **	-0.0911 *	-0.1795 **	0.2502**
Days to maturity		1.0000	0.2529 **	0.3113 **	-0.0505	-0.1445 **	0.1854**
Plant height (cm)			1.0000	0.4088 **	0.3227 **	-0.1993 **	0.2461**
Plant biomass per plot (g)				1.0000	0.1818 **	-0.2409 **	0.7580**
Hundred seed weight (g)					1.0000	0.1369 **	0.2572**
Harvest index (%)						1.0000	0.4356**

4.3.1.2 Days to maturity

Days to maturity exhibited negative direct effect (-0.073G, -0.0066P) on grain yield per plot at both levels. It had negative indirect effect *via* plant height, 100 seed weight and harvest index at genotypic level. On the other hand, positive indirect effect was noticed through days to 50 percent flowering and plant biomass per plot at both levels.

4.3.1.3 Plant height (cm)

Plant height had negative direct effect (-0.0228G, -0.0103P) on grain yield per plot. The positive indirect effect observed *via* days to flowering, plant biomass per plot and 100 seed weight at both levels. Whereas, days to maturity and harvest index exhibited negative indirect effect on grain yield per plot.

4.3.1.4 Plant biomass per plot (g)

Plant biomass per plot exhibited high positive direct effect (0.9538G, 0.9608 P) on grain yield per plot at both levels, high positive indirect effect *via* days to 50 per cent flowering at genotypic level and positive indirect effect at phenotypic level *via* 100 seed weight on grain yield per plot. Whereas, negative indirect effect was observed through plant height, days to maturity and harvest index at both levels.

4.3.1.5 Hundred seed weight (g)

Hundred seed weight had positive direct effect (0.0005G) at genotypic level, but negative direct effect (-0.0014P) at phenotypic level on grain yield per plot. Whereas, positive indirect effect was recorded *via* days to maturity at both levels. However, negative indirect effect was noticed through days to 50 percent flowering, plant height, plant biomass per plot and harvest index at both levels.

Table 14. Genotypic path analysis for seed yield and its components in RILs of *rabi* chickpea (2015-16)

Trait	Days to 50 per cent flowering	Days to maturity	Plant height (cm)	Plant biomass per plot (g)	Hundred seed weight (g)	Harvest index (%)	Grain yield per plot (g)
Days to 50 per cent flowering	0.0249	-0.0062	-0.0135	0.6449	0000	-0.0424	0.6077
Days to maturity	0.0213	-0.0073	-0.0137	0.5538	-0.0001	-0.0572	0.4968
Plant height (cm)	0.0148	-0.0044	-0.0228	0.4316	0.0001	-0.1356	0.2837
Plant biomass per plot (g)	0.0169	-0.0042	-0.0103	0.9538	0000	-0.0354	0.9207
Hundred seed weight (g)	-0.0011	0.0007	-0.0049	-0.0546	0.0005	-0.0067	-0.0660
Harvest index (%)	-0.0028	0.0011	0.0083	-0.0906	0000	0.3729	0.2889

Table 15. Phenotypic path analysis for seed yield and its components in RILs of *rabi* chickpea (2015-16)

Trait	Days to 50 per cent flowering	Days to maturity	Plant height (cm)	Plant biomass per plot (g)	Hundred seed weight (g)	Harvest index (%)	Grain yield per plot (g)
Days to 50 per cent flowering	0.0134	-0.0055	-0.0047	0.6033	0.0001	-0.0432	0.5633
Days to maturity	0.0111	-0.0066	-0.0047	0.515	0.0001	-0.054	0.4609
Plant height (cm)	0.0061	-0.003	-0.0103	0.3275	-0.0002	-0.0987	0.2214
Plant biomass per plot (g)	0.0084	-0.0035	-0.0035	0.9608	0.0001	-0.079	0.8832
Hundred seed weight (g)	-0.0005	0.0006	-0.0016	-0.0519	-0.0014	-0.01	-0.0648
Harvest index (%)	-0.0013	0.0008	0.0022	-0.1668	0000	0.4552	0.2902

4.3.1.6 Harvest index (%)

Harvest index recorded high positive direct effect (0.3729G, 0.4552P) on grain yield per plot, while positive indirect effect was noticed *via* days to maturity and plant height at both levels. Whereas, it had negative indirect effect *via* days to 50 per cent flowering and plant biomass per plot at both levels.

4.3.1.7 Grain yield per plot (g)

Days to 50 per cent flowering, plant biomass per plot, 100 seed weight and harvest index exhibited positive direct effect on grain yield per plot. Whereas, days to maturity and plant height recorded negative direct effect on grain yield per plot at both levels.

4.3.2 Path analysis for yield and yield contributing traits in chickpea RILs in *rabi* (2016-17).

Path analysis was carried out for RILs of chickpea and results are presented in Tables 16 and 17.

4.3.2.1 Days to 50 per cent flowering

Days to 50 per cent flowering exhibited positive direct effect (0.0174G, 0.0144P) on grain yield per plot at both levels. It had positive indirect effect *via* plant height and plant biomass per plot at both levels. Whereas, days to maturity and harvest index showed negative indirect effect at both levels.

4.3.2.2 Days to maturity

It exhibited negative direct effect (-0.0137G,-0.012P) on grain yield per plot at both levels. It had positive indirect effect through days to 50 percent flowering, plant height and plant biomass per plot at both levels. Whereas, harvest index had negative indirect effect at both levels.

4.3.2.3 Plant height (cm)

Plant height had positive direct effect (0.0042 G, 0.0023P) on grain yield per plot at phenotypic level. It had negative indirect effect *via* days to maturity, 100 seed weight

and harvest index at both levels. However, days to flowering and plant biomass per plot had positive indirect effect on grain yield per plot at both levels.

4.3.2.4 Plant biomass per plot (g)

Plant biomass per plot had high positive direct effect (0.899G, 0.9130P) at both levels on grain yield per plot. It had high positive indirect effect *via* days to 50 per cent flowering and plant height at both levels, while it had negative indirect effect through days to maturity, hundred seed weight and harvest index at both level.

4.3.2.5 Hundred seed weight (g)

Hundred seed weight exhibited negative direct effect (-0.0031G) on grain yield per plot at genotypic level and positive direct effect (0.0013P) at phenotypic level. It had positive indirect effect *via* days to maturity, plant height, plant biomass per plot and harvest index at both levels. Whereas, negative indirect effect was noticed through days to 50 per cent flowering at both levels.

4.3.2.6 Harvest index (%)

Harvest index had high positive direct effect (0.5767G, 0.6566P) on grain yield per plot at both levels. It had positive indirect effect *via* days to maturity at both levels. Whereas, negative indirect effect was observed *via* days to 50 per cent flowering, plant height, plant biomass per plot and 100 seed weight at both levels.

4.3.2.7 Grain yield per plot (g)

Days to 50 per cent flowering, plant height, plant biomass per plot and harvest index had positive direct effect on grain yield per plot. Plant height had negative direct effect on grain yield per plot at both levels. But, 100 seed weight had negative direct effect on grain yield per plot at phenotypic level but positive direct effect was observed at genotypic level.

Table 16. Genotypic path analysis for seed yield and its components in RILs of *rabi* chickpea (2016-17)

Trait	Days to 50 per cent flowering	Days to maturity	Plant height (cm)	Plant biomass per plot (g)	Hundred seed weight (g)	Harvest index (%)	Grain yield per plot (g)
Days to 50 per cent flowering	0.0174	-0.0091	0.0014	0.3897	0.0003	-0.1297	0.2701
Days to maturity	0.0115	-0.0137	0.0015	0.3761	0.0003	-0.1522	0.2234
Plant height (cm)	0.006	-0.005	0.0042	0.493	-0.0013	-0.1769	0.3199
Plant biomass per plot (g)	0.0075	-0.0057	0.0023	0.899	-0.0007	-0.0873	0.8151
Hundred seed weight (g)	-0.0017	0.0012	0.0018	0.1891	-0.0031	0.1022	0.2895
Harvest index (%)	-0.0039	0.0036	-0.0013	-0.1361	-0.0005	0.5767	0.4385

Table 17. Phenotypic path analysis for seed yield and its components in RILs of *rabi* chickpea (2016-17)

Trait	Days to 50 per cent flowering	Days to maturity	Plant height (cm)	Plant biomass per plot (g)	Hundred seed weight (g)	Harvest index (%)	Grain yield per plot (g)
Days to 50 per cent flowering	0.0144	-0.0063	0.0006	0.3594	0.0001	-0.1178	0.2502
Days to maturity	0.0075	-0.012	0.0006	0.2843	-0.0001	-0.0949	0.1854
Plant height (cm)	0.0041	-0.003	0.0023	0.3733	0.0004	-0.1308	0.2461
Plant biomass per plot (g)	0.0057	-0.0037	0.0009	0.913	0.0002	-0.1581	0.758
Hundred seed weight (g)	-0.0013	0.0006	0.0007	0.166	0.0013	0.0899	0.2572
Harvest index (%)	-0.0026	0.0017	-0.0005	-0.2199	0.0002	0.6566	0.4356

4. Screening of RILs of chickpea for DRR disease resistance

All the RILs (182) along with parents were screened under the laboratory condition using paper towel method to identify the reaction of the lines to the *Rhizoctonia bataticola* and also to identify the resistance sources for the disease.

The RILs were scored for the disease severity as per the 1-9 rating scale (Pande *et al.*, 2012). The RILs were grouped into different categories based on the extent of root damage caused by the disease.

4.4.1 Grouping of 182 RILs of chickpea in response to DRR infection upon artificial inoculation under laboratory condition.

Among the RILs evaluated, 13 lines showed very few and small lesions on the roots which were grouped under resistant category. About 79 RILs were found moderately resistant, lesions on the tap roots can be seen but new roots were free from the infection. About 78 RILs showed symptoms of susceptibility including many lesions on the roots but generally secondary roots free from the infection. Remaining 12 were highly susceptible, roots were completely infected and discoloured in both screenings including pooled screening (Table 18). Under controlled environment condition, parent ICCV 08305 (an average disease score of 4.33, 3.30 and 3.80) exhibited moderately resistant reaction to DRR disease as compared to highly susceptible parent BG 212 (an average disease score of above 8) during first, second time screening and pooled analysis, respectively (Table 19)

4.4.1. Analysis of variance for DRR disease incidence using paper towel method

The RILs showed significant differences for disease incidence during first and second screening and also for pooled analysis at both levels. A high broad sense heritability of 0.88, 0.99 and 0.95 were recorded for DRR disease incidence during first and second time screening and pooled analysis, respectively (Table 21).

4.5 Genetics of DRR disease resistance.

The experimental material for understanding the genetics DRR resistance consisted of 182 RILs developed from the cross BG 212× ICCV 08305. The disease severity was scored on 1- 9 scale as suggested by Pande *et al.* (2012). Phenotyping for

Table 18. Grouping of 182 RILs on their reaction to dry root rot (*Rhizoctonia*) infection using paper towel method

Scale	Category	Number of RILs		
		First screening	Second screening	Pooled screening
1	Asymtotic	0	0	0
3	Moderately resistant	13	13	13
5	Moderately susceptible	79	79	79
7	Susceptible	78	78	78
9	Highly susceptible	12	12	12

Table 19. Reaction of parents and RIL population for DRR disease

	First Screening	Second Screening	Pooled Analysis
BG 212	8.33	8	8.11
ICCV 08305	4.33	3.33	3.83
Mean	5.27	5.3	5.24
Range	0-9	00-9	0-9
SE±	0.3218	0.12	0.19
LSD (P 0.05)	0.8979	0.24	0.53

Table 20. Analysis of variance for dry root rot disease screening using paper towel method

Source of variance	First screening	Second screening	Pooled analysis
Genotype	3.30**	3.33**	3.29**
Replication	0.07	0.02	0.00
Error	0.21	0.01	0.07
Fp	<0.001	<0.001	<0.001
CV	8.65	2.28	5.15
Heritability	0.88	0.99	0.95

Table 21. Chi square Test: Paper towel method for dry root rot resistance during first screening (2016)

Category	Grouping of RILs		Observed	Expected	Chi square (Cal)	Chi square (Table)	p value
R	13	R	92	91	0.011	3.841	0.91
	79	MR					
S	78	S	90	91	0.011	3.841	
	12	HS					
	182						

Table 22. Chi square Test: Paper towel method for dry root rot resistance during second screening (2017)

Category	Grouping of RILs		Observed	Expected	Chi square (Cal)	Chi square (Table)	p value
R	13	R	92	91	0.011	3.841	0.91
	79	MR					
S	78	S	90	91	0.011	3.841	
	12	HS					
	182						

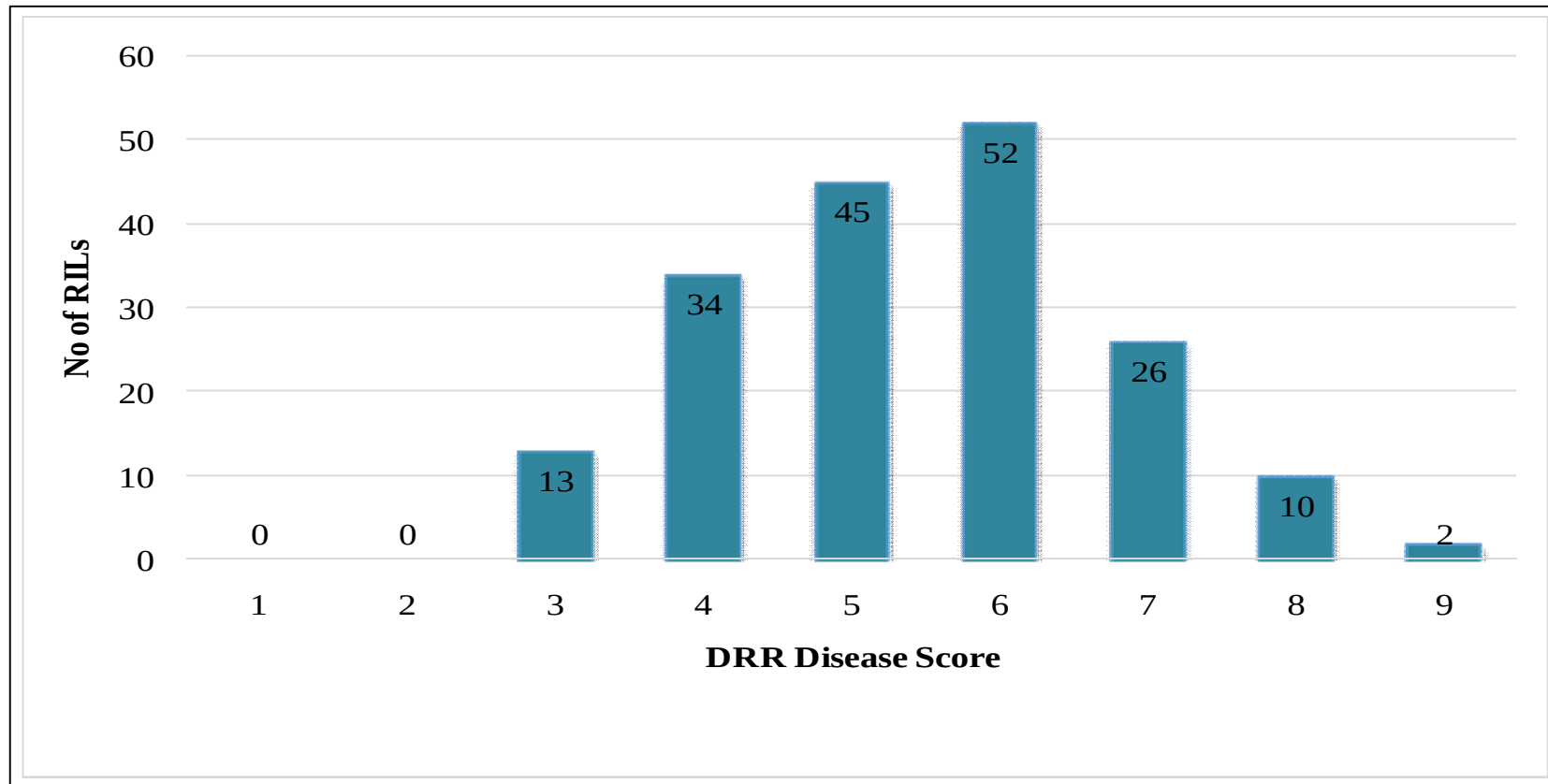


Fig. 5. Frequency distribution of DRR disease scores during first screening in RILs of cross BG 212×ICCV 08305

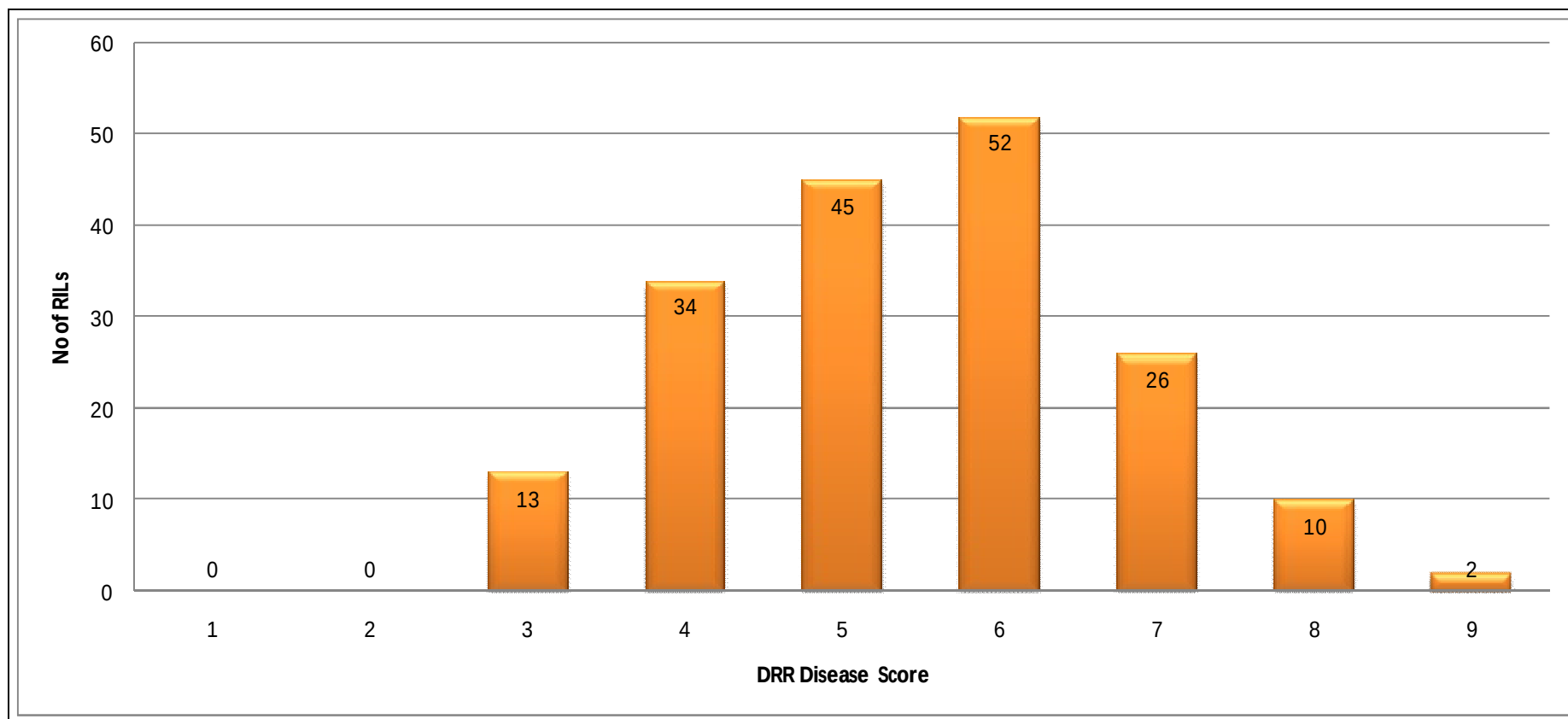


Fig. 6. Frequency distribution of DRR disease scores during second screening in RILs of cross BG 212×ICCV 08305

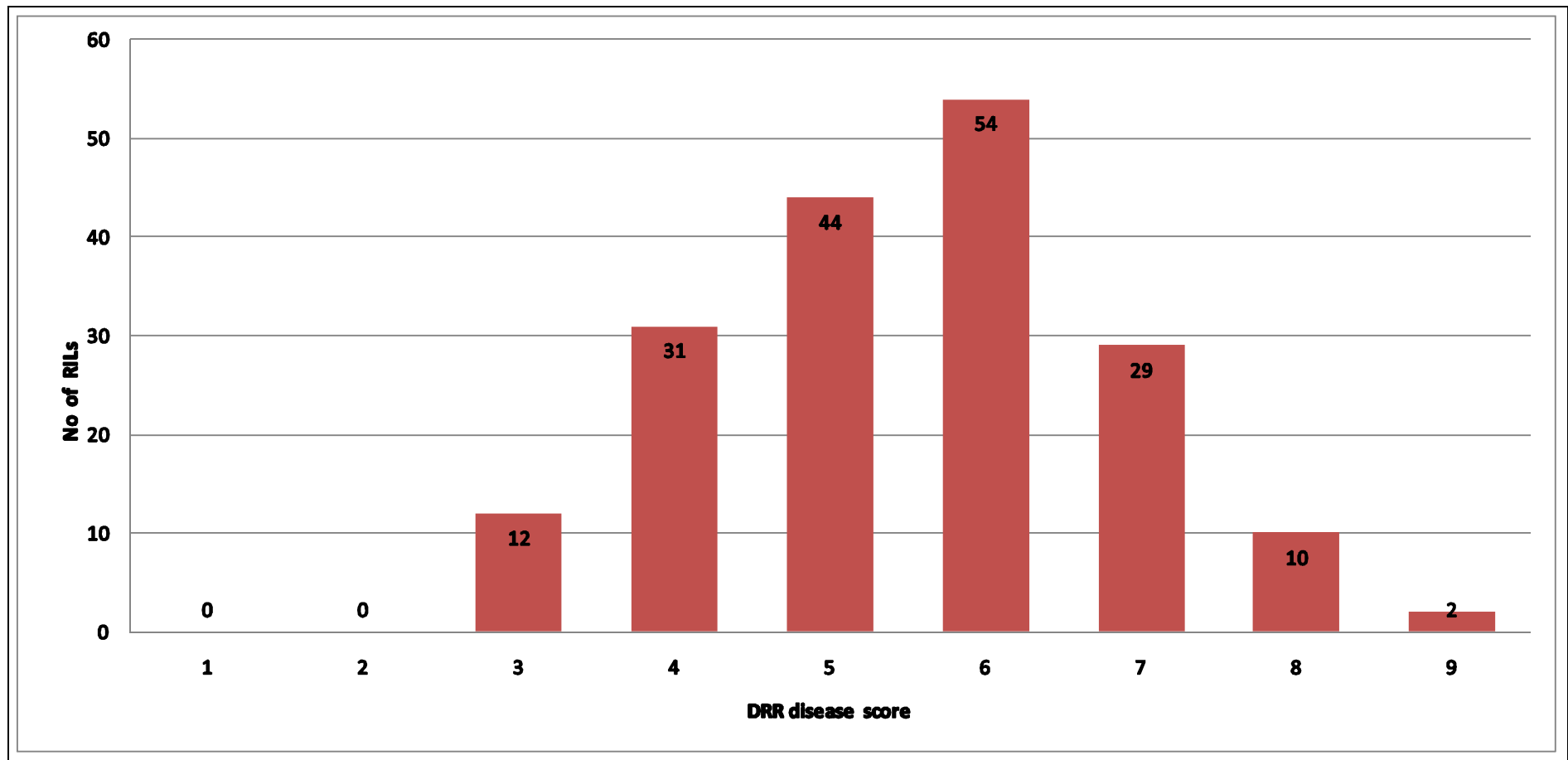


Fig. 7. Frequency distribution of DRR disease scores during pooled screening in RILs of cross BG 212×ICCV 0830



**Un-inoculated
(BG-212)**



**Inoculated
(BG-212)**



**Un-inoculated
(ICCV 08305)**



**Inoculated
(ICCV 08305)**

Disease incidence of inoculated and uninoculated parents using paper towel method

Plate 1. DRR disease scoring of RILs using paper towel method

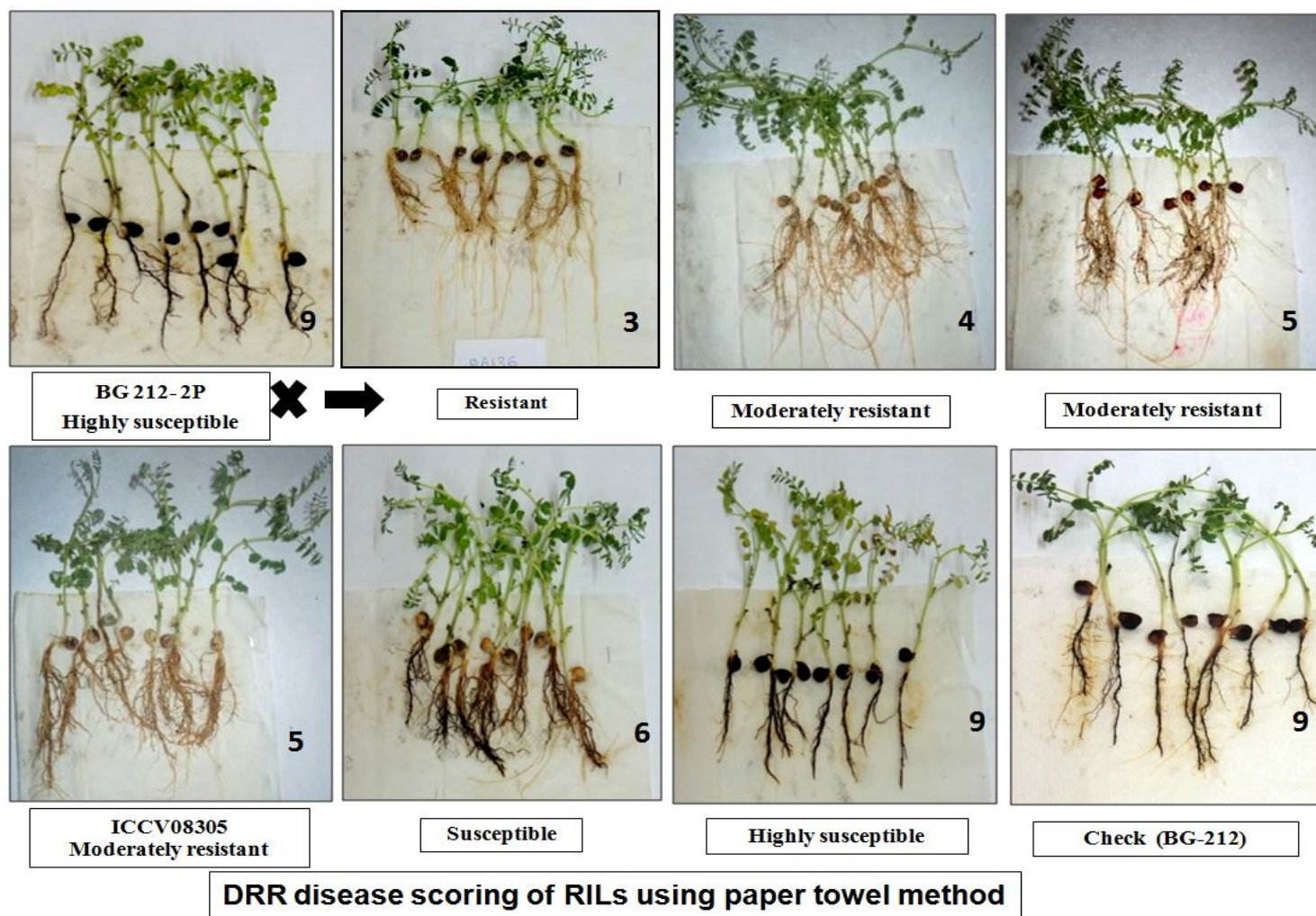


Plate 2. DRR disease incidence of inoculated and uninoculated RILs using paper towel method

Table 23. Relative comparison of resistant and susceptible chickpea RILs for yield, yield attributing traits and DRR disease incidence in *rabi* (2015-16)

Trait	Category	Mean	Variance	P(T<=T)One Tail	P(T<=T)Two Tail	P Value
DF	R	51.43	86.79	0.13	0.72	NS
	S	49.81	97.31			
DM	R	101.54	122.82	0.09	0.18	NS
	S	99.27	135.91			
PH	R	39.72	22.23	0.42	0.83	NS
	S	39.57	21.33			
BM	R	342.77	7875.70	0.36	0.72	NS
	S	337.92	8332.91			
GY	R	165.92	2120.61	0.33	0.66	NS
	S	168.88	2112.54			
HSW	R	19.56	20.44	0.14	0.28	NS
	S	30.73	29.32			
HI	R	48.67	33.53	0.02	0.04	NS
	S	50.30	24.77			
DI	R	4.22	0.48	<0.001	<0.001	S
	S	6.32	0.64			

Table 24. Relative comparison of resistant and susceptible chickpea RILs for yield, yield attributing traits and DRR disease incidence in *rabi* (2016-17)

Trait	Category	Mean	Variance	P(T<=T)One Tail	P(T<=T)Two Tail	P Value
DF	R	45.10	100.07	0.18	0.35	NS
	S	43.70	106.88			
DM	R	96.50	12.03	0.06	0.13	NS
		95.80	6.81			
PH	R	46.16	54.40	0.25	0.49	NS
	S	46.83	32.21			
BM	R	409.11	8133.63	0.45	0.91	NS
	S	410.59	5907.42			
GY	R	214.47	2487.11	0.38	0.76	NS
	S	212.33	2137.04			
HSW	R	18.97	29.39	0.03	0.06	NS
	S	20.43	25.58			
HI	R	52.77	41.34	0.29	0.59	NS
	S	52.20	58.54			
DI	R	4.23	0.50	<0.001	<0.001	S
	S	6.29	0.83			

DRR was carried out under laboratory condition using paper towel method. The data was collected from each individual RIL with two replications, which was subjected to goodness of fit with the expected genetic ratio using chi-square test. The segregation for disease resistance and susceptibility is presented in Tables 21 and 22.

In F₉, the segregation pattern of 1:1 (1 Resistant: 1 Susceptible) was observed and it showed very good fit with the expected ratio and chi-square values were non-significant at n-1 degrees of freedom (n= number of classes). The frequency distribution of RILs for DRR disease incidence (recorded on 1-9 scale) in first, second and pooled screening depicted a normal distribution (Figures 5, 6 and 7).

4.6 Relative comparison of resistant and susceptible chickpea RILs for yield, yield attributing traits and DRR disease incidence in both *rabi* 2015 and 2016.

The disease screening was done under controlled environmental condition using paper towel method. The results of t-test showed a non-significant difference between resistant and susceptible RILs for yield and yield attributing traits, whereas significant difference was found between the resistant and susceptible RILs for the disease incidence (Table 23 and 24).

4.7 Top performing DRR resistant RILs for agronomic traits

Among 182 RILs, 13 were found resistant and 79 as moderately resistant to DRR. Some of these lines were superior in agronomic performance (Table-25). These include ICCRIL 14-0133 for early flowering and early maturity; ICCRIL 14-0145 for plant height; ICCRIL 14-0151 for grain yield and plant biomass; ICCRIL 14-0119 for plant biomass and grain yield; ICCRIL 14-0154 for 100-seed weight; and ICCRIL 14-0152 for harvest index during 2015.

During 2016, the RILs with superior performance include ICCRIL14-133 for early flowering; ICCRIL14-032 for early maturity; ICCRIL14-80 for plant height; ICCRIL14-69 for grain yield and plant biomass; ICCRIL14-21 for 100-seed weight; and ICCRIL14-141 for harvest index.

Table 25. Mean performance of DRR resistant and moderate resistant RILs for yield and other agronomic traits of chickpea (*rabi* 2015-16 and *rabi* 2016-17)

Traits	2015			2016		
Days to flowering	RIL no	Days to flowering	Disease score	RIL no	Days to flowering	Disease score
	ICCRIL14-0133	36	5	ICCRIL14-0133	33	5
	ICCRIL14-0116	37	4	ICCRIL14-0002	33	5
	ICCRIL14-0139	38	5	ICCRIL14-0105	33	3
	ICCRIL14-0018	38	4	ICCRIL14-0072	33	4
	ICCRIL14-0154	38	4	ICCRIL14-0116	33	4
Days to maturity	RIL no	Days to maturity	Disease score	RIL no	Days to maturity	Disease score
	ICCRIL14-0133	82	5	ICCRIL14-0154	91	4
	ICCRIL14-0018	84	4	ICCRIL14-0105	92	3
	ICCRIL14-0109	85	5	ICCRIL14-0115	92	4
	ICCRIL14-0117	85	4	ICCRIL14-0181	93	3
	ICCRIL14-0139	85	5	ICCRIL14-031	93	5
Plant height (cm)	RIL no	Plant height (cm)	Disease score	RIL no	Plant height (cm)	Disease score
	ICCRIL14-0145	54.89	3	ICCRIL14-080	66.44	5
	ICCRIL14-0080	51.11	5	ICCRIL14-0110	65.44	5
	ICCRIL14-0110	50.44	5	ICCRIL14-0120	60.89	4
	ICCRIL14-0120	48.56	4	ICCRIL14-0132	60.56	4
	ICCRIL14-0074	47.44	5	ICCRIL14-088	57.11	3
Plant biomass per plot (g)	RIL no	Plant biomass per plot (g)	Disease score	RIL no	Plant biomass per plot (g)	Disease score
	ICCRIL14-0119	600.00	4	ICCRIL14-0069	610.00	3
	ICCRIL14-0043	527.33	5	ICCRIL14-0106	603.33	4
	ICCRIL14-0151	514.00	4	ICCRIL14-0118	578.33	4

	ICCRIL14-0106	512.33	4	ICCRIL14-0172	576.67	5
Grain yield per plot (g)	RIL no	Grain yield per plot (g)	Disease score	RIL no	Grain yield per plot (g)	Disease score
	ICCRIL14-0151	292.33	4	ICCRIL14-0069	381.67	3
	ICCRIL14-0152	281.33	3	ICCRIL14-0058	336.67	3
	ICCRIL14-0119	267.33	4	ICCRIL14-0124	300.83	4
	ICCRIL14-0043	260.00	5	ICCRIL14-0067	288.33	5
	ICCRIL14-0106	242.33	4	ICCRIL14-0123	286.67	5
Hundred seed weight (g)	RIL no	Hundred seed weight (g)	Disease score	RIL no	Hundred seed weight (g)	Disease score
	ICCRIL14-0154	33.62	4	ICCRIL14-0021	34.03	5
	ICCRIL14-0021	32.93	5	ICCRIL14-0154	32.69	4
	ICCRIL14-0035	31.94	5	ICCRIL14-0065	30.11	5
	ICCRIL14-0065	30.96	5	ICCRIL14-0031	29.39	5
	ICCRIL14-0146	30.91	4	ICCRIL14-0079	28.53	3
Harvest Index (%)	RIL no	Harvest Index (%)	Disease score	RIL no	Harvest Index (%)	Disease score
	ICCRIL14-0152	59.87	3	ICCRIL14-0141	63.67	3
	ICCRIL14-0147	58.34	3	ICCRIL14-0147	63.10	3
	ICCRIL14-0151	57.02	4	ICCRIL14-0145	62.76	3
	ICCRIL14-0031	56.38	5	ICCRIL14-0069	62.60	3
	ICCRIL14-0021	56.34	5	ICCRIL14-0108	62.46	3

4.8 Genetic map construction

The cost effective SNP genotyping platform, comprising of 50,590 high quality non-redundant SNPs tiled on to Affymetrix® Axiom® genotyping array was used for genotyping the RIL population. Sequence alignment and SNP calling identified 50,590 SNPs which contained polyhigh resolution 16,211 (32.04%) SNPs. Upon processing and filtering of SNPs, all SNP markers monomorphic across the parental genotypes were removed. All SNP markers with more than 10% missing SNP calls were removed before further analysis. As a result, a total of 13,110 SNP markers were used for further analysis. Out of these, a total of 13,110 SNPs were found highly polymorphic among the parental genotypes of DRR mapping population. High density genetic map was constructed containing 13,110 SNP markers distributed across eight linkage groups with a total map length of 1224.11 cM having an average inter-marker distance of 0.09 cM (Table-26, Figure 8). Further, the SNPs with redundant data across the RILs were excluded by retaining one SNP per locus. As a result, in the Genetic map a total of 5,052 SNPs having a total map length of 1244.11 cM with an average inter-marker distance of 0.09 cM were used for further QTL analysis. (Table 27 and Figure 9). The map length of eight LGs ranged from 89.93cM (CaLG08) to 230.77cM (CaLG04) with an average of 1530.01 cM per LG. The inter-marker distance ranged from 0.15 cM (CaLG04) to 0.86 cM (CaLG03) with an average marker density of 0.24 cM per LG.

4.9 QTL analysis for DRR resistance and yield attributing traits in RILs of chickpea.

4.9.1 QTL analysis for DRR resistance

In the present investigation, a consistent minor QTL was found on CaLG04 with flanking marker Ca4_4561482 and Ca4_4509561 explaining 5.87% of phenotypic variation (LOD 3.39) with 0.35 additive effect during first screening. QTL identified was same during second screening, explaining 7.41% phenotypic variation (LOD: 4.25) with 0.39 additive effect. Same QTL was observed during pooled analysis also, explaining 6.11% of phenotypic variation (LOD: 4.03) with 0.37 additive effect. On the same genomic region (CaLG04), another minor QTL was identified explaining 6.76% of phenotypic variation (LOD: 3.91) having 0.39 additive effect (Table 28 and Figures 10, 11 and 12).

Table 26. List of SNPs used in the construction of linkage map

CaLG	Mapped markers	Map length (cM)	Inter-marker distance
CaLG01	3,118	195.17	0.06
CaLG02	637	102.33	0.16
CaLG03	145	103.03	0.71
CaLG04	4,896	230.77	0.05
CaLG05	288	118.5	0.41
CaLG06	1,852	188.03	0.10
CaLG07	1,808	196.35	0.11
CaLG08	366	89.93	0.25
Total	13,110	1,224.11	0.09

Table 27. List of SNPs used in the construction of linkage map

CaLG	Mapped marker	Map length (cM)	Inter-marker distance
CaLG01	1,089	195.17	0.18
CaLG02	349	102.33	0.29
CaLG03	120	103.03	0.86
CaLG04	1,544	230.77	0.15
CaLG05	216	118.5	0.55
CaLG06	733	188.03	0.26
CaLG07	816	196.35	0.24
CaLG08	185	89.93	0.49
Total	5,052	1224.11	0.24

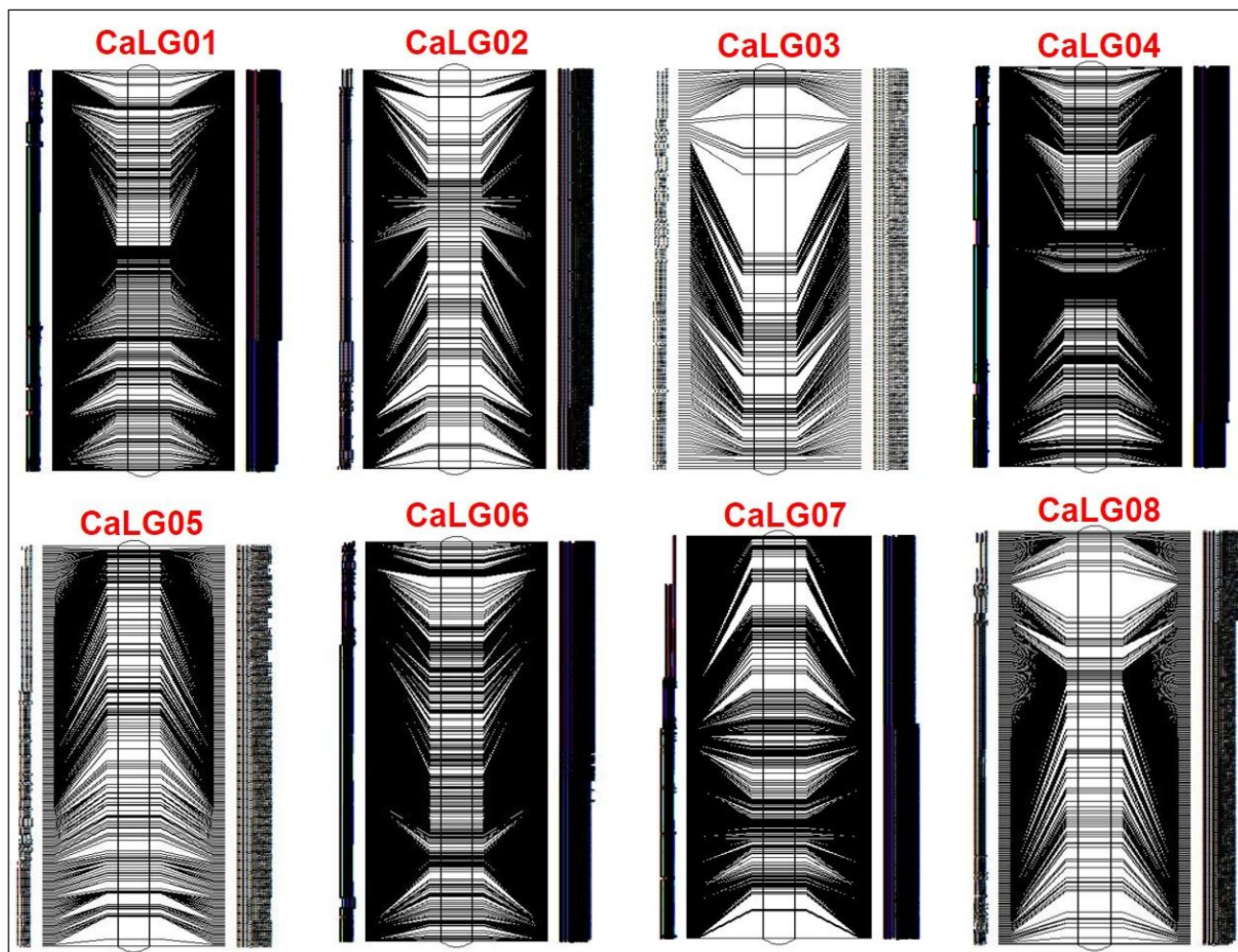


Fig. 8. High density genetic map of chickpea developed from the cross BG 212 $\times$ ICCV 08305 containing 13,110 SNP markers distributed across eight linkage groups with a total map length of 1224.11 cM having average inter-marker distance of 0.09 cM

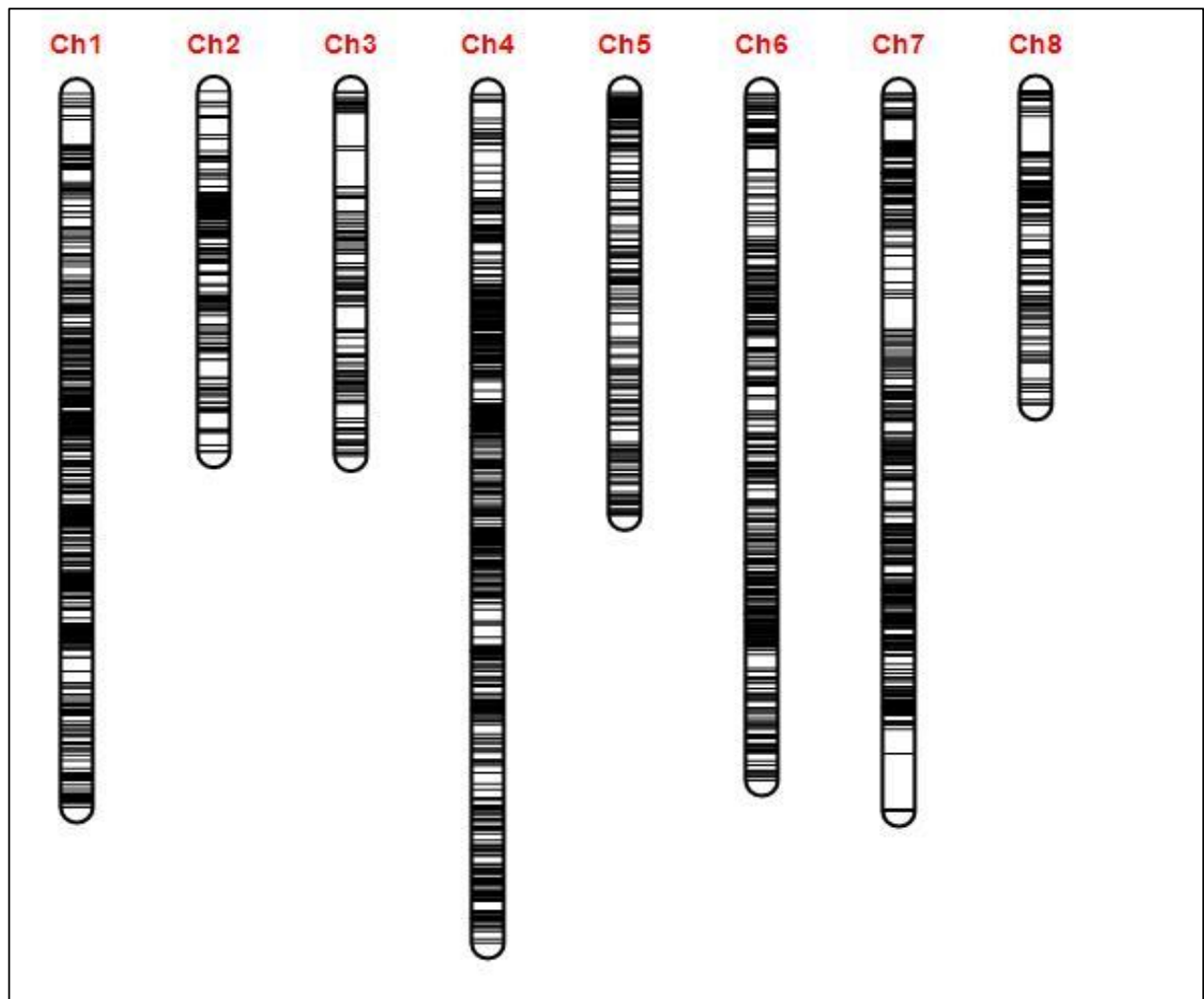


Fig. 9. High density genetic map of chickpea developed from the cross BG 212 × ICCV 08305 containing 5052 SNP markers distributed across eight linkage groups with a total map length of 1224.11cM

Table 28. QTL for dry root rot disease in RILs of cross BG212× ICCV 08305 using paper towel method

Trait Name	Chromosome	Position	Left Marker	Right Marker	LOD	PVE (%)	Add	Left CI	Right CI
First Screening									
DRR	4	205	Ca4_4561482	Ca4_4509561	3.3933	5.8712	0.3538	204.5	205.5
Second Screening									
DRR	4	137	Ca4_15856966	Ca4_15544997	3.917	6.7614	-0.3758	136.5	138.5
DRR	4	205	Ca4_4561482	Ca4_4509561	4.2596	7.4154	0.3917	204.5	205.5
Pooled Analysis									
DRR	4	205	Ca4_4561482	Ca4_4509561	4.03	6.1127	0.3759	204.5	205.5

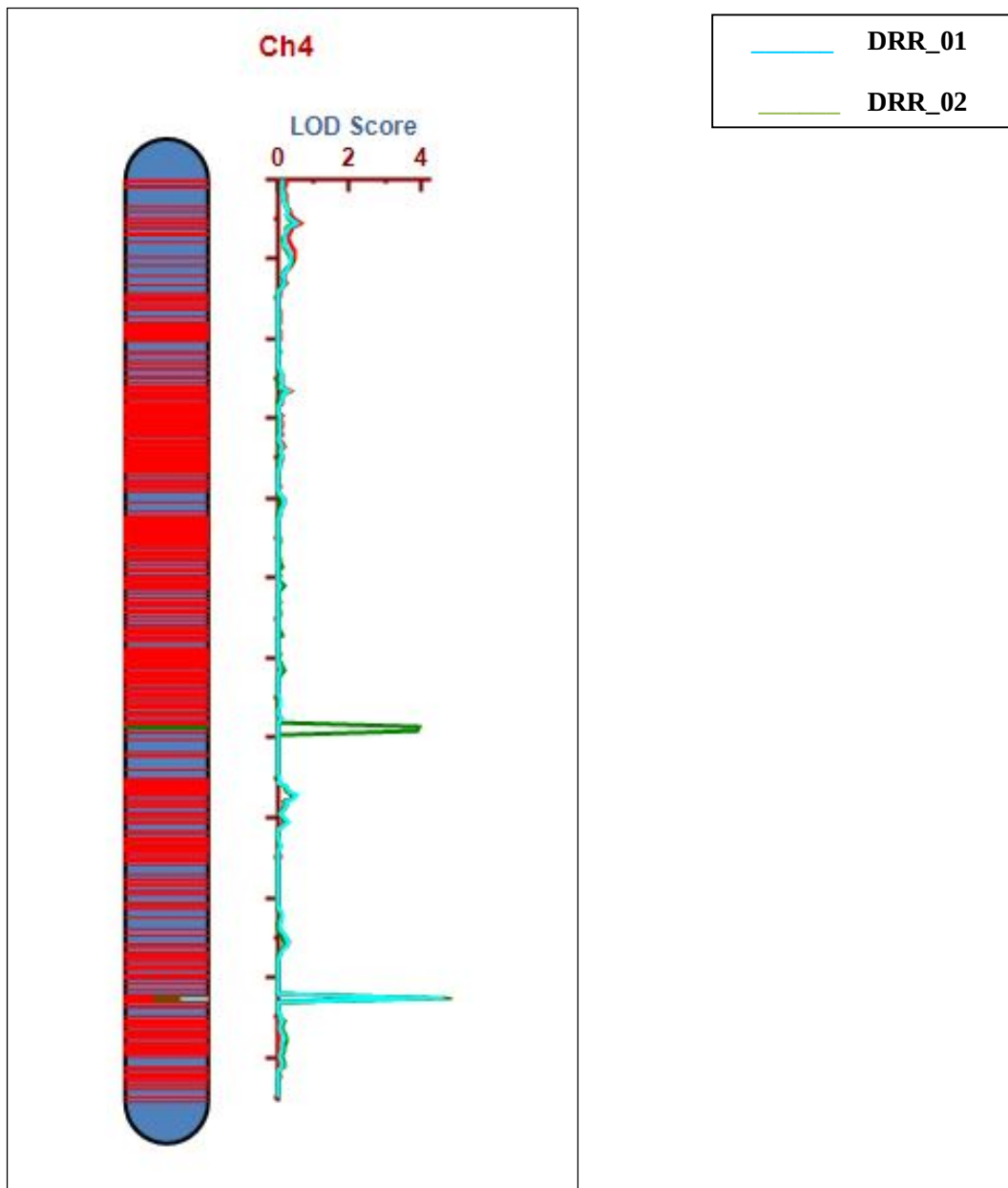


Fig. 10. QTL identified for Dry root rot resistance on CaLG04

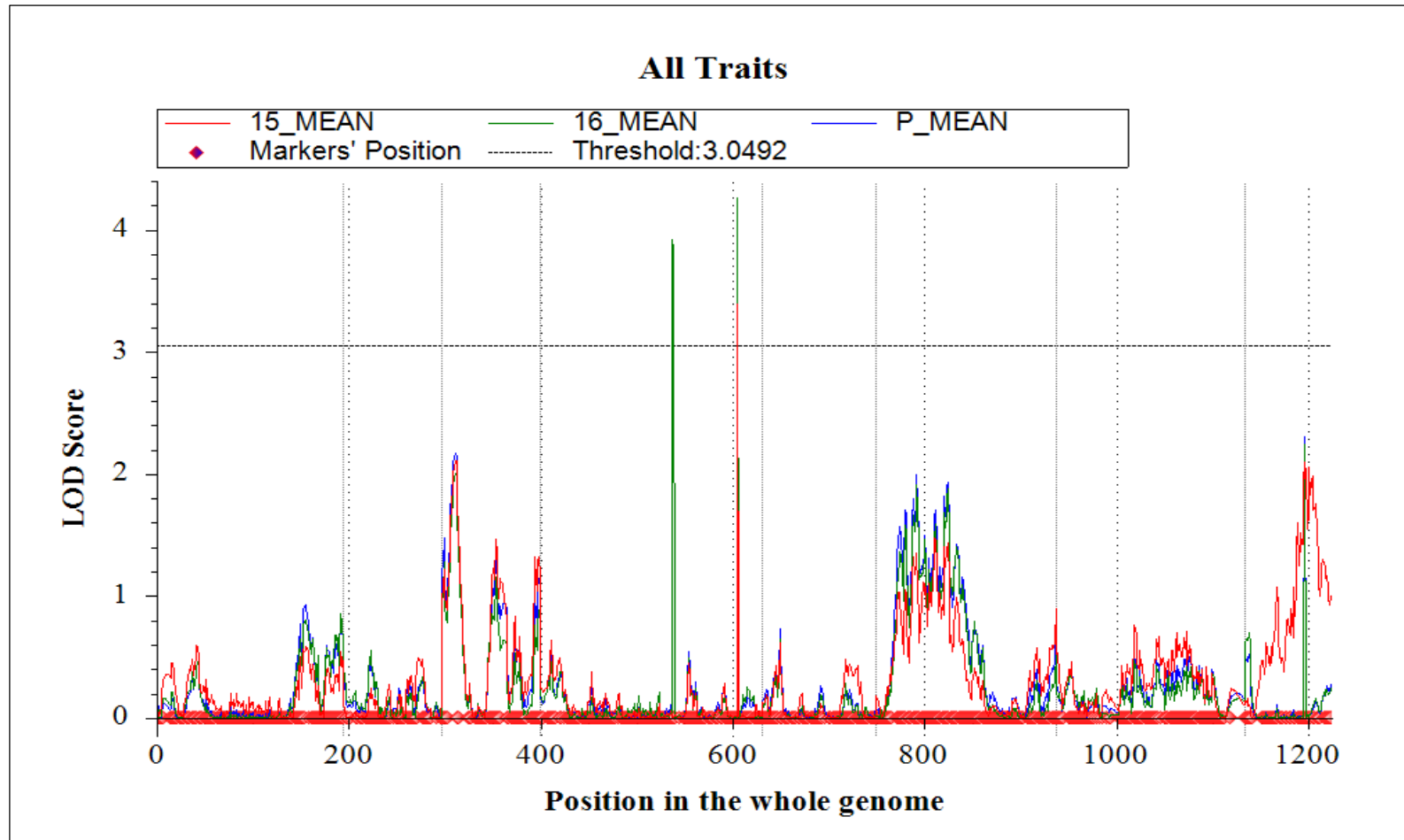


Fig. 11. Likelihood of odds ratio (LOD) curves obtained by composite interval mapping for quantitative trait loci (QTL) mapped for DRR resistant genes in RIL population (BG 212× ICCV 08305)

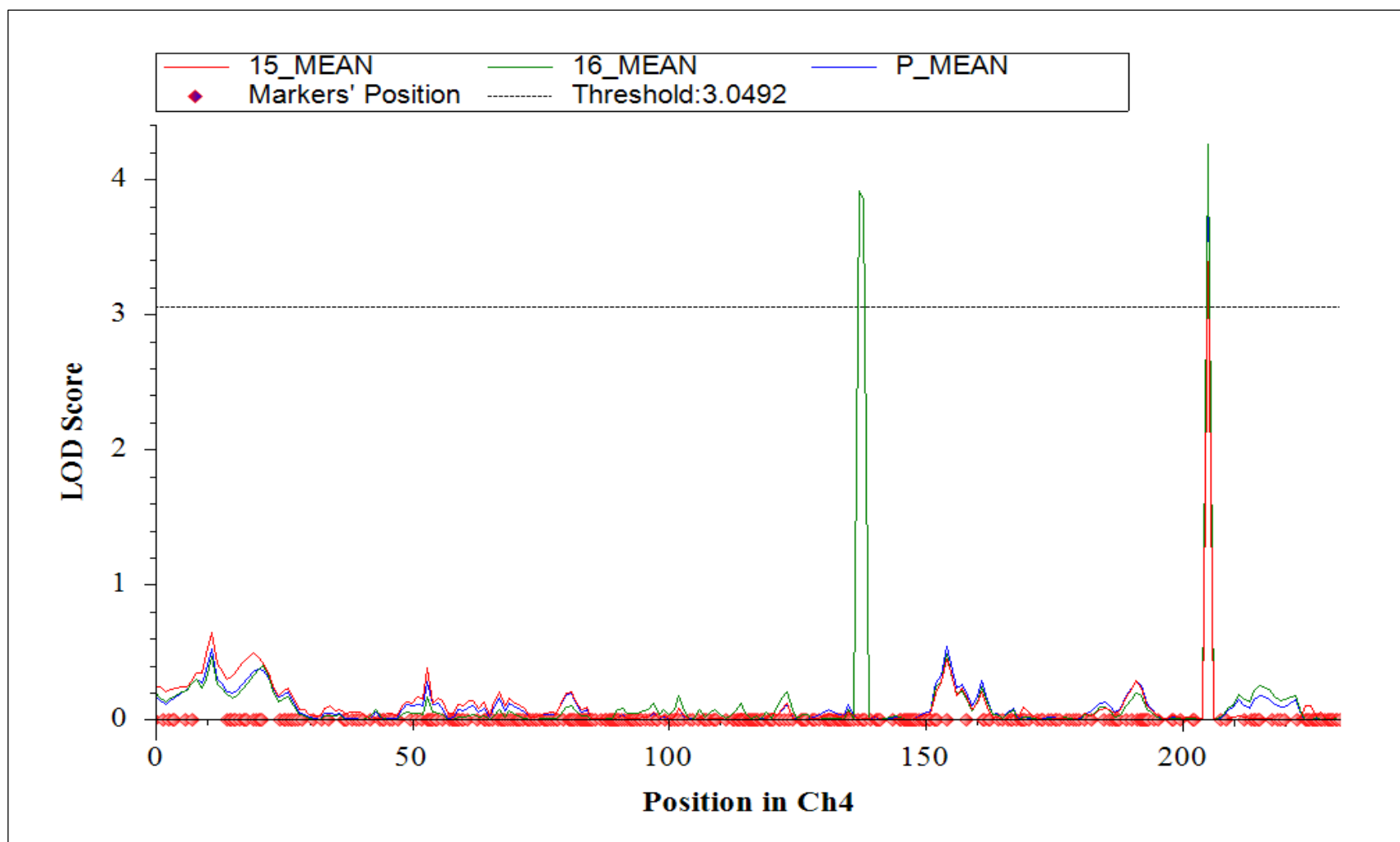


Fig. 12. Likelihood of odds ratio (Lod) curves obtained by composite interval mapping for quantitative trait loci (QTL) mapped for DRR resistant genes on chromosome number 4 in RIL population (BG 212× ICCV 08305)

4.9.2 QTLs for yield and yield contributing traits in RILs of chickpea.

4.9.2.1 QTLs for yield and yield contributing traits in RILs of chickpea during *rabi* (2015-16).

QTLs for yield and yield contributing traits of chickpea are presented below (Table 29 and Figure 13).

4.9.2.1.1 Days to 50 per cent flowering¹

For days to flowering, three QTLs were identified and mapped on the linkage group CaLG05 with LOD value ranging from 3.23 to 29.72 explaining phenotypic variation ranging from 3.94% to 52.3%. All the three QTLs together accounted for 60.81% of phenotypic variation for days to 50 per cent flowering. The highest additive effect for this trait was 6.28.

4.9.2.1.2 Days to maturity

QTL analysis resulted in identification of two QTLs for this trait on linkage group CaLG05 and CaLG08. The LOD score value ranged from 4.94 to 26.18 explaining 6.11% to 45.6 % of phenotypic variation. The QTLs together explained 51.71 % phenotypic variation for days to maturity. The additive effect contributed for this trait was 7.58.

4.9.2.1.3 Plant height (cm)

For plant height, four QTLs were identified and mapped on chickpea genome. Two QTLs on CaLG05, one on CaLG03 and another on CaLG04. The LOD score ranged from 4.75 to 22.61 and phenotypic variation ranged from 4.06% to 24.55% for this trait. The highest additive effect for the plant height was 2.73.

4.9.2.1.4 Plant biomass per plot (g)

Single major QTL was identified on the CaLG05 with flanking marker Ca5_35884449 and Ca5_35848057 having LOD score 15.84 explaining phenotypic variation 35.66%, which was having high additive effect 50.62.

Table 29. QTL for yield and yield related traits in RILs during *rabi* chickpea (2015-16)

Trait	Chromosome	Position	Left Marker	Right Marker	LOD	PVE (%)	Add	Left CI	Right CI
Days to 50 per cent flowering	5	7	Ca5_23071253	Ca5_35848057	29.72	52.30	6.28	82.50	83.50
	5	118	Ca5_29986923	Ca5_30124197	3.54	4.57	1.84	117.50	118.00
	5	83	Ca5_35884449	Ca5_23069087	3.23	3.94	1.71	6.50	7.50
Days to maturity	5	80	Ca5_37055203	Ca5_36751936	26.18	45.60	7.58	79.50	80.50
	8	19	Ca8_10705587	Ca8_12046409	4.94	6.11	2.77	18.50	19.50
Plant height (cm)	5	83	Ca5_35884449	Ca5_35848057	22.61	24.55	2.73	82.50	83.50
	5	77	Ca5_37887670	Ca5_37520917	8.46	7.67	-1.52	75.50	77.50
	3	2	Ca3_21515505	Ca3_21545929	6.78	5.98	1.35	0.00	2.50
	4	184	Ca4_7543422	Ca4_7033820	4.75	4.06	-1.11	182.50	187.50
Plant biomass per plot (g)	5	83	Ca5_35884449	Ca5_35848057	15.84	35.66	50.62	82.50	83.50
Grain yield per plot (g)	5	80	Ca5_37055203	Ca5_36751936	9.73	16.95	16.55	79.50	80.50
	5	43	Ca5_40925660	Ca5_41372147	6.13	9.75	12.51	41.50	43.50
	1	164	Ca1_3856206	Ca1_3349198	4.59	7.18	-10.73	162.50	164.50
	4	80	Ca4_36720956	Ca4_36761322	4.17	6.64	10.41	77.50	82.50
	5	115	Ca5_31025511	Ca5_30921590	3.79	5.83	9.66	114.50	115.50
Hundred seed weight (g)	7	33	Ca7_6743102	Ca7_6756070	14.35	11.57	-1.89	32.50	33.50
	8	38	Ca8_7508869	Ca8_7698349	3.23	2.30	-0.84	37.50	40.50

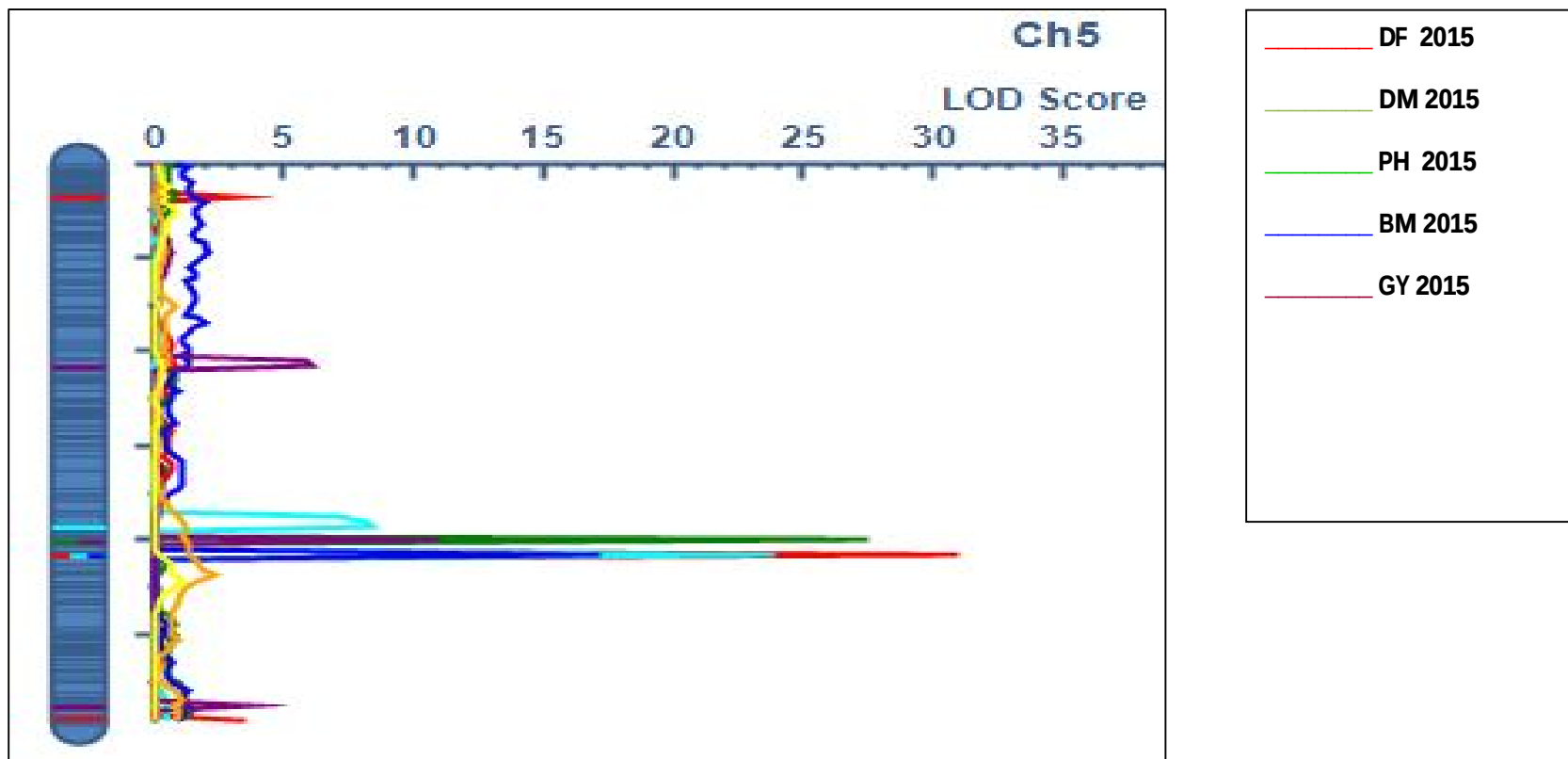


Fig. 13. QTL Mapping for yield and other agronomic characters of chickpea during *rabi* 2015-16

4.9.2.1.5 Grain yield per plot (g)

The QTL mapping resulted in the identification of five QTLs. Three QTLs on CaLG05, one on CaLG01 and another QTL on CaLG04. The LOD value ranged from 3.79 to 9.73 with phenotypic variation ranging from 5.83 to 16.95%. All the five QTLs together accounted for 46.35% of phenotypic variation. The highest additive effect was 16.55.

4.9.2.1.6 Hundred seed weight (g)

Two QTLs were found on CaLG07 and CaLG08 with LOD score ranging from 3.23 to 14.35 explaining phenotypic variation ranging from 2.3 to 11.57% with additive effects of -1.89 and -0.84, respectively.

4.9.2.2 QTLs for yield and yield attributing traits in RILs of chickpea during *rabi* (2016-17).

QTLs for yield and yield attributing traits in RILs of chickpea are presented below (Table 30; Figures 14 and 15).

4.9.2.2.1 Days to flowering

For days to flowering, two QTLs were found on CaLG05 with LOD score ranging from 3.76 to 35.18 explaining phenotypic variation ranging from 4.18 to 59.96%. The total phenotypic variation for this trait was 64.14% with a high additive effect of 6.88.

4.9.2.2.2 Days to flowering

Single major QTL was found on CaLG05 for days to maturity with flanking marker Ca5_35884449 and Ca5_35848057. The LOD score was 18.31 with phenotypic variation of 37.46% and an additive effect of 1.57.

4.9.2.2.3 Plant height (cm)

Single major QTL was found on CaLG05 with flanking marker Ca5_36747984 and Ca5_36430169 having LOD score 13.66 explaining phenotypic variation about 27.55% with high additive effect of 41.

Table 30. QTLs for yield and yield related traits of RILs during *rabi* chickpea (2016-17)

Trait	Chromosome	Position	Left Marker	Right Marker	LOD	PVE (%)	Add	LeftCI	RightCI
Days to 50 per cent flowering	5	83	Ca5_35884449	Ca5_35848057	35.18	59.96	6.88	82.50	83.50
	5	55	Ca5_44057830	Ca5_44299261	3.76	4.18	1.81	53.50	56.50
Days to maturity	5	83	Ca5_35884449	Ca5_35848057	18.31	37.46	1.57	82.50	83.50
Plant height (cm)	3	2	Ca3_21515505	Ca3_21545929	14.53	22.28	2.39	0.00	2.50
	4	156	Ca4_13024647	Ca4_12969201	10.69	16.28	-2.05	155.50	156.50
	5	82	Ca5_36747984	Ca5_36430169	11.71	17.10	2.10	81.50	82.50
	7	36	Ca7_7241923	Ca7_7288245	3.39	4.48	-1.06	35.50	36.50
Plant biomass per plot (g)	5	82	Ca5_36747984	Ca5_36430169	13.66	27.55	41.50	80.50	82.50
Grain yield per plot (g)	5	82	Ca5_36747984	Ca5_36430169	5.52	13.03	16.80	80.50	82.50
Hundred seed weight (g)	4	153	Ca4_13331455	Ca4_13184656	68.50	48.86	-3.52	152.50	153.50
	1	190	Ca1_822344	Ca1_724001	35.35	15.24	-1.94	189.50	190.50
	8	32	Ca8_8887172	Ca8_8744480	19.68	7.10	-1.33	31.50	32.50
	3	5	Ca3_22180448	Ca3_22168145	5.86	1.69	0.65	4.50	5.50
	7	56	Ca7_9763309	Ca7_9972588	4.53	1.28	-0.57	54.50	60.50
	6	161	Ca6_5722981	Ca6_5543821	3.98	1.12	-0.53	160.50	161.50
	4	113	Ca4_22682019	Ca4_29327001	3.77	1.05	-0.51	111.50	113.50
Harvest Index (%)	5	83	Ca5_35884449	Ca5_35848057	6.44	9.50	-2.28	82.50	191.50
	1	190	Ca1_822344	Ca1_724001	4.06	5.77	-1.76	189.50	52.50
	5	52	Ca5_43728352	Ca5_43744254	3.64	5.22	1.68	51.50	83.50

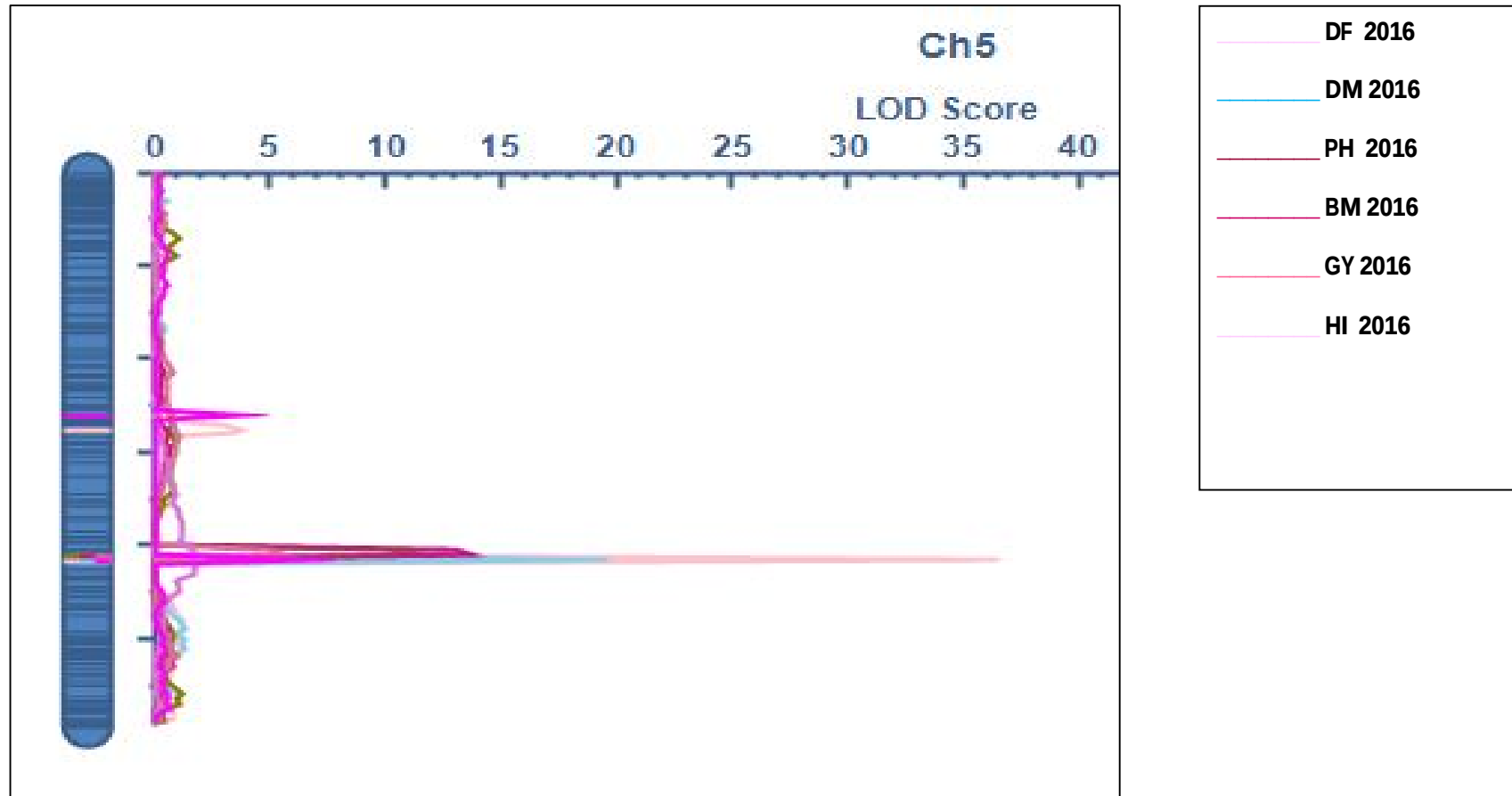


Fig. 14. QTL Mapping for yield and other agronomic characters of chickpea during rabi 2016-17

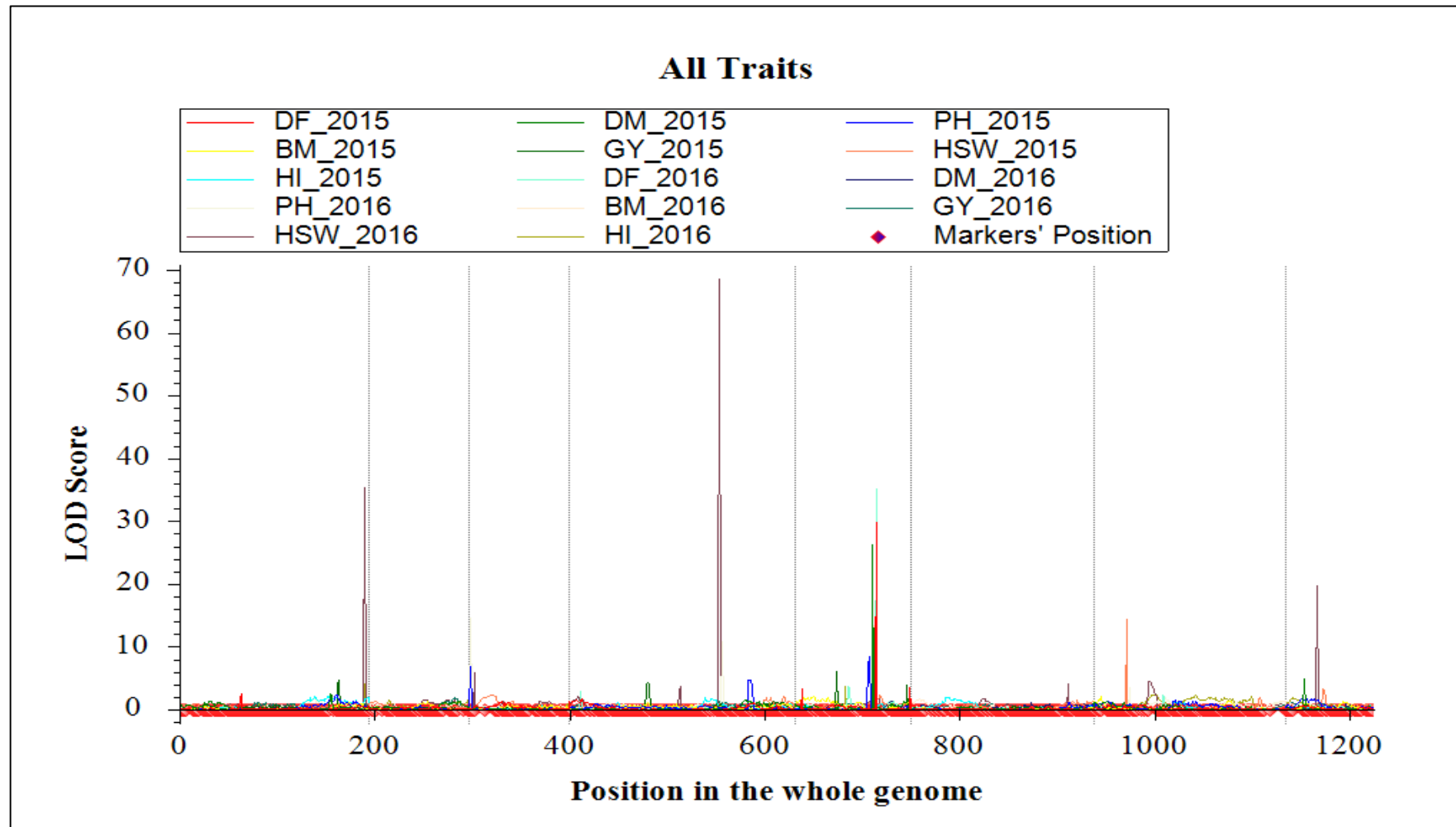


Fig. 15. Likelihood of odds ratio (LOD) curves obtained by composite interval mapping for quantitative trait loci (QTL) mapped for yield and other traits in RIL population (BG 212× ICCV 08305

4.9.2.2.4 Grain yield per plot (g)

Single major QTL was found on CaLG05 with flanking marker Ca5_36747984 and Ca5_36430169 having LOD score 5.52 with phenotypic variation of 13.03% and additive effect of 16.8.

4.9.2.2.5 Hundred seed weight (g)

Total seven QTLs were found on CaLG04, CaLG01, CaLG08, CaLG03, CaLG07 and CaLG06 having LOD score value ranging from 6.44 to 68.5. The phenotypic variation ranging from 9.55% to 48.86% with a total phenotypic variation of 76.34%.

4.9.2.2.6 Harvest index (%)

For harvest index, three QTLs were detected where two on chromosome CaLG05 and one on CaLG01 with LOD score value ranging from 3.64 to 6.44 explaining phenotypic value 5.22 to 9.5%. The total phenotypic value accounted was about 20.49%.

4.9.2.3 Genomic region identified on CaLG05

A promising genomic region identified on CaLG05 in this study harbouring major QTLs for the five traits viz., days to flowering, days to maturity, plant height, plant biomass per plot, grain yield per plot and harvest index. During 2015, co-localized QTLs were found for days to flowering which explained phenotypic variation of 3.94% (LOD 3.23), plant height which explained 24.55% (LOD 22.61) and plant biomass per plot having phenotypic variation 35.66% (LOD 15.84) with flanking marker Ca5_35884449 and Ca5_23069087. During 2016, co-localized QTLs were found for days to flowering which explained phenotypic variation (LOD 35.18), days to maturity having 37.46% phenotypic variation (LOD 18.31) and harvest index with phenotypic variation of 9.5% (LOD 6.44) on position 83 with flanking marker Ca5_35884449 and Ca5_23069087.

Another co-localized QTL was found on CaLG05 for plant height which explained 17.1% phenotypic variation (LOD 11.71), plant biomass per plot with phenotypic variation of 27.55% (LOD 13.66) and also for grain yield per plot which explained phenotypic

Table 31. Co-localized QTLs found for yield and its components in RILs of chickpea across the year

Year	Trait	Chromosome	Position	Left Marker	Right Marker	LOD	PVE (%)	Add	Left CI	Right CI
2015	Days to 50 per cent flowering	5	83	Ca5_35884449	Ca5_23069087	3.23	3.94	1.71	6.5	7.5
	Plant height(cm)	5	83	Ca5_35884449	Ca5_35848057	22.61	24.55	2.73	82.5	83.5
	Plant biomass per plot (g)	5	83	Ca5_35884449	Ca5_35848057	15.84	35.66	50.62	82.5	83.5
2016	Days to flowering	5	83	Ca5_35884449	Ca5_35848057	35.18	59.96	6.88	82.5	83.5
	Days to maturity	5	83	Ca5_35884449	Ca5_35848057	18.31	37.46	1.57	82.5	83.5
	Harvest Index (%)	5	83	Ca5_35884449	Ca5_35848057	6.44	9.5	-2.28	82.5	191.5
2016	Plant height(cm)	5	82	Ca5_36747984	Ca5_36430169	11.71	17.1	2.1	81.5	82.5
	Plant biomass per plot (g)	5	82	Ca5_36747984	Ca5_36430169	13.66	27.55	41.5	80.5	82.5
	Grain yield (g)	5	82	Ca5_36747984	Ca5_36430169	5.52	13.03	16.8	80.5	82.5

Table 32. Consistent QTLs in RILs of chickpea across the years

Trait	Year	Chromosome	Position	Left Marker	Right Marker	LOD	PVE (%)	Add	Left CI	Right CI
Days to 50 per cent flowering	2015	5	83	Ca5_35884449	Ca5_23069087	3.23	3.94	1.71	6.5	7.5
	2016	5	83	Ca5_35884449	Ca5_35848057	35.18	59.96	6.88	82.5	83.5
Plant height(cm)	2015	3	2	Ca3_21515505	Ca3_21545929	6.78	5.98	1.35	0	2.5
	2016	3	2	Ca3_21515505	Ca3_21545929	14.53	22.28	2.39	0	2.5

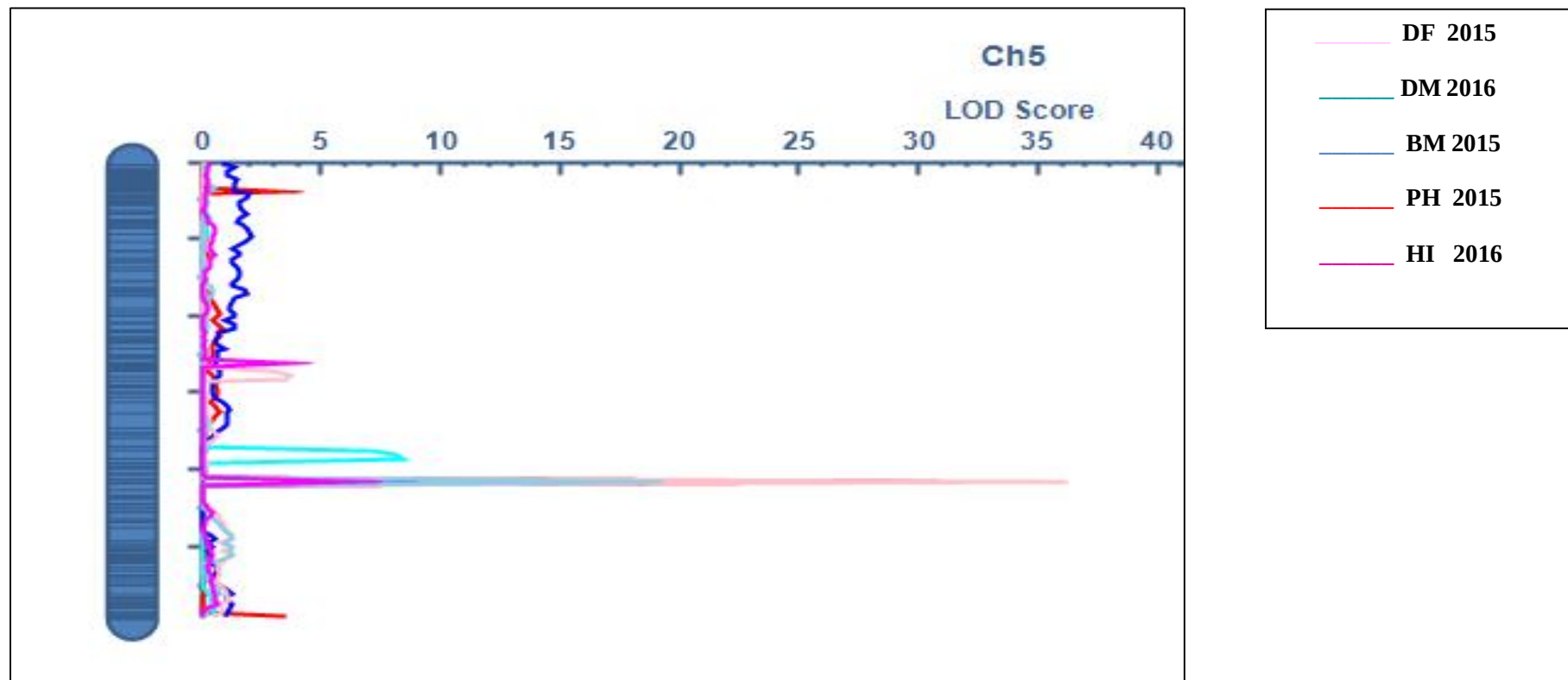


Fig. 16. Co-localized QTLs found for yield and its components in RILs of chickpea across the year

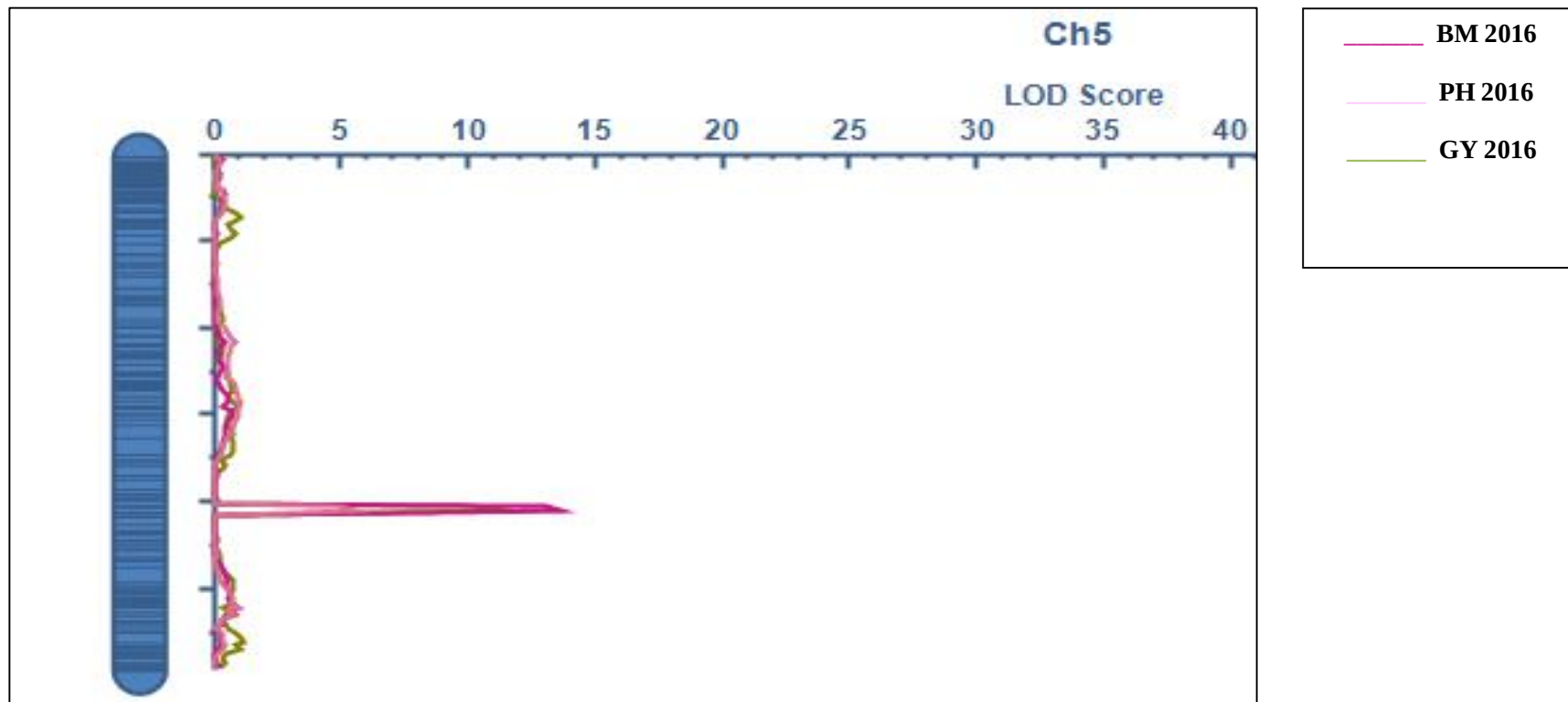


Fig. 17. Co-localized QTLs found for yield and its components in RILs of chickpea during *rabi* 20

variation of 13.03% (LOD 5.52) with flanking marker Ca5_36747984 and Ca5_36430169 (Table 31; Figure 16 and 17).

4.9.2.4 Consistent QTLs identified on linkage group.

A consistent QTL was found for days to flowering in both seasons with flanking marker Ca5_35884449 and Ca5_23069087 explaining phenotypic variation of 3.94% with LOD score 3.23 during *rabi* 2015 and again same QTL was found during 2016 explaining phenotypic variation of 59.96 % (LOD 35.18) (Table 32 and Figure 18).

Plant height had consistent QTL in both seasons with flanking marker Ca3_21515505 and Ca3_21545929 explaining about 5.98% of phenotypic variation (LOD 6.78) during 2015 and 22.28% of phenotypic variation (LOD 14.53) during 2016 respectively (Figure 19).

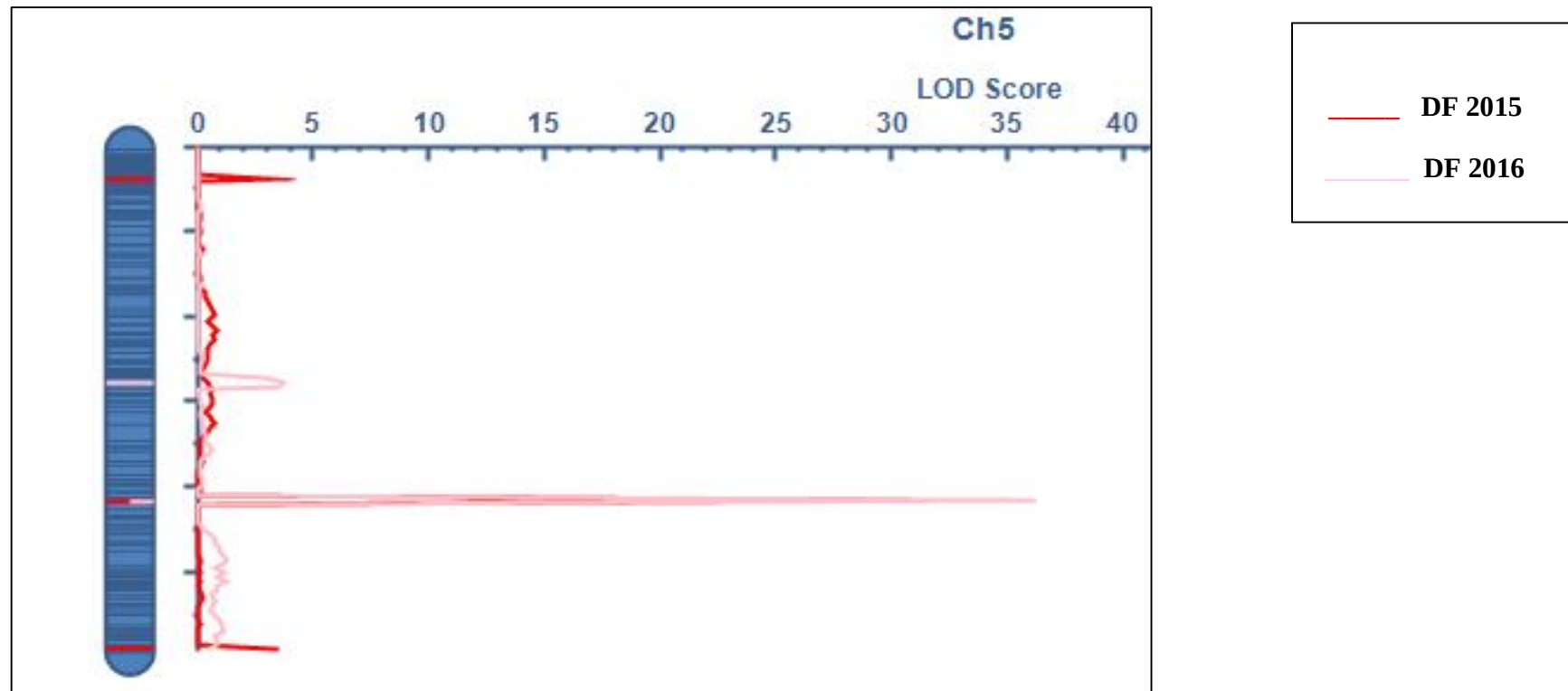


Fig. 18. Consistent QTLs found for Days to 50 per cent flowering in RILs of chickpea across the years

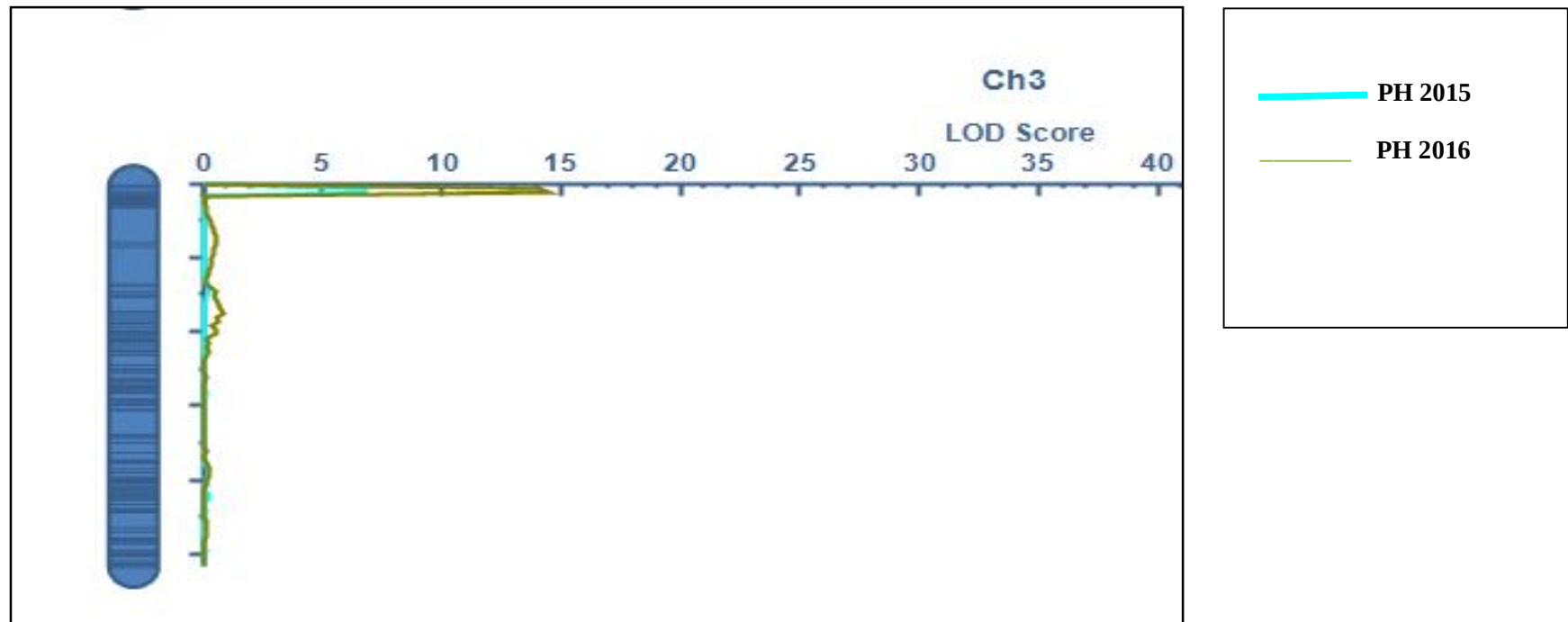


Fig. 19. Consistent QTLs found for plant height in RILs of chickpea across the year

Discussion

V. DISCUSSION

Dry root rot caused by *Rhizoctonia bataticola* (Taub.) Butler is an emerging disease of chickpea that poses severe constraint to chickpea production. The present study with 182 RILs along with parents was aimed at unraveling the genetics of resistance, identification of SNP markers linked to DRR resistance and detection of relationship between yield attributing traits and DRR disease incidence. The results of the present investigation are discussed along with relevant literature under the following headings.

5.1 Genetic variability parameters for seed yield and its attributing traits in RILs of chickpea

5.1.1 Analysis of variance

Genetic variability in yield attributing traits is an important pre-requisite for an efficient crop improvement programme. The selection technique for any breeding programme depends upon the nature and magnitude of variability. A wide range of variation was generated in RIL population as compared to their parents. Hence, assessment of genetic variability for yield and its contributing traits in RILs was an important target of the present investigation. Analysis of variance revealed that, 182 RILs along with parents exhibited highly significant differences for all the characters like days to 50 percent flowering, days to maturity, plant height, plant biomass per plot, grain yield per plot, 100-seed weight and harvest index in both seasons and also in pooled analysis. Analysis of variance showed highly significant differences for disease incidence during both the screenings and also for pooled analysis at both levels. This indicates that the RILs were genetically diverse for most of the traits under study and these are suitable for genetic studies. Similar findings on variation in RIL population of chickpea were reported by Qureshi *et al.* (2004), Sidramappa *et al.* (2008), Gul *et al.* (2013) and Joshi (2014).

5.1.2 Genetic variability parameters for yield and its attributing traits in RILs of *rabi* chickpea.

Analysis of variance may not reveal the absolute variability and this could be accessed through standardizing the phenotypic and genotypic variances by obtaining coefficients of variability. Thus, the components of variation such as genotypic coefficient of variation (GCV) and phenotypic coefficient of variation (PCV) were computed. Further it is essential for selection to separate out the environmental influence from the total variability. This indicates the accuracy with which a genotype can be identified by its phenotypic performance. The estimates of heritability alone fail to indicate the response to selection. Therefore, the heritability estimates appeared to be more meaningful when accompanied by estimates of genetic advance. The genetic advance as per cent means (GAM) was also estimated. One of the ways in which variability is assessed through a simple approach is by examining range values where, RILs showed wide range of values for most of the traits. Wide range of variation provides an ample scope for selection of superior and desirable RILs by the breeder for further improvement of these traits.

High values of PCV and GCV were obtained for plant biomass per plot, grain yield per plot and 100-seed weight in both seasons indicating the presence of high degree of variability and low environmental influence on the expression of these traits. Hence, these characters can be relied upon simple selection. Magnitude of GCV and PCV was found moderate for days to 50 per cent flowering, days to maturity and harvest index which could be due to conscious selection towards seed yield and for DRR resistance, thereby, reducing the extent of variability and enhancement of homogeneity. Whereas, plant height exhibited moderate GCV and PCV in *rabi* 2015 and low during *rabi* 2016 that could be due to high environmental influence and selection for this trait will not be very effective.

The effectiveness of selection for any yield component depends not only on the amount of variability, but also how much of it is readily amenable for improvement. The coefficient of variation indicates only the extent of variation and does not discriminate the variability into heritable and non-heritable. Heritability value alone cannot provide information on the amount of progress that would result from selection of best individuals. Johnson *et al.* (1955) reported that heritability estimates along with genetic gain would be more useful than the former alone in predicting the effectiveness of selecting the best

individuals. Therefore, it is essential to consider the predicted genetic advance along with heritability estimate as a tool in the selection programme for better efficiency in the selection.

In the present study, high PCV, GCV and heritability coupled with high genetic advance as per cent mean was observed for days to flowering, plant height, plant biomass per plot, grain yield per plot and 100-seed weight in both seasons except harvest index. This indicates that these traits are controlled by additive gene action and environmental influence is low for these traits. Here, directional phenotypic selection could be effective for genetic improvement. These traits could be further improved by applying pedigree selection. These results are in agreement with the reports of Mishra *et al.* (1972), Agarwal *et al.* (1985), Sharma *et al.* (1990), Mishra *et al.* (1991), Kumar *et al.* (1991), Jahagirdar *et al.* (1994), Rao *et al.* (1994), Mathur and Mathur (1996), Patil and Phandis (1996), Wanjari *et al.* (1996), Sudheer and Ram (1997), Kumar *et al.* (1999), Jeena and Arora (2000), Arshad *et al.* (2002), Sable *et al.* (2002), Burli *et al.* (2004), Jeena *et al.* (2005), Patel and Babbar (2005), Saleem *et al.* (2005), Joshi and Bapu (2008), Thakur *et al.* (2008), Deb and Khaleque (2009), Vaghela *et al.* (2009), Borate *et al.* (2010) and Devaraja *et al.* (2011) in chickpea for these traits.

High heritability coupled with high genetic advance was observed for harvest index during *rabi* 2015, which was moderate during *rabi* 2016. This could be due to both additive and non-additive gene action. The high heritability exhibited might be due to favorable influence of environment rather than the genotype and selection for such traits may not be rewarding. The variation in gene action could be due to use of plant material belonging to different generations *viz.*, F₇ during *rabi* 2015 and F₁₀ during *rabi* 2016.

5.2 Association studies for seed yield and its attributing traits in RILs of chickpea.

In any crop improvement programme, it becomes necessary to have simultaneous selection of more than one character, especially, a complex character such as yield which is influenced by many other contributing traits. Correlation analysis provides the information on nature and magnitude of association of different component characters with seed yield. Correlation of characters serves as a measure and forms the basis of selection as it gives strength and direction of relationship between the characters studied. Selection for yield will

be effective only when it is considered along with its yield components rather than relying on yield alone. When a breeder applies selection pressure for a trait, it also brings about a simultaneous change in all characters associated with it. So, it is a matter of great importance to the plant breeders to find out as to which of the characters are correlated with yield and also how they are associated among themselves.

In the present study, genotypic correlation coefficient was slightly higher than their corresponding phenotypic correlation coefficient for the majority of the traits indicating an inherent association between the traits and less influence of environment on the association of these traits. These associations could be the result of pleiotropic effect or linkage between the genes governing respective traits.

Grain yield had highly significant positive association with days to 50 per cent flowering, days to maturity, plant height, plant biomass per plot and harvest index except 100-seed weight in both seasons, which indicated that these traits are important for effective improvement in yield through selection based on these traits. Hundred seed weight had positive association with grain yield during *rabi* 2016, while the association was negative during *rabi* 2015. It could be due to environmental influence or difference in sowing date by one month or difference in filial generation of the plant material. Inter-relationship among the agronomic traits was strong. So, simultaneous selection can be practiced based on these traits. Earlier, studies also reported the association of seed yield with 100-seed weight and days to 50 per cent flowering (Maloo and Sharma, 1987 and Saleem *et al.*, 2002), 100-seed weight and harvest index (Jivani and Yadavendra, 1988), hundred seed weight and days to maturity (Wadud and Yaqoob, 1989), hundred seed weight, days to flowering and days to maturity (Jain *et al.*, 1992; Sarvalia and Goyal, 1994), biomass and hundred seed weight (Arora and Kumar, 1994; Muhammad *et al.*, 2004 and Meena *et al.*, 2006), hundred seed weight and harvest index (Khorgade, 1995), hundred seed weight, days to maturity and harvest index (Talebi, 2007), hundred seed weight, harvest index, days to flowering and days to maturity, (Bhavani, 2008, Zena, 2008 and Akhtar, 2011), plant biomass and harvest index (Singh, 2007), harvest index (Ozveren and Anlarsal 2010), hundred seed weight (Kumar *et al.*, 1999; Arshad *et al.*, 2002; Jeena *et al.*, 2005; Saleem *et al.*, 2005; Obaidullah *et al.*, 2006;

Singh and Sandhu, 2008; Bhavani *et al.*, 2008; Borate and Dalvi, 2010; Gohil and Patel, 2010 and Akhtar *et al.*, 2011) in chickpea.

5.3 Path coefficient analysis for seed yield and agronomic traits in RILs of chickpea

The correlation coefficient indicates the relationship existing between pair of characters. But, a dependent character is a product of the interaction of many mutually associated component characters and change in any one component will disturb whole network of cause and effect system. The path coefficient analysis also measures the relative importance of causal factors involved. This is simply standardized regression analysis, where in total correlation value is subdivided into causal scheme. Li (1956) emphasized the importance of path diagram which facilitates the understanding of the nature of cause and effect system. The path analysis suggested by Dewey and Lee (1959) helps to understand thoroughly these correlations further by throwing more light on the way in which component traits contribute to yield and identifying important component traits.

Path coefficient analysis revealed that, days to 50 percent flowering, plant biomass per plot, 100-seed weight and harvest index had positive direct effect on grain yield, indicating that true relationship between these traits with grain yield and direct selection for these traits will be rewarding for crop improvement. Similar results of positive direct effect were noticed for 100-seed weight (Arshad *et al.*, 2002; Jeena *et al.*, 2005; Saleem *et al.*, 2005; Obaidullah *et al.*, 2006; Amandeep *et al.*, 2008; Bhavani *et al.*, 2008; Borate and Dalvi, 2010; Gohil and Patel, 2010; Akhtar *et al.*, 2011; Talebi and Karami, 2011), 100-seed weight and harvest index (Jivani and Yadavendra, 1988; Khorgade, 1995; Talebi, 2007; Bhavani, 2008, Zena, 2008 and Akhtar, 2011), biomass and 100 seed weight (Arora and Kumar, 1994; Muhammad *et al.*, 2004), plant biomass and harvest index (Singh, 2007) and harvest index (Ozveren and Anlarsal, 2010).

Days to maturity exhibited negative direct effect on grain yield, but had significant positive association with grain yield. The high positive association could be mainly because of high indirect effect *via* plant biomass per plot and days to 50 per cent flowering in both seasons. Similarly, 100-seed weight had negative direct effect on grain yield but had significant positive association with grain yield. It could be mainly due to positive indirect

effect *via* plant biomass per plot, days to maturity, plant height and harvest index. So, indirect selection for grain yield per plot through biomass, plant height and harvest index could be effective for selection of RILs for higher grain yield.

5.4 Screening of chickpea RILs for resistance to DRR disease incidence

Efficient screening technique and selection method will help in the development of resistant cultivars. Several screening techniques have been developed for DRR at ICRISAT such as “paper towel screening technique and sick pot technique” (Nene *et al.* 1981 and Pande *et al.* 2012). Majority of the studies were done at seedling level (Pande *et al.* 2006 and Gupta *et al.* 2012). The resistant sources can be utilized in chickpea breeding to develop varieties that are resistant or tolerant to multiple diseases (Gurha *et al.* 2003). But, Search for specific resistance to DRR in chickpea has been unsuccessful so far as none of the lines showed consistent resistant reaction to DRR (Sharma *et al.* 2016). Moderate resistance to DRR has also been identified at ICRISAT in chickpea germplasm and breeding lines (Pundir *et al.*, 1988) that led to the development of moderately resistant varieties (Dua *et al.*, 2001). Few germplasm lines such as ICCV 05530, ICCV 08305, ICCV 05529, ICCV05532, ICCV 07117 and ICCV 07112 have been identified at ICRISAT which exhibited moderate level of resistance to DRR (Sharma, unpublished). Among these, ICCV 08305 was used as male parent for development of RILs which were used in the present study to identify markers linked to the DRR resistance.

In the present investigation, 182 RILs of chickpea along with the parents were screened under laboratory condition using paper towel method against *Rhizoctonia bataticola*. Among the 182 RILs, 13 showed resistant reaction followed by 79, 78 and 12 RILs with moderately resistant, susceptible and highly susceptible reaction, respectively. Several germplasm and breeding lines were evaluated using these two screening techniques by various researchers such as, Nene *et al.* (1981), Reddy *et al.* (1990), Baker and Ahmed (1991), Bekele Hunde Paul and Hailu Tefera (1992), Gupta (1995), Oad *et al.* (1995), Gurha *et al.* (2003) and Ashraf *et al.* (2005). Gupta and Babbar (2006), Al Shareef *et al.* (2009), Gupta *et al.* (2012) also reported resistance to DRR in some genotypes of chickpea. Iftikhar and Ilyas (2000) screened 108 chickpea germplasm cultivars and found only one chickpea

line (ICCV 97112) resistant to DRR. Among 211 mini core accessions, ICC 1710 and ICC 2242 were found resistant to dry root rot (Pande *et al.*, 2006). Jayalakshmi *et al.* (2008) reported that four genotypes (GCP-101, GBM-2, GBM-6 and ICCV-10) were found tolerant. Few resistant sources for DRR have also been reported by Gangwar *et al.* (2002), Prajapati *et al.* (2003), Pande *et al.* (2006), Gupta *et al.* (2012b) and Khan *et al.* (2013). Resistance identified so far in improved chickpea germplasm and breeding lines as well as wild accessions and its relatives under field and greenhouse conditions needs to be reconfirmed through epidemiologically sound, repeatable resistance screening technique using the existing variability in *R. bataticola* isolates. Studies on host plant resistance and inheritance of resistance to DRR suggested that the screening techniques for this disease need confirmation of resistance under controlled environment and field.

5.5 Genetics of resistance to dry root rot disease

Plant breeders need to tailor the plant type in a way that the crop can cope with biotic stresses such as DRR disease and yet provides a sustainable and economic yield. This can be achieved by employing various genetic approaches. Understanding the genetics, behaviour of host and pathogen in the process of disease development and host–pathogen relationship are crucial for reliable breeding programmes aimed for disease resistance. The knowledge about the variation in host and pathogen is a pre-requisite for a successful breeding programme (Porta-Puglia *et al.*, 1996). Development of disease resistant varieties calls for screening and evaluating the available germplasm for resistance to diseases as a first step to understand the genetics of resistance for successful disease resistance breeding programme. Introgression of genetic resistance into cultivars is generally considered as an important and successful component in preventing and controlling diseases.

Grouping the RILs, the frequency distribution of RILs for DRR disease incidence (recorded on 1-9 disease score), depicted a normal distribution indicating that resistance to DRR was quantitative in nature. When RILs were grouped into two category, the segregation pattern of 1 Resistant (score >1 to <5): 1 Susceptible (score >5 to <9) was observed, which gave a very good fit to expected 1:1 ratio the chi- square values were non-significant indicating that a major gene segregating F₉ RIL population for the DRR disease resistance

gene had attained homozygosity. Eventhough, the segregation pattern fits mendelian segregation ratio, it is difficult to conclude on the number of genes governing DRR resistance at this stage. Earlier, Rao and Haware (1987) observed that all F₁ plants of the resistant × susceptible crosses were resistant, while in F₂ generation the segregation was 3 resistant: 1 susceptible indicating monogenic inheritance, with resistance being dominant over susceptibility. Talekar (2014) screened a set of 520 chickpea accessions and study revealed that a high level of resistance in three entries, namely PG 06102, BG 2094 and IC 552137, the resistance was under monogenic control and resistance being dominant over susceptibility..

5.6 Relative comparison of resistant and susceptible RILs in chickpea for yield, yield contributing traits and for DRR disease incidence

Sick plot not available to screen for DRR resistance and to find out the relative comparison between resistant and susceptible RILs for yield and other agronomic traits along with the disease incidence. Therefore, crop was grown in solarized condition and disease screening was done in the controlled environment condition. There was no such difference in the performance of resistant and susceptible RILs for yield traits and other agronomic traits. But, significant difference was found between the RILs for the disease incidence.

5.7 Superior RILs for yield and other traits

During *rabi* 2015, ICCRIL14- 60 and ICCRIL14-160 (35 days) were found as early flowering lines. Similarly, ICCRIL14-133 (82 days) matured early, ICCRIL-78 (292.33g) recorded the highest grain yield per plot, ICCRIL14-119 (600g) had the highest plant biomass per plot, ICCRIL14-13 (35.34g) recorded maximum 100-seed weight and ICCRIL14-52 exhibited the highest harvest index (59.87%).

During *rabi* 2016, ICCRIL14-62 (32 days) flowered early, ICCRIL14-32 matured early, ICCRIL14-56 (635g) recorded the maximum plant biomass per plot, ICCRIL14-69 (381.67g) recorded the maximum grain yield per plot, ICCRIL14-13 (37.73g) recorded the maximum 100-seed weight and ICCRIL14-33 recorded the highest harvest index (63.74%).

5.8 QTL analysis for DRR resistance and other agronomic traits in RILs of chickpea

5.8.1 QTL analysis for DRR resistance

Rapid advancement was seen in development of genome map of chickpea and molecular mapping of genes/QTL controlling agronomically important traits such as drought avoidance root traits (Chandra *et al.*, 2004 and Gaur *et al.*, 2008), fusarium wilt resistance (Mayer *et al.*, 1997; Ratnaparkhe *et al.*, 1998; Tullu *et al.*, 1998; Winter *et al.*, 2000 and Sharma *et al.*, 2004) and ascochyta blight resistance (Santra *et al.*, 2000; Tekeoglu *et al.*, 2002; Collard *et al.*, 2003; Flandez- Galvez *et al.*, 2003; Udupa and Baum 2003; Cho *et al.*, 2004; Lichtenzveig *et al.*, 2005; Cobos *et al.*, 2006; Taran *et al.*, 2007 and Kottapalli *et al.*, 2008) and Botrytis grey mould (Anuradha *et al.*, 2011). But, there was no report available on mapping of QTL conferring resistance to DRR. Thus, this study was conducted to identify marker linked to DRR resistance, to study the genomic region conferring resistance and to map QTL for DRR resistance in chickpea.

Identification of DNA markers linked to DRR disease resistance is important to initiate marker-assisted introgression of genomic region/s into chickpea cultivars. The marker polymorphic survey among the genotypes with contrasting response to particular traits provides preliminary information on their possible linkage with the traits. In the present study, a consistent minor QTL was found on CaLG04 with flanking marker Ca4_4561482 and Ca4_4509561 explaining 5.87% of phenotypic variation (LOD 3.39) with 0.35 additive effect during first screening. Same QTL was identified during second screening explaining 7.41% of phenotypic variation (LOD: 4.25) with -0.37 additive effect. During pooled analysis, the QTL for DRR resistance observed was same explaining 6.11% of phenotypic variation (LOD: 4.03) with 0.37 additive effect. On the same genomic region (CaLG04) another minor QTL was identified explaining 7.41% of phenotypic variation (LOD: 4.25) having 0.39 additive effect with flanking marker Ca4_15856966 and Ca4_15544997. The presence of common flanking markers indicates the same genomic regions. Hence, it can be considered as relatively stable and apparently common QTL. Additive effect indicates that the resistance gene is contributed by male parent ICCV08305. The lower phenotypic contribution explained by the QTLs may be an indication of the possible involvement of additional unidentified loci which have not been detected, may be because of the small

population size (Dholakia *et al.*, 2003). The identified genomic region and the SNP markers linked will be further utilized for mining candidate genes involved in DRR resistance. To our knowledge, this is the first report on identification and mapping of QTL for DRR resistance in chickpea and therefore these QTLs may be considered as novel. Earlier, an attempt was made by Talekar (2014) who used 520 chickpea accessions to identify DNA markers linked to genes conferring DRR resistance and revealed that a screen of 381 SSR primer pairs realized 52 which were informative between L 550 and PG 06102. A bulk segregant analysis approach was applied to the DNAs of L 550 x PG 06102 F₂ progeny, showed potential marker/ trait linkage for four SSR primer pairs. A linkage analysis based on 129 F₂ individuals and these four SSR markers showed that two markers (ICCM 0299 and ICCM 0120b) co-segregated with resistance to DRR. Recently, Desai *et al.* (2017) found two SSR (GSSR 106 and TR 19) and two RAPD (A05 and A18) markers showed distinct polymorphism between DRR resistance and five susceptible genotypes, which can be used to investigate possible linkage with DRR resistance.

5.8.2 QTL for yield and other agronomic traits

For days to flowering, three QTLs were found on CaLG05 accounting for 3.94%-52.3% of phenotypic variation and two QTLs on genomic region CaLG05 explaining phenotypic variation ranging from 4.18%-59.96%. Although, in the present study, QTLs were found on LG5 but their locations on the linkage groups were different from Cho *et al.*, 2002; Anbessa *et al.*, 2006; Lichtenzveig *et al.*, 2006; Cobos *et al.*, 2007, 2009; Radhika *et al.*, 2007; Aryamanesh *et al.*, 2010; Hossain *et al.*, 2010; Gowda *et al.*, 2011; Rehman *et al.*, 2011; Hiremath *et al.*, 2012; Vadez *et al.*, 2012; Jamalabadi *et al.*, 2013; Varshney *et al.*, 2014; Upadhyaya *et al.*, 2015; Daba *et al.*, 2016 and Mallikarjuna *et al.*, 2017 suggesting the presence of different genes controlling of flowering time or it may be due to different population or markers or linkage map analysis.

For days to maturity, two QTLs were found on LG5 explaining 45.6 % phenotypic variation (LOD 26.18) and LG8 explaining 6.11% phenotypic variation (LOD 4.94) during 2015. Single major QTL was found on LG5 explaining 37.46 % phenotypic variation (LOD 18.31) during 2016. Similar study was done by Pushpavalli *et al.* (2015) who identified four

stable QTLs for days to maturity, each on CaLG05 (with flanking markers CaM0463 and ICCM272) and on CaLG08 (CKAM1903 and CKAM0343).

For plant height, four QTLs were found in both the seasons. During 2015, two QTLs were found on LG5, one on LG3 and another on LG4 accounted for a total phenotypic variation of 42.26 %. During 2016, four QTLs were found on LG3, LG4, LG5 and LG7 with total phenotypic variation of 60.14%. Similarly, Aryamanesh *et al.* (2010) reported a QTL mapped on LG3 for plant height. Varshney *et al.* (2014) reported five QTLs for plant height (CaLG03, CaLG04, CaLG05, CaLG06 and CaLG08). However, both populations share at least one QTL for plant height on CaLG08 that contributed 14.73 % of phenotypic variation (ICCRIL03) to 31.32 % phenotypic variation (ICCRIL04). Kujur *et al.* (2016) identified six major genomic regions underlying six robust QTLs (11.5 - 21.3% of phenotypic variation with 35.8% combined phenotypic variation) governing plant height traits in chickpea. Kale *et al.* (2015) mapped QTLs for 100-seed weight and plant height on CaLG04, while a QTL for days to flowering on CaLG08.

During 2015, single major QTL was identified for plant biomass per plant on LG5 with LOD 15.84 having 35.66% of phenotypic variation. During 2016, a single major QTL was identified on LG5 (LOD 13.66) with 27.55% of phenotypic variation. For grain yield, five QTLs were identified three on LG5, one on LG1 and another on LG4 during 2015. During 2016, a single major QTL was identified on LG5 with 27.55 % phenotypic variation. Pushpavalli *et al.* (2015) identified four yield QTLs: in 2010, in the salinity treatment one major QTL was identified on CaLG07 explaining 17% of phenotypic variation; during 2011, one major QTL in the salinity treatment that explained 12% of phenotypic variation was also identified on CaLG05, whereas, one major QTL (16% of phenotypic variation) and one minor QTL (8% of phenotypic variation) were identified on each of CaLG05 and CaLG07 in the control treatment.

For hundred seed weight, two QTLs were identified during 2015 on LG7 explaining 11.57% of phenotypic variation (LOD 14.35) and another on LG8 with 2.3 % of phenotypic variation (LOD 3.23). During 2016, total seven QTLs were found two on LG4 and one each on LG1, LG8, LG3, LG7 and LG6 explaining 76.34% of total phenotypic variation. It is

interesting to note that several studies reported the presence of QTL for 100-seed weight on CaLG04 in chickpea (Abbo *et al.*, 2005; Cho *et al.*, 2002; Cobos *et al.*, 2007; Radhika *et al.*, 2007; Hossain *et al.*, 2010; Varshney *et al.*, 2014; Jaganathan *et al.*, 2015; Verma *et al.*, 2015 and Pushpavalli *et al.*, 2015).

During 2015, two QTLs were identified one major QTL on LG7 and another on LG8 for 100-seed weight. During 2016, three minor QTLs were found -two on LG5 and another one on LG1. Pushpavalli *et al.* (2015) identified nine QTLs for harvest index across years and treatments. Rehman (2009) identified four QTLs for harvest index on LG1, LG3, LG4 and LG8. Hence, concentrating on these linkage groups in molecular breeding programs could speed up the improvement in these traits.

5.8.3 Key genomic region associated yield traits

Two potential genomic regions controlling majority of yield related traits were identified on CaLG05 and CaLG03. The genomic region harbouring QTL was identified on genomic region CaLG05 for days to flowering, plant height and plant biomass with flanking marker Ca5_35884449 and Ca5_23069087 during 2015. During 2016, the genomic region on CaLG05 for the traits like plant height, plant biomass per plot and grain yield per plot with flanking marker Ca5_36747984 and Ca5_36430169 was identified. A consistent QTL was found for plant height on genomic region CaLG03 having flanking markers Ca3_21515505 and Ca3_21545929, explaining 5.98% to 22.28% of phenotypic variation in both seasons. The resistant parent ICCV 08305 is contributing the desirable alleles for all the QTLs found in the two genomic regions on CaLG05 and CaLG03.

The presence of co-localized QTLs for these significant traits along with high LOD value suggests a tight linkage or pleiotropy. These co-localized QTLs could be very useful in the improvement of these traits to develop high yielding chickpea cultivars. The presence of transgressive segregants and positive allele contribution by both the parents confirmed the fact that the sensitive parent has also beneficial alleles for enhancement of seed yield. Fine mapping of the target genomic region will help in resolving the issues of pleiotropy and tight linkage.

Consistent QTL was found for plant height on genomic region CaLG03 having flanking markers Ca3_21515505 and Ca3_21545929 explaining 5.98-22.28% of phenotypic variation with LOD 6.78-14.53 observed in both seasons. Another consistent QTL was found for days to flowering on genomic region CaLG05 having flanking markers Ca5_35884449 and Ca5_23069087 explaining 3.94-59.96% of phenotypic variation with LOD 3.23-35.18 observed in both seasons. The resistant parent ICCV 08305 is contributing the desirable alleles for all the QTLs found in the two genomic regions on CaLG05 and CaLG03.

During *rabi* 2015, moderately resistant ICCRIL14-151 and resistant ICCRIL14-152 lines recorded the highest grain yield per plot (292.33g) and (281.33g) and the highest harvest index (59.87%) and (57.02%), respectively. Moderately resistant ICCRIL14-119 and ICCRIL14-154 exhibited high plant biomass per plot (600g) and the maximum 100-seed weight (33.62g), respectively. During *rabi* 2016, ICCRIL14-69 was found as moderately resistant for DRR and recorded the highest grain yield (381.67g) and high biomass (610g). ICCRIL14-58 was found as resistant source for DRR with high grain yield (336.67g).

5.9 Suggested future research

The present study focused on genetic variability present in the RIL population under field conditions and suggested some traits which can be considered in selecting chickpea RILs for use in the breeding. The study also identified the genomic regions controlling DRR resistance genes in chickpea. However, much more needs to be done and the opportunities for future research are summarized below:

- The identified genomic regions with linked SNP markers can be further utilized for mining candidate genes and can also be deployed for introgressing DRR resistance trait into elite chickpea cultivars through marker-assisted selection (MAS) to develop resistant cultivar.
- RIL population used in this study was phenotyped in the field in one location across two seasons. This population should be further tested in multi-locations to identify stable QTLs.
- Two genomic regions containing several major QTLs responsible for yield traits in chickpea can be validated through functional genomics approach to identify more candidate genes for enhancement of yield in chickpea.

- The SNP markers found to be linked with DRR resistance need to be validated in the background of other population to use them in marker-assisted selection.

Summary and conclusions

VI. SUMMARY AND CONCLUSIONS

Recent years have witnessed significant progress in the development of genomic resources in chickpea that has led to identification of molecular markers for genes/quantitative trait loci (QTLs) controlling several agronomic traits in chickpea. The present research entitled “Molecular mapping of dry root rot resistance genes in chickpea (*Cicer arietinum* L.)” was carried out to identify molecular markers linked to DRR resistance genes in chickpea. To identify the genomic regions conferring resistance to DRR, 182 RILs were developed from a cross between a susceptible line BG 212 and moderately resistant breeding line ICCV 08305. This study provides insight into the genetic control of resistance to DRR in chickpea which can subsequently help in developing chickpea cultivars with resistance to DRR. The findings from the research are briefly summarized below:

High values of PCV and GCV for plant biomass per plant, grain yield per plot and 100-seed weight in both seasons indicated low environmental influence on the expression of these characters. Hence, these characters can be relied upon for simple phenotypic selection.

High heritability coupled with genetic advance as per cent of mean were recorded for harvest index in *rabi* 2015 and days to flowering, plant height, plant biomass per plot, grain yield per plot and 100-seed weight in *rabi* 2016 indicating additive gene action for these traits. Phenotypic selection in early generations will be effective to improve these traits.

Grain yield per plot had highly significant positive association with days to 50 per cent flowering, days to maturity, plant biomass per plot, 100-seed weight and harvest index in both seasons and at both levels, indicating that these traits are important for yield and selection can be done for these traits. Whereas, non- significant association was found between disease incidence and other agronomic traits including yield.

Path coefficient analysis revealed that days to 50 per cent flowering, plant biomass per plot, hundred seed weight and harvest index had positive direct effect on grain yield indicating direct selection for these traits will be rewarding for the improvement of grain yield per plot.

Analysis of variance (ANOVA) revealed that the 182 RILs along with the parents exhibited highly significant differences for all characters in both seasons including the pooled analysis indicating the useful genetic diversity in the material under study. DRR screening results in the two screenings and the pooled analysis of variance showed highly significant differences for disease incidence.

When the RILs were classified into two categories, Resistant (score >1 to <5) and Susceptible (score >5 to <9) was observed a good fit to 1R:1S ratio that with a non-significant chi-square value, indicating that the segregation of a major gene had attained homozygosity.

A high density genetic map with 13,110 loci spanning 1,224.11 cM with an average inter-marker distance of 0.09 cM was developed. A consistent minor QTL which was found on genomic region CaLG04 with flanking marker Ca4_4561482 and Ca4_4509561 explaining 5.87% of phenotypic variation (LOD 3.39) with 0.35 additive effect during first screening. Same QTL was identified explaining 7.41% of phenotypic variation (LOD 4.25) with 0.39 additive effect during second screening. Same QTL was found on LG4 having phenotypic variation of 6.11% (LOD 4.03) with 0.37 additive effect in pooled analysis as well. Another minor QTL was found on CaLG04 with 6.76% phenotypic variation (LOD 3.91) with -0.37 of additive effect during second screening. Additive effect indicates that the resistance gene was contributed by male parent ICCV08305.

Two potential genomic regions CaLG05 and CaLG03 were identified using ICIM software. The genomic region on CaLG05 harboring QTL for days to flowering, plant height and plant biomass per plot was identified during 2015. Similarly, during 2016 the genomic region CaLG05 was observed for the traits like plant height, plant biomass per plot and grain yield per plot. The presence of co-localized QTL for these significant traits along with high LOD value suggested tight linkage or pleiotropy. Fine mapping of the target genomic region will help in resolving the issues of pleiotropy and tight linkage.

Consistent QTL was found for plant height on genomic region CaLG03 with phenotypic variation of 22.28% and 5.98 % in both seasons. Moreover, the resistant parent

ICCV 08305 is contributing the desirable alleles for all the QTLs found in the two genomic regions on CaLG05 and CaLG03. The presence of transgressive segregants and positive allele contribution by both the parents confirmed the fact that the sensitive parent has also beneficial alleles for enhancement of seed yield.

During 2015, moderately resistant ICCRIL14-151 and resistant ICCRIL14-152 lines recorded highest grain yield per plot (292.33g and 281.33g) and these lines also exhibited high harvest index (59.87% and 57.02%, respectively). Moderately resistant lines ICCRIL14-119 and ICCRIL14-154 exhibited high plant biomass per plot (600 g) and maximum 100-seed weight (33.62g), respectively. During 2016, ICCRIL14-69 was found as moderately resistant for DRR and recorded the highest grain yield per plot (381.67g) and high plant biomass per plot (610g). ICCRIL14-58 was found as resistant for DRR with high grain yield per plot (336.67g).

The present intra-specific map will be helpful for mapping and tagging of the genes. The identified genomic regions with linked SNP markers will be further utilized for simultaneous selection at an early stage, mining candidate genes and can be deployed for introgressing dry root rot resistant trait into elite chickpea cultivars through marker-assisted selection (MAS) to develop resistant cultivars.

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Appendices

APPENDIX – A

Table 1. Weekly average weather data from October 2015 to March 2016 at ICRISAT, Patancheru

Week	Rain	Evap	Max Temp (%)	Min Temp (%)	Rel Humidity1 (%)	Rel Humidity2 (%)	Wind Velocity	Solar Radiation	Bright Sunshine
44	10	29.69	31.51	19.73	92.14	48.57	4.08	16.8	7.41
45	0	36.5	31.53	17.05	86.43	38	4.62	16.92	8
46	0	36.1	30.8	14.5	88.56	36.85	3.99	17.53	7.08
47	0.3	30.69	29.76	18.37	84.85	54.28	5.96	14.68	6.63
48	2.2	27.8	30.8	17.62	89.29	42.42	5.56	15.21	7.93
49	0	27.1	29.94	14.57	90	44	4.63	15.07	6.74
50	0	29.39	32.71	17.03	91.56	36.42	5	15.13	7.96
51	0	34.79	32.68	16.39	91.85	32.57	6	16.67	8.97
52	0	38.7	30.48	9.63	83.87	28.75	3.49	17.03	9.14
1	0	34.29	30.91	11.42	91.43	37.28	4.04	17.03	9.57
2	0	33	29.76	11.1	87	39.14	4.54	16.69	8.57
3	0.2	37	29.89	16.53	80.56	40.71	8.31	15.26	6.4
4	0	31.6	29.94	14.4	78.14	38.85	4.88	15.99	7.2
5	0	41.2	34.2	12.5	77.56	27.42	3.5	17.96	9.64
6	0	44.39	33.53	16.73	80.7	29.71	6.57	16.8	8.07
7	0	57	33.57	18.1	77.14	29	8.23	17.91	8.42
8	0	59.89	36.29	18	73.56	28.14	7.63	19.03	9.06
9	0	58.29	33.89	20.85	79.87	39.63	7.96	16.78	6.05
10	0	63.29	35.82	20.3	68.85	32	7.56	18.58	6.49
11	0	66.09	37.28	21.94	71.56	24	7.79	19.96	8.39

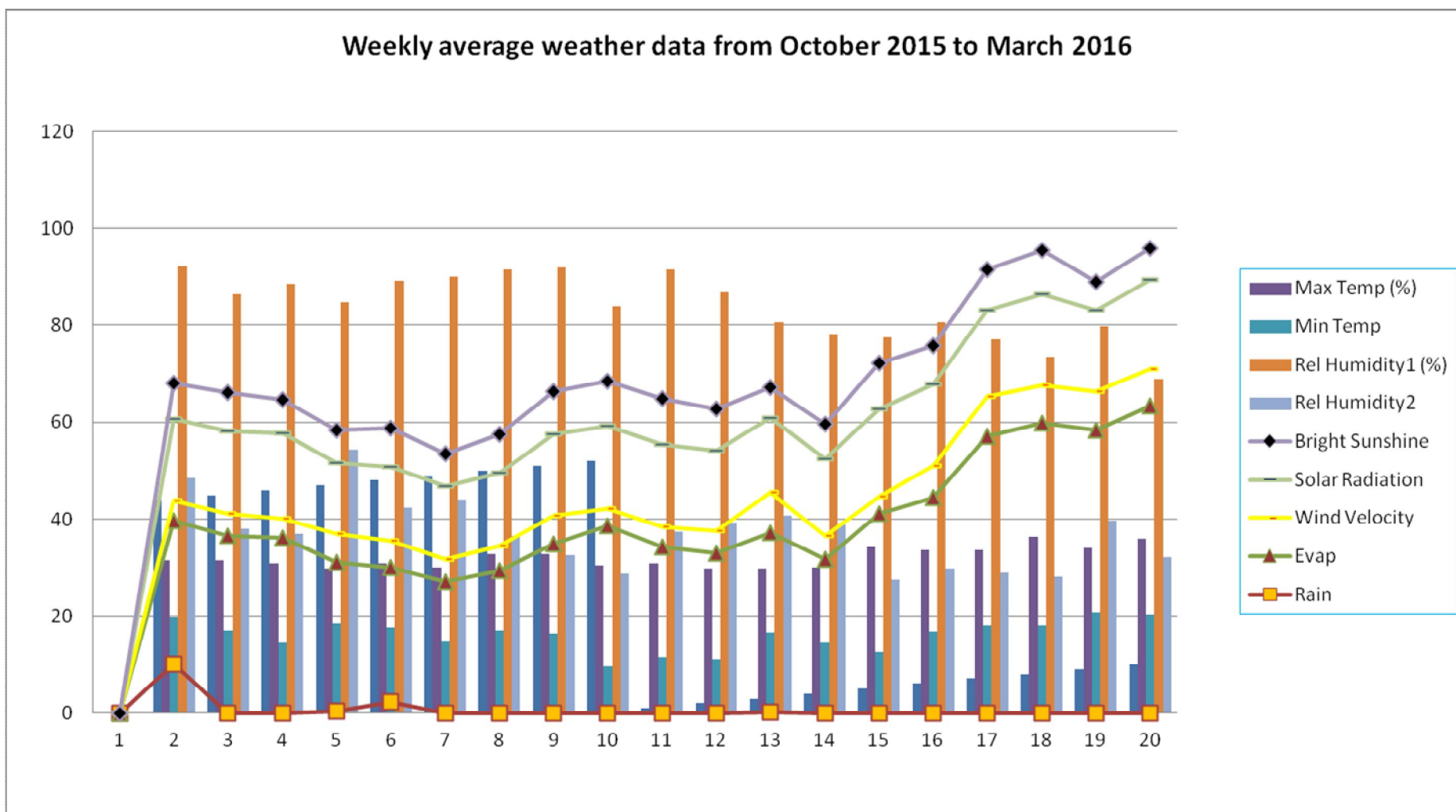


Fig. 20. Weekly average weather data from October 2015 to March 2016 at ICRISAT, Patancheru

APPENDIX – B

Table 2. Weekly average weather data from October 2016 to March 2017 at ICRISAT, Patancheru

Week	Rain	Evap	Max Temp (%)	Min Temp (%)	Rel Humidity1 (%)	Rel Humidity2 (%)	Wind Velocity	Solar Radiation	Bright Sunshine
44	0.2	27.19	30.48	20.19	86.43	47.42	3.79	16.62	7.2
45	0	25.3	29.76	14.19	84.29	33.85	2.89	18.1	8.74
46	0	25.8	29.91	16.57	86.14	42.71	3.89	15.53	6.7
47	0	27.69	29.76	11.28	88	30.85	2.83	17.42	8.35
48	0	26.5	30.94	10.73	87.7	28.57	2.29	17.14	8.41
49	0	22.19	29.1	14.6	89.56	47.28	4.19	14.24	6.73
50	1	22.1	28.05	14.18	85.43	46.28	4.67	13.28	6.42
51	0	27.39	29.23	9.89	90	31.42	2.87	17.17	9.1
52	0	28.69	29.3	9.8	88.87	29.5	3.16	16.73	9.22
1	0	26.39	28.71	10.81	88.14	29.42	3.56	16.76	9.28
2	0	27	28.82	13.83	85.43	37.14	5.11	15.31	7.87
3	0	31.1	28.23	12.92	83.85	36	6.05	16.44	7.76
4	0	34.5	29.89	15.96	82.85	36.71	5.79	16.1	7.79
5	0	35.6	31.46	13.1	85.43	31.42	4.75	18.23	8.97
6	0	37.79	32.29	15.19	77	29.14	5.11	18.5	9.14
7	0	43.29	30.94	14.43	79.7	30	7.11	18.8	8.92
8	0	45.29	34.99	14.75	73	19.85	4.04	21.03	10.31
9	0	52.5	34.39	14.19	64.56	18	5.81	20.78	9.24
10	0	49.1	34.89	19.62	76.7	30	6.99	18.64	6.3
11	0.6	46.39	33.89	18.73	73.85	30.57	6.16	19.69	6.46

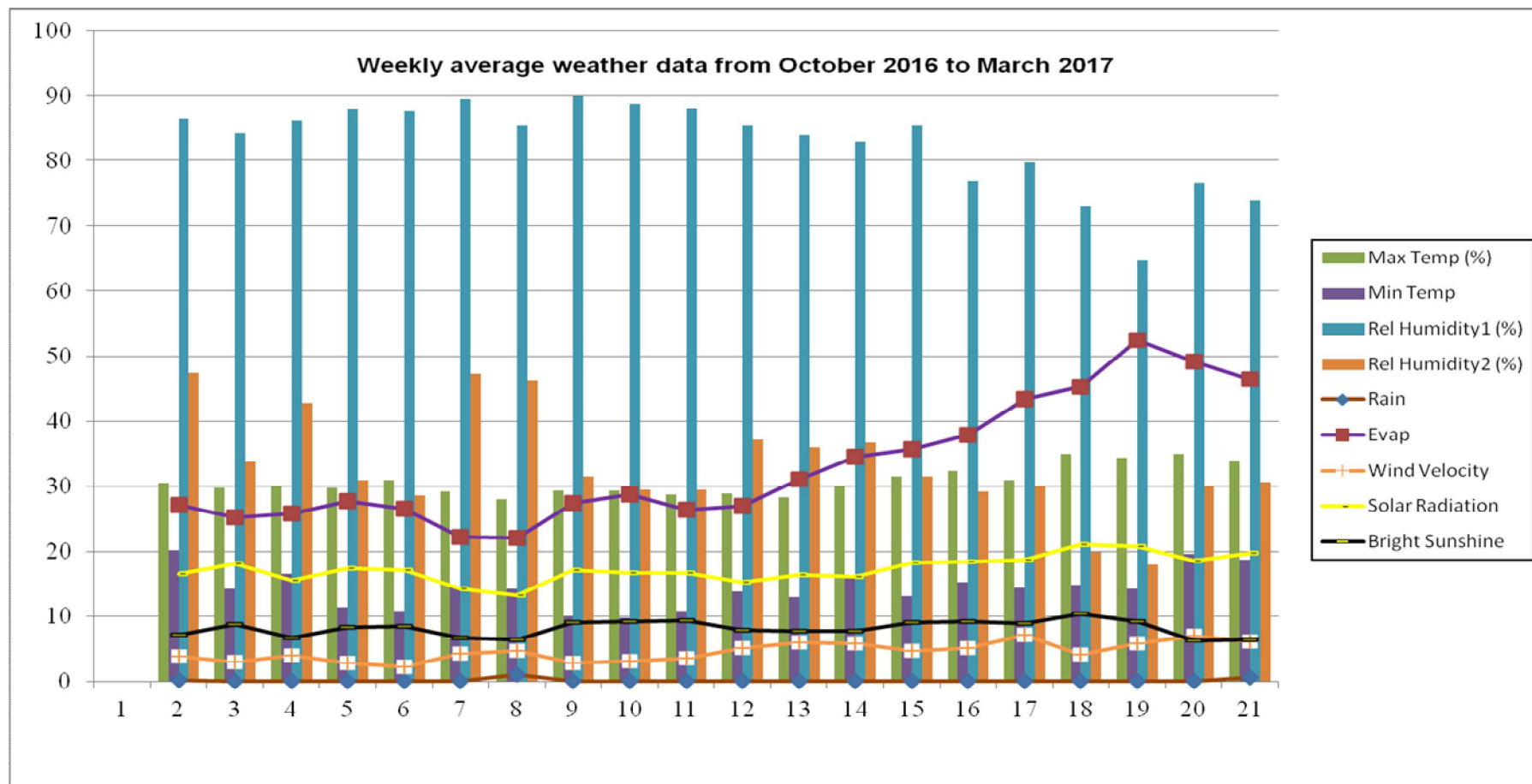


Fig. 20. Weekly average weather data from October 2015 to March 2016 at ICRISAT, Patancher

MOLECULAR MAPPING OF DRY ROOT ROT RESISTANCE GENES IN CHICKPEA (*Cicer arietinum* L.)

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2018

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ABSTRACT

Dry root rot (DRR) caused by *Rhizoctonia bataticola* (Taub.) Butler [Synonyms: *Macrophomina phaseolina* (Tassi) Goid] is an emerging disease of chickpea gaining importance in the changed scenario of climate. To identify the genomic regions conferring resistance to DRR, 182 Recombinant Inbred Lines (RILs) were developed from the cross between a susceptible line BG 212 and moderately resistant line ICCV 08305. The RILs were screened in the laboratory using "Paper Towel Method". Thirteen RILs were found as resistant, whereas 79, 78 and 12 RILs showed moderate resistance, susceptible and highly susceptible reaction, respectively. The frequency distribution for DRR disease incidence depicted a normal distribution which showed quantitative nature. RILs were classified in two categories (resistance and susceptible), exhibited a ratio of 1R:1S which gives good fit with the expected ratio and chi-square value was non-significant. The RIL population was genotyped with cost-effective SNP genotyping platform, comprising of 50,590 high quality non-redundant SNPs tiled on to Affymetrix[®] Axiom[®] genotyping array. Out of the 50,590 SNPs, 13,110 were highly polymorphic and used for genetic map construction. As a result, a high density genetic map with 13,110 loci spanning 1,224.11 cM with an average inter-marker distance of 0.09 cM was developed. A consistent minor QTL was found for DRR on CaLG04 explaining phenotypic variation ranging from 5.87-7.41%. Two potentially important genomic regions were identified on CaLG05 and CaLG03 contains major loci for days to 50% flowering, plant height, plant biomass per plot and grain yield per plot. Resistant ICCRIL14-69 recorded high grain yield per plot (292.33g), plant biomass per plot (610g) and maximum harvest index (62.6%) during 2016. The identified genomic regions with linked SNP markers will be further utilized for introgressing DRR resistant trait into elite chickpea cultivars through Marker-Assisted Selection (MAS) to develop dry root rot resistant cultivars.